



Certified Lean Six Sigma Green Belt (CLSSGB)

Course Code: C481

Conducted by Tertiary Infotech Academy Pte Ltd · UEN: 201200696W

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COURSE ADMINISTRATION

Welcome & Housekeeping

About the Trainer



Your Trainer

General Trainer template —
to be completed by the trainer

NAME

TITLE / DESIGNATION

QUALIFICATIONS

AREAS OF EXPERTISE

TRAINING & INDUSTRY EXPERIENCE

CONTACT

About the Trainer



Dr. Alfred Ang

Principal Trainer
Tertiary Infotech Academy Pte. Ltd.

ROLE

Principal Trainer, Tertiary Infotech Academy Pte. Ltd.

QUALIFICATIONS

PhD; Certified Lean Six Sigma Black Belt practitioner and trainer.

DELIVERS

Courses on Lean Six Sigma, quality management and data analytics.

EXPERIENCE

Process improvement across manufacturing, service and technology sectors.

FOUNDER

Founder and lead instructor at Tertiary Infotech / Tertiary Courses.

Let's Know Each Other

- Your name, organisation and role.
- Your experience with process improvement, quality or data analysis.
- Have you completed a Yellow Belt or run an improvement project before?
- One process at work that frustrates you — we may use it as your course scenario.

Ground Rules

1

Set your mobile phone to silent mode.

2

Participate actively — no question is too small.

3

Mutual respect: agree to disagree.

4

One conversation at a time.

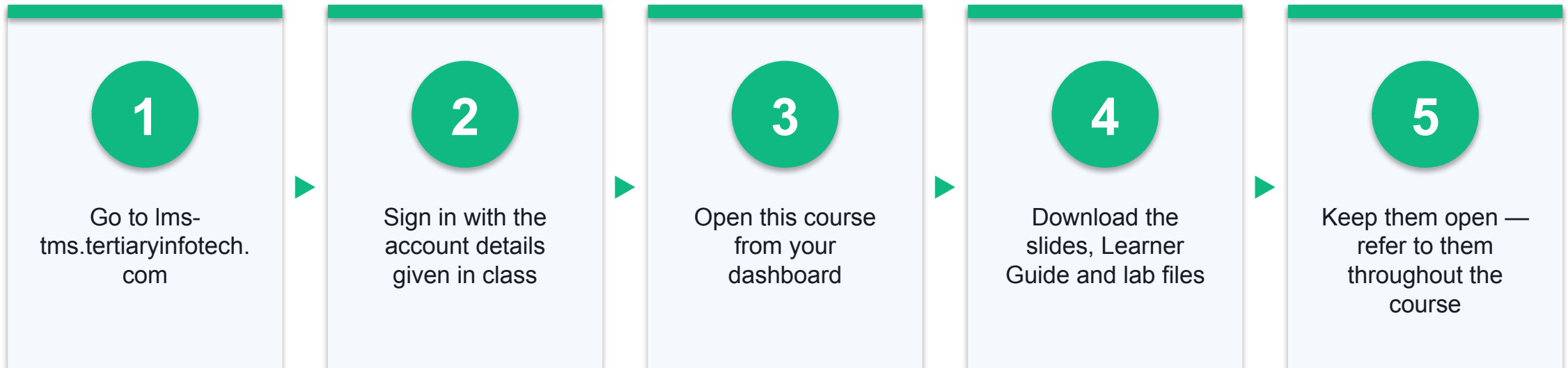
5

Be punctual; return from breaks on time.

6

Full attendance — attend all four days to get the most from the course.

Download Your Course Material



Your Interactive Toolkit

1

SIPOC & Process Map Builder

alfredang.github.io/sipoc — SIPOC, pain points, swimlane and handoff table (Labs 6-7).

2

5 Whys

alfredang.github.io/5whys — drill from symptom to an actionable root cause (Lab 16).

3

Fishbone Diagram

alfredang.github.io/fishbone — Ishikawa cause-and-effect by 5M+E (Lab 16).

4

Pareto Chart

alfredang.github.io/paretochart — the team brainstorms and votes, the chart builds live (Labs 15, 18).

5

NovaSPC

alfredang.github.io/novaspc — run charts, SPC charts and capability from your CSV (Labs 13, 14, 23).

6

No install needed

All five run in the browser. No licence, no setup — just open the link.

Your Lab Data

1

One dataset, twenty-five labs

4,200 Northwind orders from one month — 357 of them late (8.5%). Every lab cuts the same data.

2

It reconciles end to end

Your Lab 2 sigma level, your Lab 15 Pareto and your Lab 25 before/after all trace to those same orders — the numbers always agree.

3

Excel, in every lab folder

28 workbooks. Open labs/NN - <Title>/data/ and the file is there, with the column guide built into the sheet.

4

One consistent baseline

Every lab quotes the same 4,200 orders and 357 late, so your figures reconcile across the whole course.

5

Real statistics, not toy numbers

A genuine special cause on day 19, one unreliable Gage appraiser, a real regression at $r = 0.88$ — the data rewards doing the analysis properly.

6

Bring your own tool

Excel, Google Sheets, LibreOffice or NovaSPC — the workbooks open anywhere.

Lesson Plan — Days 1 & 2

Day 1

- Day 1 — Six Sigma foundations, the Green Belt role and the Define phase
 - Welcome · Introductions
 - Foundations: quality, Lean, Six Sigma, $Y = f(X)$
 - Sigma level, DPMO, COPQ, belt roles, DMAIC
 - Labs 1-2 — project selection, $Y = f(X)$, baseline sigma
 - DEFINE: VOC, affinity, Kano, CTQ trees
 - Charter, problem statement, SIPOC, stakeholders
 - Labs 3-6 — VOC/Kano, CTQ tree, charter, SIPOC

Day 2

- Day 2 — Measure — process mapping, sampling, MSA and baseline capability
 - MEASURE: process mapping, swimlanes, 8 wastes
 - VSM, takt time, lean metrics
 - Data types, sampling, sample size calculation
 - MSA / Gage R&R — accuracy, repeatability, reproducibility
 - Yield, DPU, DPO, DPMO, RTY, baseline capability
 - Labs 7-13 — mapping, VSM, sampling, MSA, metrics

Lesson Plan — Days 3 & 4

Day 3

- Day 3 — Analyze — root cause, hypothesis testing, correlation and regression
 - ANALYZE: common vs special cause, run charts
 - Pareto, stratification, boxplots
 - Fishbone, 5 Whys, multi-voting
 - Hypothesis testing: H_0/H_a , α , p-values, test selection
 - Type I & II errors, correlation, regression
 - Labs 14-18 — run charts, Pareto, root cause, hypothesis tests

Day 4

- Day 4 — Improve and Control
 - Welcome back · Day 3 recap
 - IMPROVE: solution generation, benchmarking
 - Solution selection matrix, 5S, poka-yoke, FMEA, DOE
 - CONTROL: SPC, control chart selection, C_p/C_{pk}
 - Out-of-control rules, control plan, SOP, handover
 - Labs 19-25 — selection matrix, FMEA, SPC, control plan
 - Revision · Course wrap-up and next steps

Learning Outcomes

1

LO1 — Lead & scope

Define the problem, scope the work and charter the project.

2

LO2 — Map & baseline

SIPOC, detailed process maps and value stream maps.

3

LO3 — Valid measurement

Sampling, sample size, MSA/Gage R&R and capability.

4

LO4 — Analyse statistically

Pareto, run charts, hypothesis testing, regression.

5

LO5 — Prove root cause

Fishbone, 5 Whys and statistical evidence.

6

LO6 — Select & de-risk

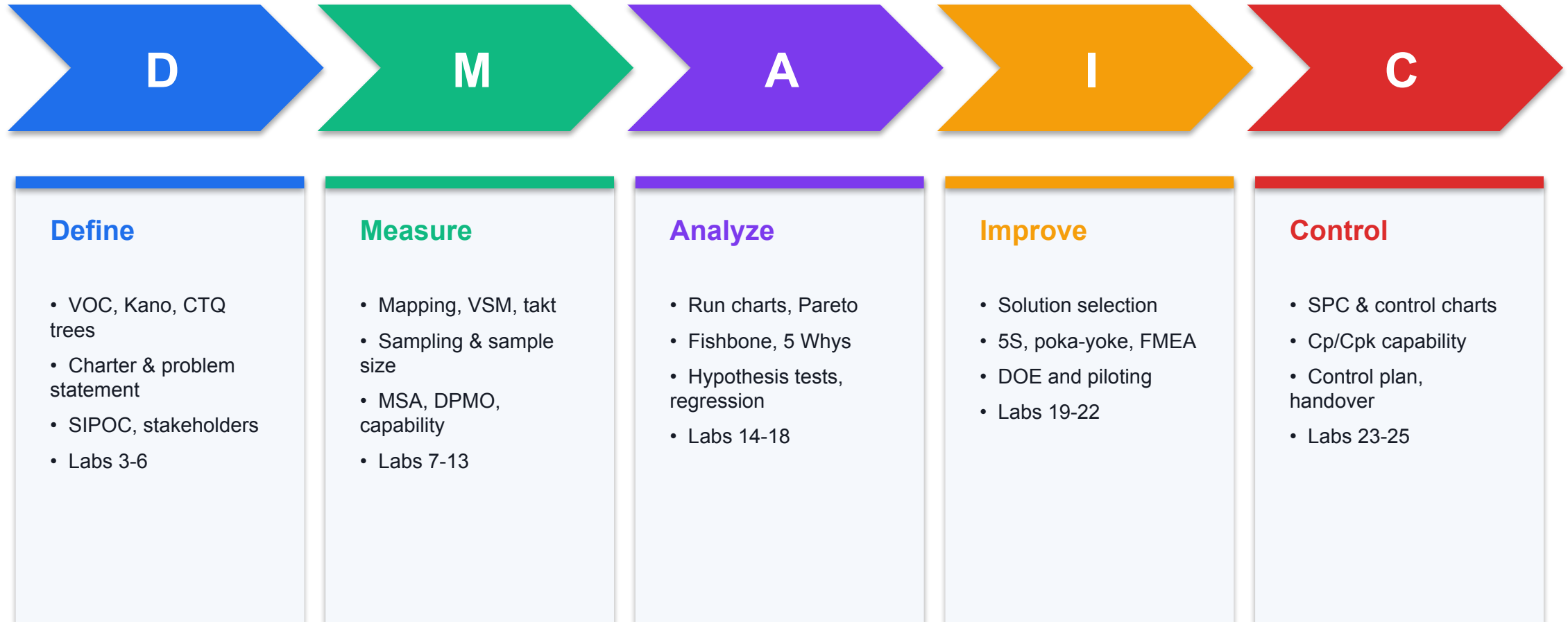
Selection matrices, FMEA and DOE before piloting.

7

LO7 — Sustain the gain

SPC control charts, Cp/Cpk and the control plan.

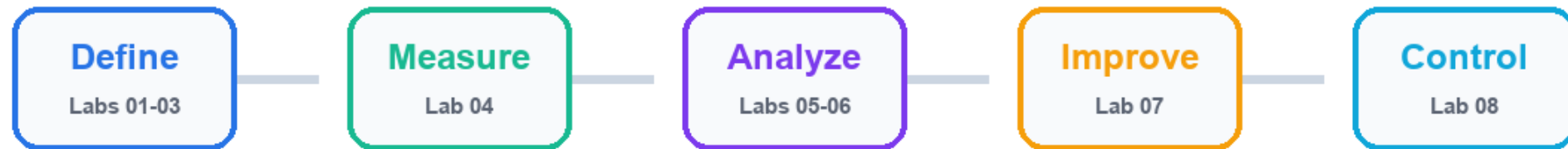
Course Outline — We Follow DMAIC End to End



The DMAIC Roadmap

CLSSGB DMAIC Lab Roadmap

Each lab builds one artifact for the final A3 project story.



Every topic and lab in this course sits inside one of these five phases.

FOUNDATIONS

Six Sigma Foundations & The Green Belt Role

Quality - Lean - Six Sigma - $Y = f(X)$ - Belt roles - COPQ - Project selection - DMAIC

Three Common Answers - Only One Survives

Defect-free

The zero-defect answer

- Sounds strong, but defect-free of WHAT?
- A part with no scratches can still be late
- Says nothing about what the buyer valued
- Verdict: incomplete

Meets specification

The engineering answer

- Better - it is measurable and auditable
- But WHO wrote the spec, and from what?
- A wrong spec, met perfectly, is still failure
- Verdict: necessary, not sufficient

Meets customer expectations

The Six Sigma answer

- The customer defines what good looks like
- The spec is then derived from that voice
- Covers delivery, service and cost, not just parts
- Verdict: this is the one we use

Quality in a Nutshell - Four Obligations

1

Understand requirements

Capture what the customer actually needs - Voice of the Customer, not assumption.

2

Design to satisfy them

Translate needs into specific, measurable CTQ requirements the process can be built around.

3

Deliver consistently

Consistency is the hard part - variation, not the average, is what customers feel.

4

Improve continuously

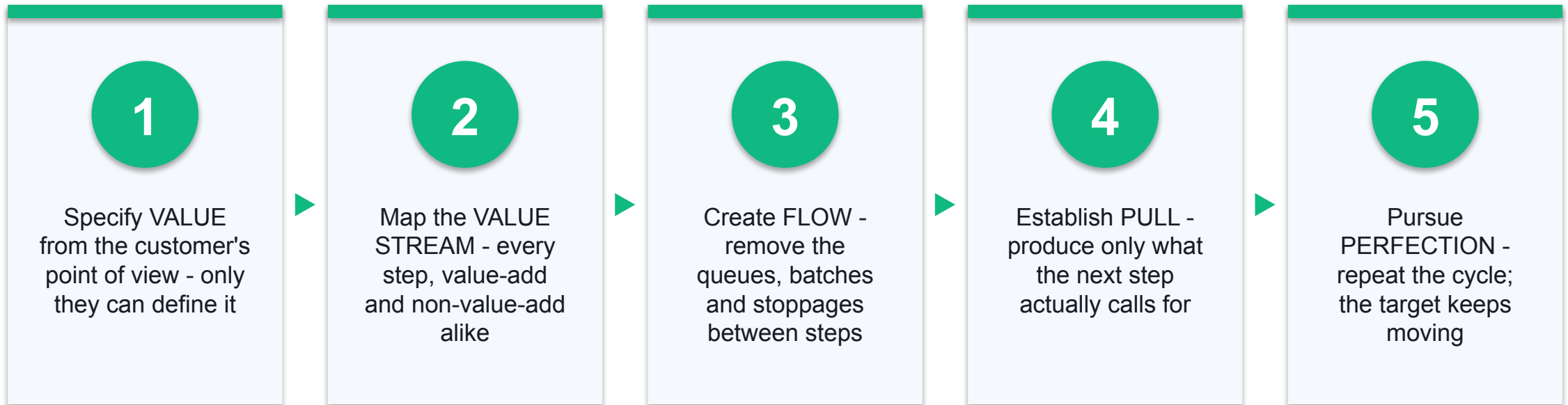
Requirements move. A process that stands still is quietly getting worse.

THE RUNNING SCENARIO

Northwind Retail Distribution Centre

Our running case for four days: order fulfilment, 12,400 orders a month, promised in 48 hours. Today 18 per cent arrive late and customers are leaving. Every tool gets applied here.

The Five Lean Principles



Lean at Northwind - What It Would See

1

Waiting

Picked orders sit 4.2 hours on the staging floor waiting for the carrier sweep.

2

Motion

Fast-moving SKUs stored at the far aisle - pickers walk 11 km per shift.

3

Over-processing

Every order double-checked, including the 92 per cent that were never in doubt.

4

Inventory

Three weeks of packaging stock tying up cash and floor space.

Why Variation - Not the Average - Is the Enemy

The average lies

Both DCs average 46 hours

- Northwind: 46 hours average delivery
- Southport: 46 hours average delivery
- Identical on the monthly report
- Completely different customer experience

Northwind

Average 46 h, spread 28 to 71 h

- 18 per cent of orders breach the 48 h promise
- Customers cannot plan around it
- Expediting cost SGD 340k a year
- High variation - low capability

Southport

Average 46 h, spread 43 to 49 h

- Under 1 per cent breach the promise
- Customers trust the date given
- Almost no expediting
- Low variation - high capability

Lean + Six Sigma = Lean Six Sigma

Lean

- Speed, flow and lead time
- Removes waste and non-value-add steps
- Fast, visual, shop-floor tools
- Weak on statistical proof



Six Sigma

- Accuracy, capability and variation
- Removes defects and inconsistency
- Rigorous data and statistics
- Can be slow on obvious waste



LEAN SIX SIGMA

Do the right things fast AND right - waste removal with statistical proof, run through DMAIC

When Each Half Does the Heavy Lifting

1

Long lead time, few defects

Lean leads - map the value stream, cut queues and batch sizes.

2

Short lead time, many defects

Six Sigma leads - find the Xs driving variation and control them.

3

Both problems at once

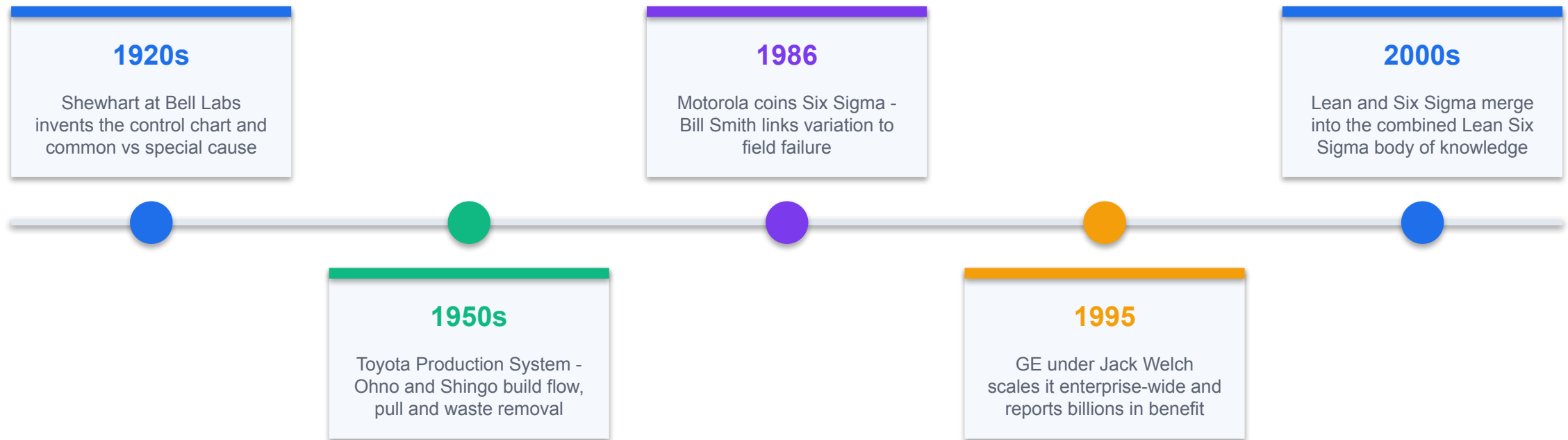
Lean first to expose the real process, then Six Sigma on what remains.

4

Neither is clear yet

Measure first. Never choose the toolkit before you have baseline data.

How We Got Here - A Short History



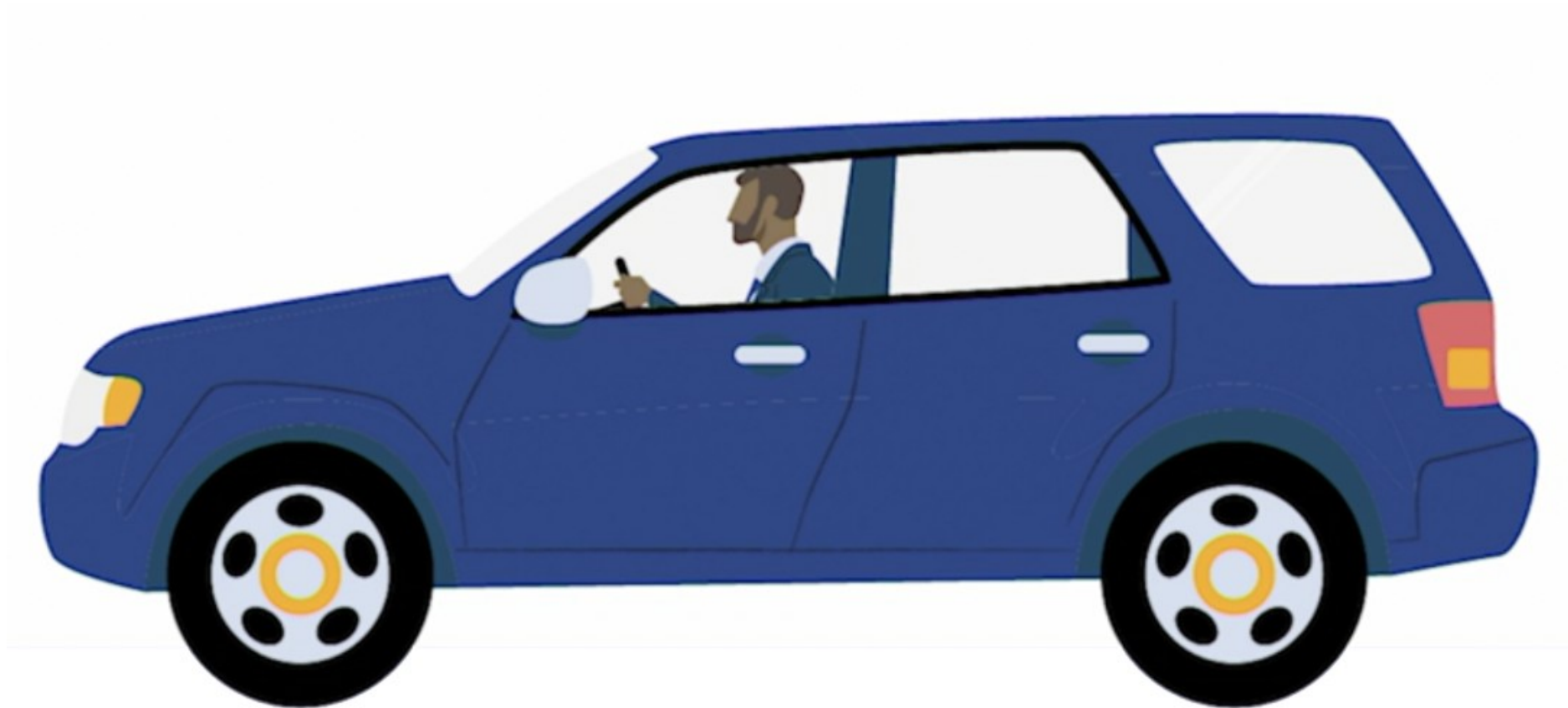
FOUNDATIONS 6 - THE CENTRAL MODEL

$$Y = f(X)$$

The output Y is a function of the process inputs X. You cannot manage Y directly - you can only find, verify and control the vital few Xs that drive it.

THE CENTRAL MODEL

The $Y = f(X)$ Model



Y is the dependent output the customer feels; the X s are the independent inputs you control.

Y and X at Northwind

THE Y - OUTPUT

- Y = order delivered within 48 hours
 - Dependent - it is an OUTCOME
 - Measured after the fact
 - Cannot be changed by instruction
 - Shouting at the Y does nothing
 - This is what the customer feels

THE Xs - INPUTS

- Candidate Xs driving that Y
 - Pick path length and pick errors
 - Carrier sweep schedule and cut-off time
 - Order-entry accuracy at the front end
 - Stock accuracy in the pick face
 - Shift staffing versus order arrival profile

Three Kinds of X - Know Which You Have

1

Controllable X

You can set it and hold it - cut-off time, slot layout, batch size. This is where solutions live.

2

Noise X

Real but not economically controllable - weather, traffic, customer order mix. Design robustness against it.

3

Critical X

A controllable X that is statistically proven to move Y. The vital few you will control in phase C.

Classical Quality vs Six Sigma Quality

CLASSICAL VIEW ON QUALITY 99% GOOD

About 1,000 defected bulbs
manufactured per hour

Unsafe air inhaled for almost 15
minutes each day

Two short or long unplanned
stoppage at most major railway
stations each day

180,000 lost couriers each year



SIX SIGMA VIEW ON QUALITY 99.99966% GOOD

Eight defected bulbs manufactured
per hour

Unsafe air inhaled for per minute
every seven months

One short or long unplanned
stoppage every five years

61 lost couriers each year

Left - inspect and sort after the fact. Right - understand and control the Xs before the defect exists.

Classical vs Six Sigma - Side by Side

Classical Quality

Detection and containment

- Inspect the output, sort good from bad
- Goal is conformance to specification
- Defect cost is treated as unavoidable
- Quality owned by the QA department
- Reacts to the symptom

Six Sigma Quality

Prevention and variation reduction

- Study the process, control the Xs
- Goal is reduced variation around target
- Defect cost is treated as recoverable waste
- Quality owned by the process owner
- Removes the root cause

The Focus of Six Sigma

"There is an underlying cause for every effect to occur"

Effects

Causes

$$Y = f(X)$$

Dependent

Independent

For every Y to occur there could be one or more X's

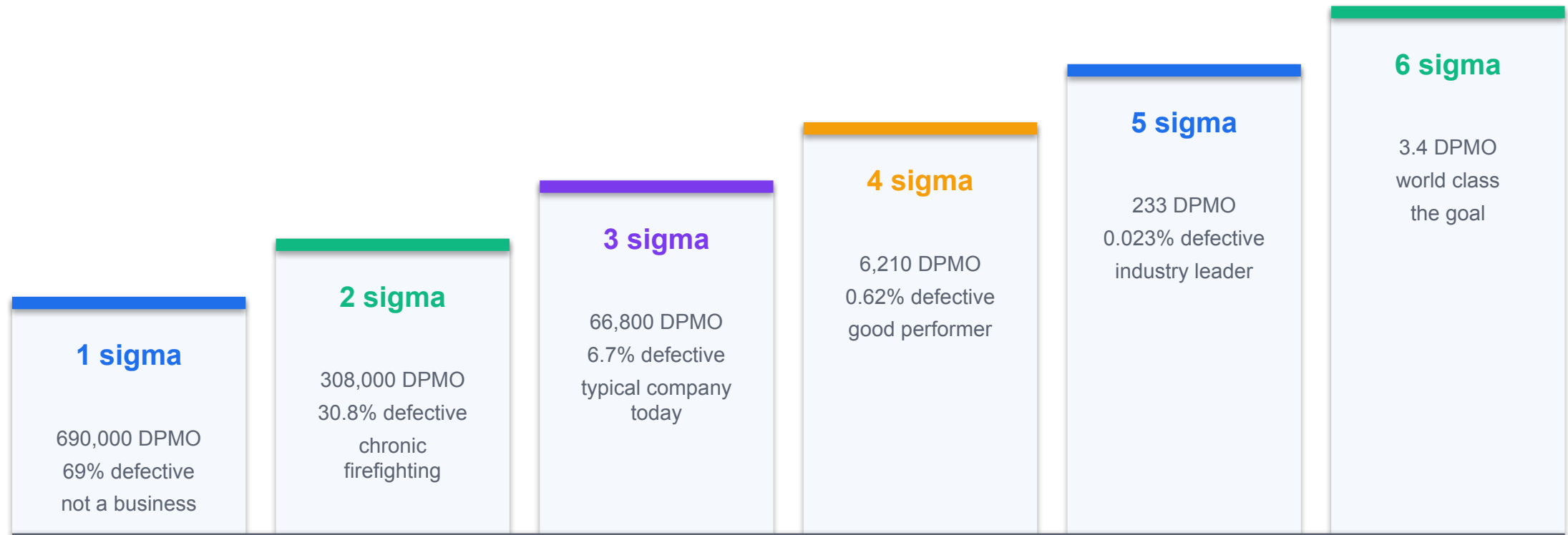
Getting late to office

*Getting up late
Unusually heavy traffic
Drop kids to school
Cook food before leaving
Flat tire*

"The focus of Six Sigma is on the Causes and NOT on the Effect"

Move the mean onto target AND shrink the spread - both are required for capability.

Sigma Level to DPMO - The Verified Table



Most organisations run between 3 and 4 sigma. Each step up is roughly a ten-fold cut in defects.

Why 3.4 and Not 2 Per Billion - The 1.5 Sigma Shift

1

The short-term picture

A perfectly centred process with six sigma between mean and spec gives about 2 defects per BILLION.

2

Processes drift

Over weeks and months the mean wanders - tool wear, staff changes, supplier lots, seasonality.

3

Motorola's allowance

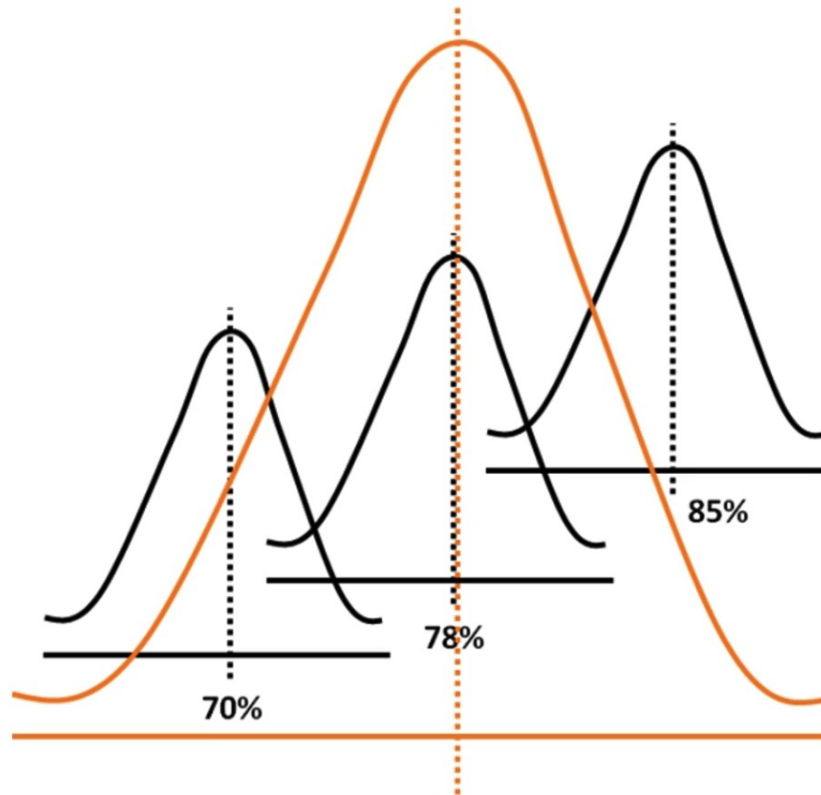
Empirically that long-term drift is about 1.5 sigma, so the effective margin is 4.5 sigma.

4

The result

4.5 sigma in the tail gives 3.4 DPMO. That is why six sigma means 3.4, not 0.002.

The 1.5 Sigma Long-Term Shift



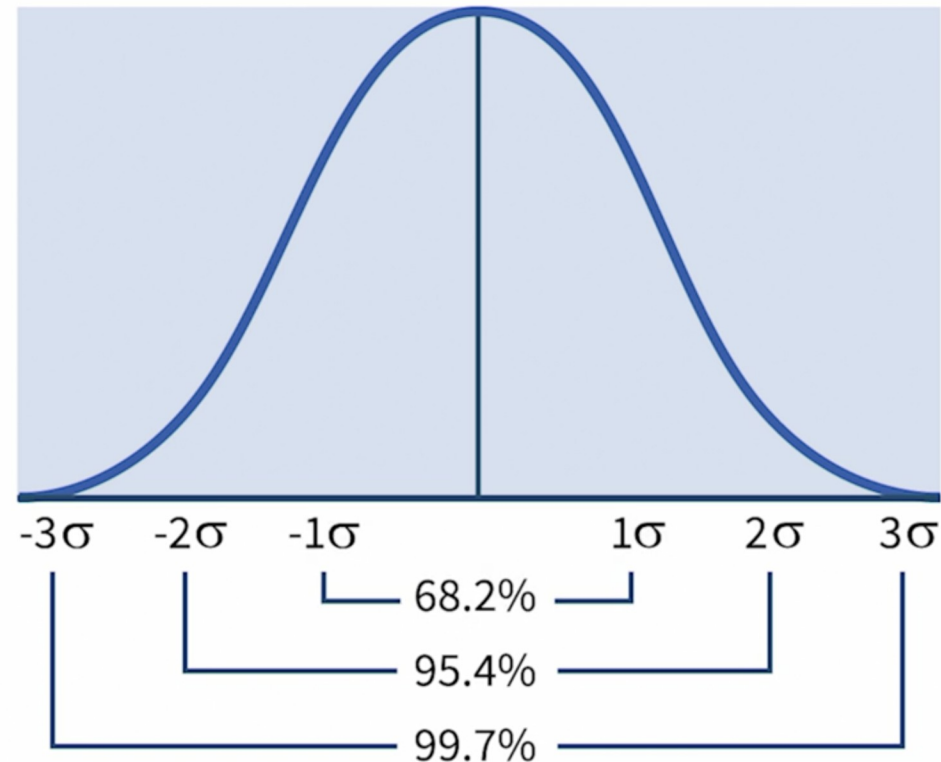
$$Z_{ST} = 1.5 + Z_{LT}$$

$$6 = 1.5 + 4.5$$

Short-term capability is measured over a snapshot; long-term performance absorbs the drift.

WHAT 'SIX SIGMA' LITERALLY MEANS

Six Standard Deviations Between Mean and Spec



The name is geometric - six standard deviations must fit between the process mean and the nearest limit.

Process Capability at Different Sigma Levels

Sigma Level	DPMO
2 σ	308,537
3 σ	66,807
4 σ	6,210
5 σ	233
6 σ	3.4



Most industries operate at
3 Sigma level



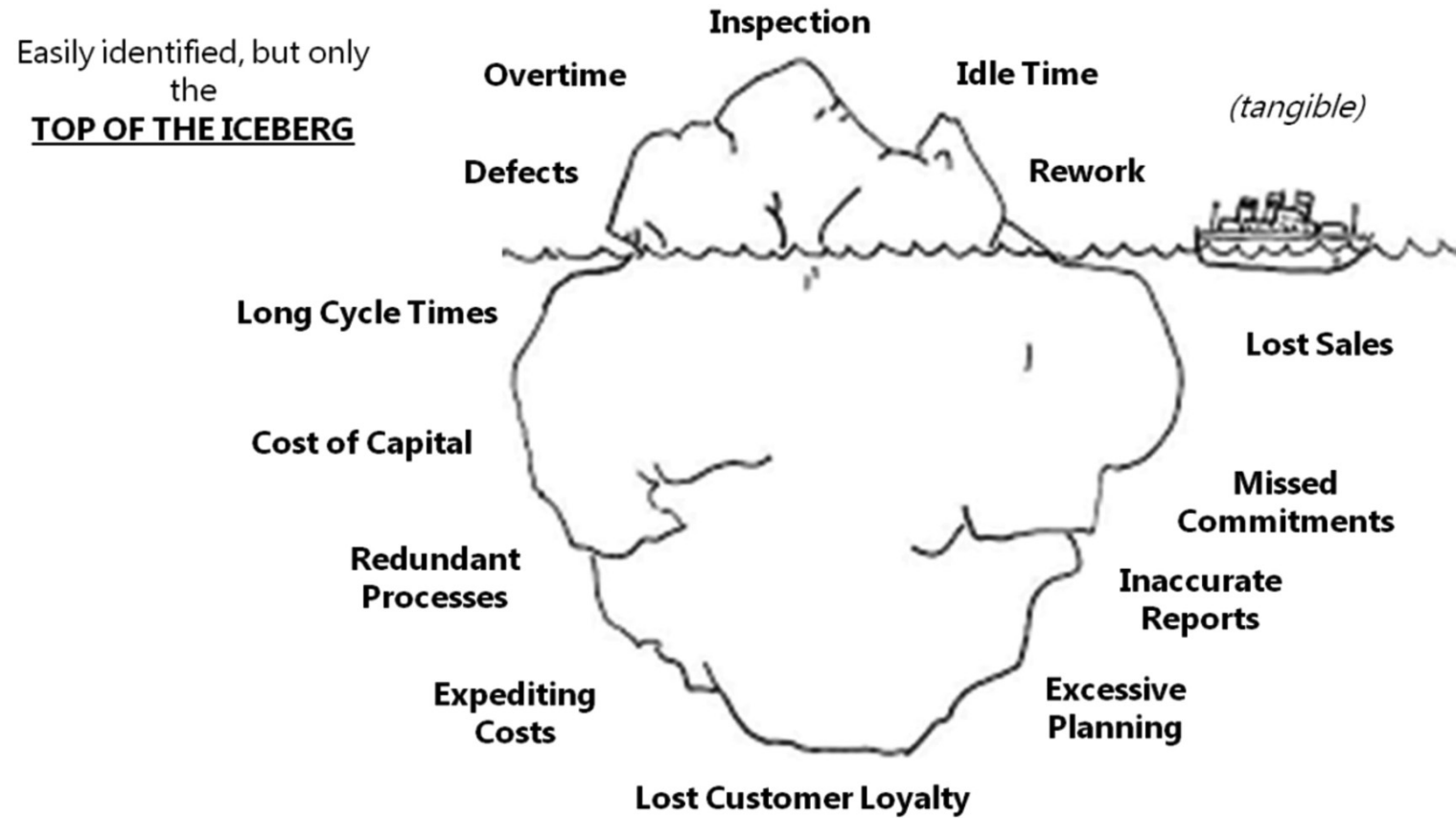
Automobile
Industry

Best in Class
Companies

Same spec limits, different spreads - capability is the ratio of customer tolerance to process voice.

VISIBLE VS HIDDEN COST

The COPQ Iceberg



What accounting reports is the tip. The submerged costs are larger and rarely on anyone's ledger.

Above the Waterline and Below It

Visible COPQ

Measured, budgeted, argued over

- Scrap and written-off stock
- Rework and re-picking
- Warranty and credit notes
- Returns processing and freight
- Inspection and sorting labour

Hidden COPQ

Real money, no cost code

- Lost customers and lost future orders
- Expediting and premium freight
- Overtime to recover the schedule
- Excess inventory held as a safety net
- Damaged reputation and lost bids

COPQ at Northwind - Building the Business Case

1

Expedited freight

SGD 340,000 a year in premium carrier charges to rescue late orders.

2

Service credits

SGD 185,000 in contractual late-delivery credits issued to key accounts.

3

Recovery overtime

SGD 96,000 in weekend shifts run purely to clear the late backlog.

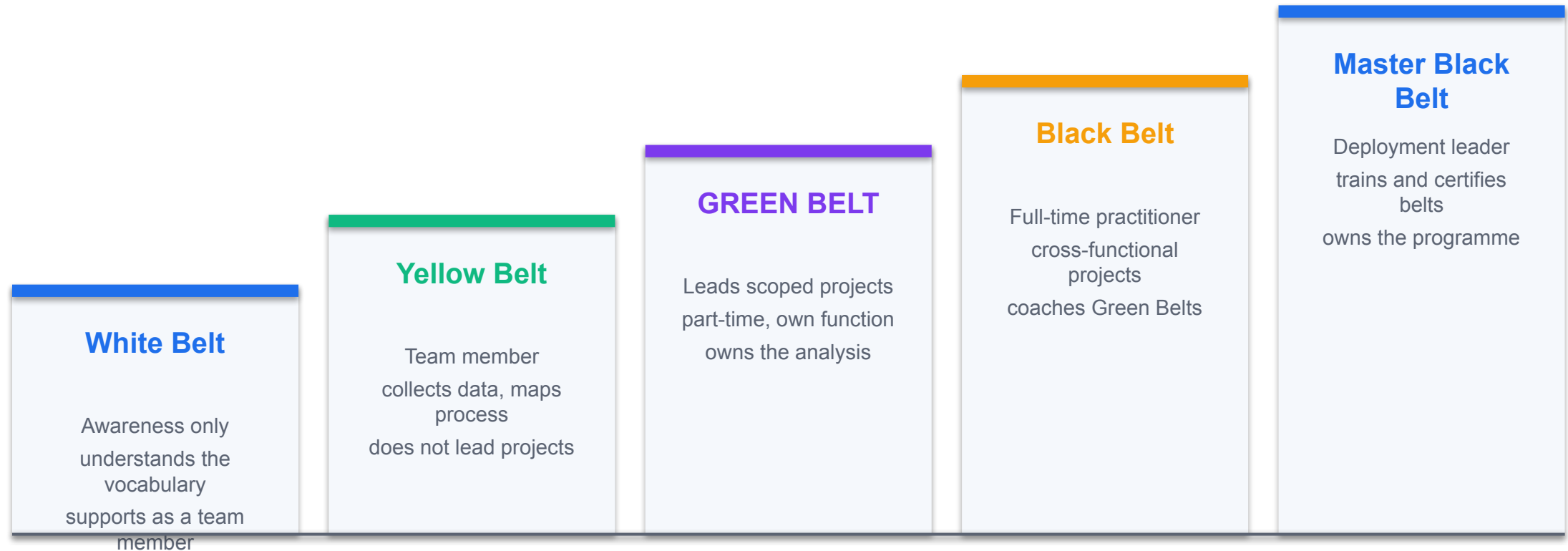
4

Churned accounts

Two accounts lost last year - SGD 1.2m of annual revenue, entirely hidden.

THE PATHWAY

The Six Sigma Belt Pathway



Alongside the belts sit the Champion and Sponsor - executives who select projects, fund them and remove barriers.

Green Belt vs Yellow Belt - Be Precise

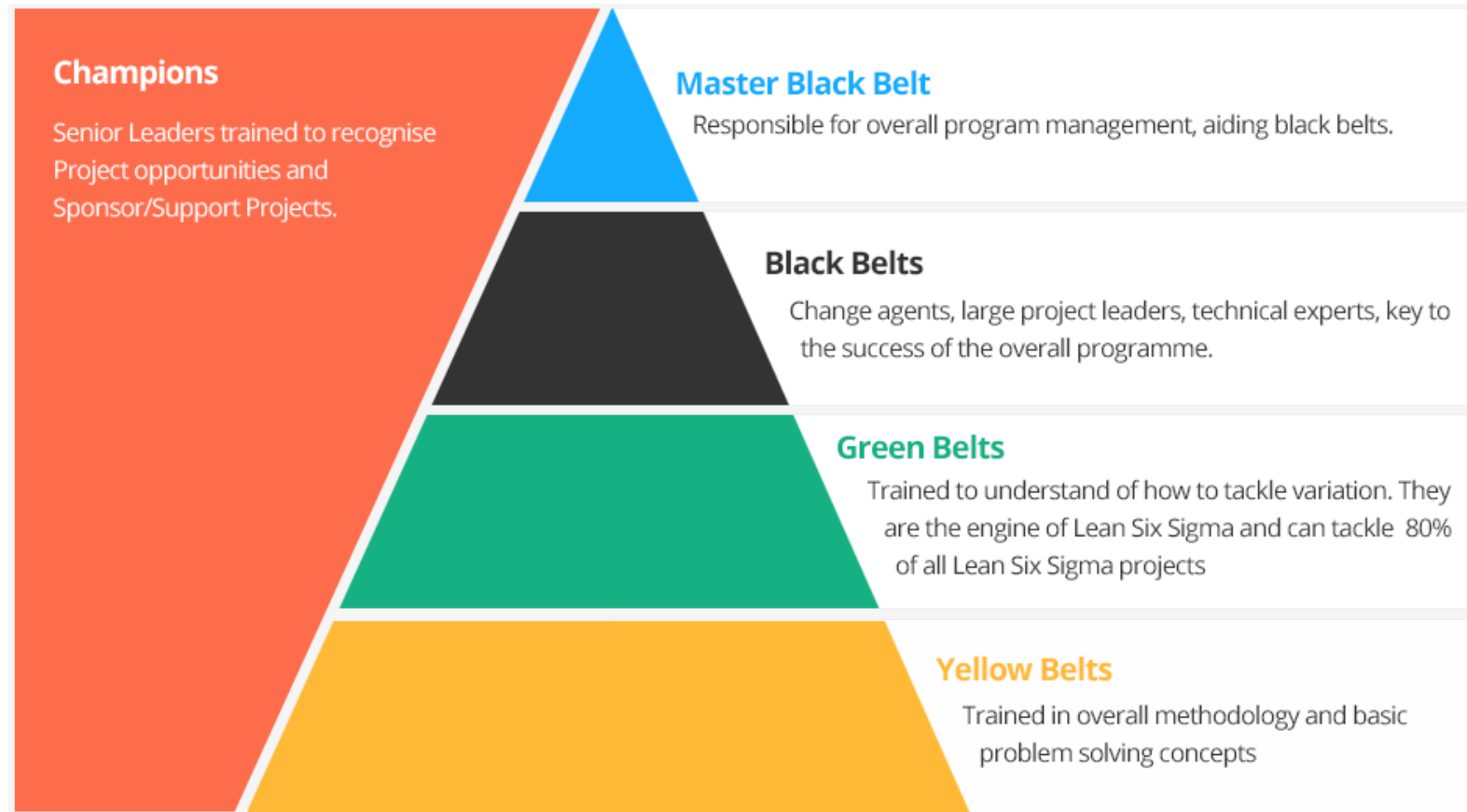
YELLOW BELT

- A Yellow Belt
 - Contributes to someone else's project
 - Collects data as instructed
 - Helps build the process map
 - Uses basic tools - checksheet, Pareto
 - Does not own the charter or the analysis
 - Does not face the tollgate panel

GREEN BELT

- A GREEN BELT
 - LEADS a scoped DMAIC project end to end
 - Owns the charter, scope and problem statement
 - OWNS the statistical analysis - MSA, capability, hypothesis tests
 - RUNS the tollgate reviews with the Champion
 - Selects and defends the vital few Xs
 - Hands over a control plan to the process owner

Six Sigma Roles and Responsibilities



Champion selects and funds; Black Belt coaches; Green Belt delivers; process owner sustains.

The Non-Belt Roles You Must Work With

1

Champion

Senior leader who selects the project, approves the charter and clears organisational barriers.

2

Sponsor

Owns the business area and the benefit. Signs off the financial validation with Finance.

3

Process Owner

Runs the process day to day and inherits your control plan. Must be on the team from day one.

4

Finance Representative

Independently validates claimed savings. Without their sign-off the benefit is an opinion.

Traits of a Good Green Belt Project

1

Clear, measurable Y

The output can be counted or timed today - no new measurement system required first.

2

Cause is genuinely unknown

If everyone already knows the fix, do not run DMAIC. Just do it.

3

Scoped to weeks, not years

One process, one site, one product family. 3 to 6 months of part-time effort.

4

Real, quantified pain

A COPQ or customer number attached. If you cannot size it, you cannot defend it.

5

Data is available

Historical data exists or can be collected inside the project timeline.

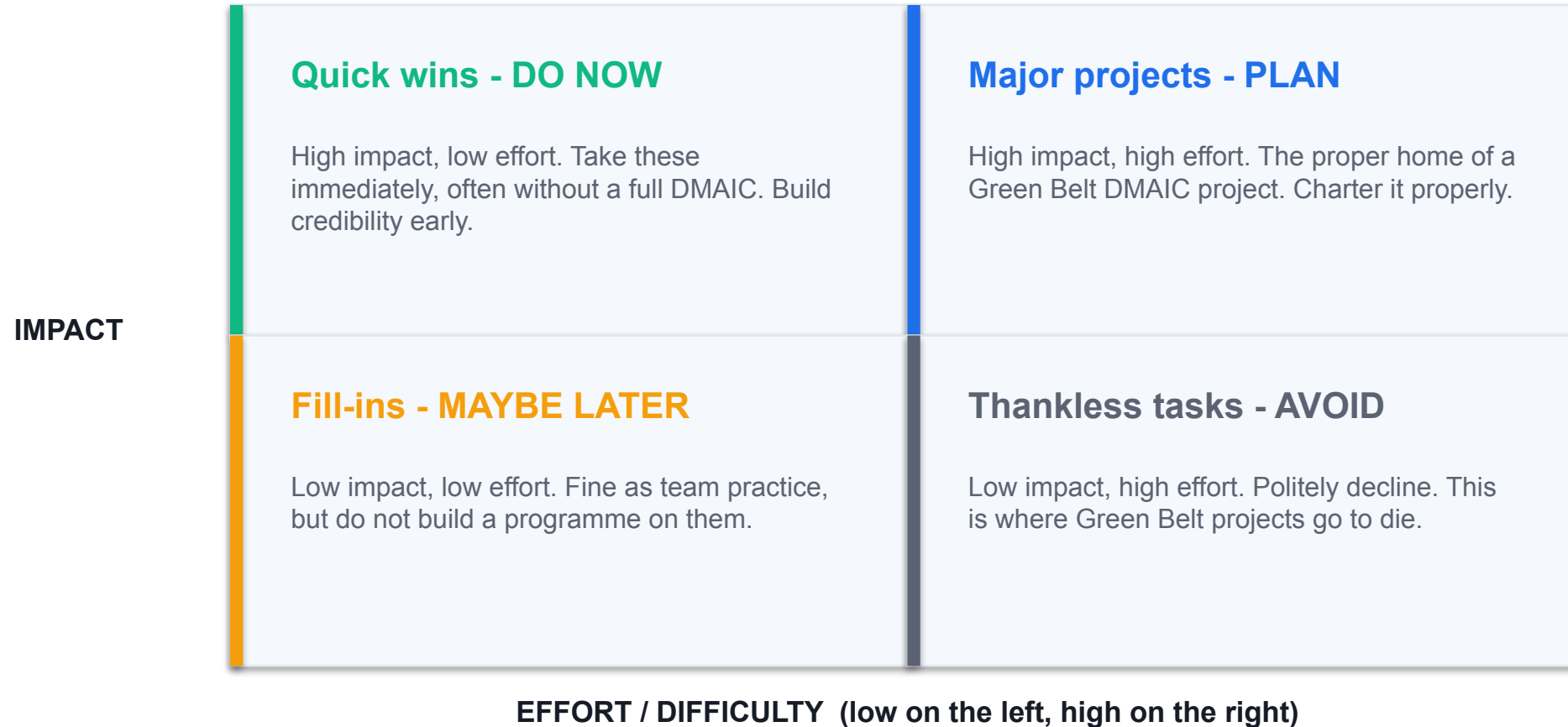
6

Willing process owner

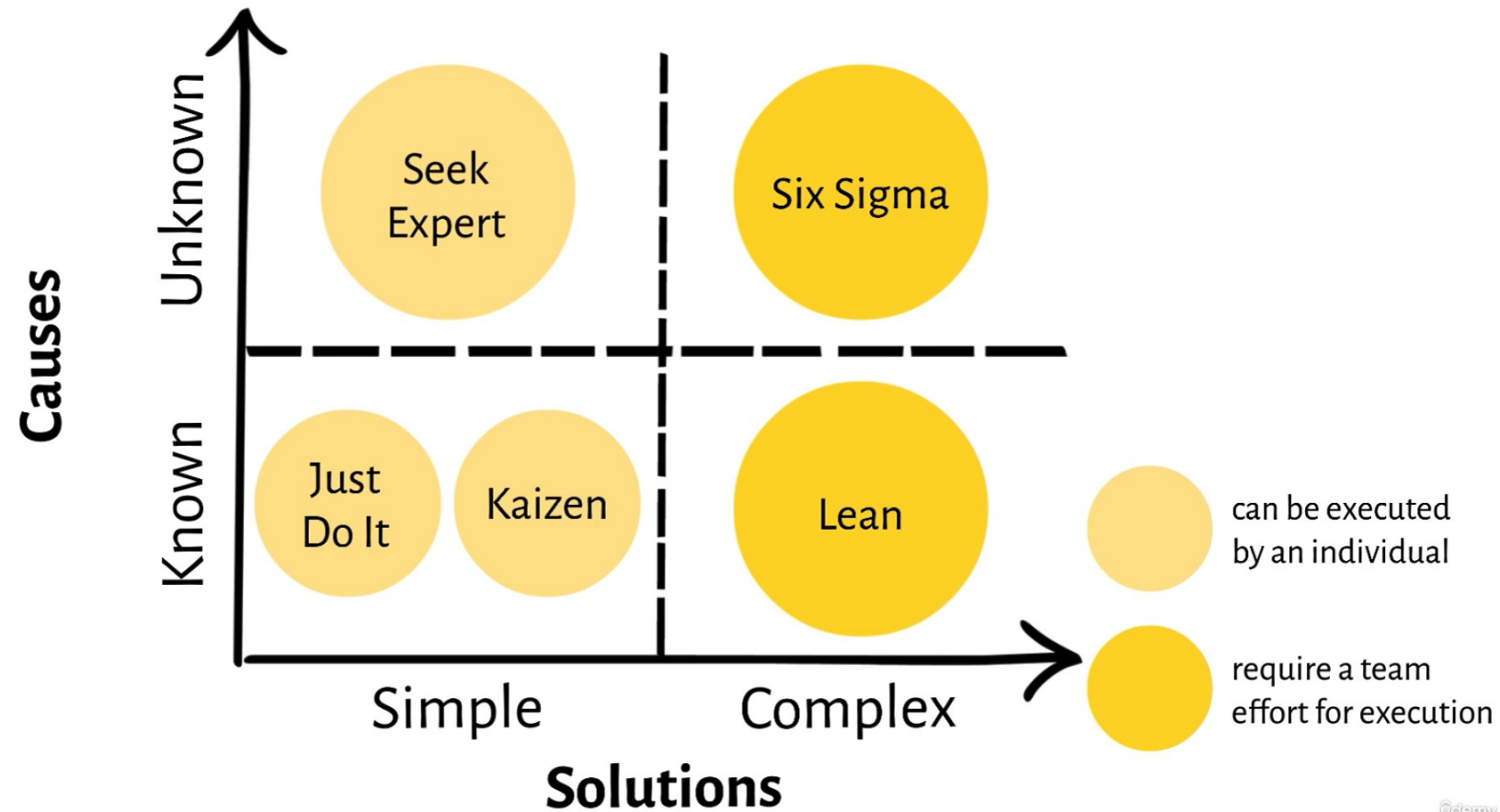
Someone will inherit the controls. No owner means no sustained gain.

THE PRIORITISATION MATRIX

Prioritising Candidates - Impact vs Effort



The Cause and Solution Matrix



Ask two questions first - do we know the cause, and do we know the solution?

The Three Reference Points

1

Baseline

What the process delivers TODAY, measured over a representative period. Northwind: 82 per cent on time.

2

Entitlement

The best the CURRENT process has ever sustained, with no new investment. Northwind: 91 per cent in March.

3

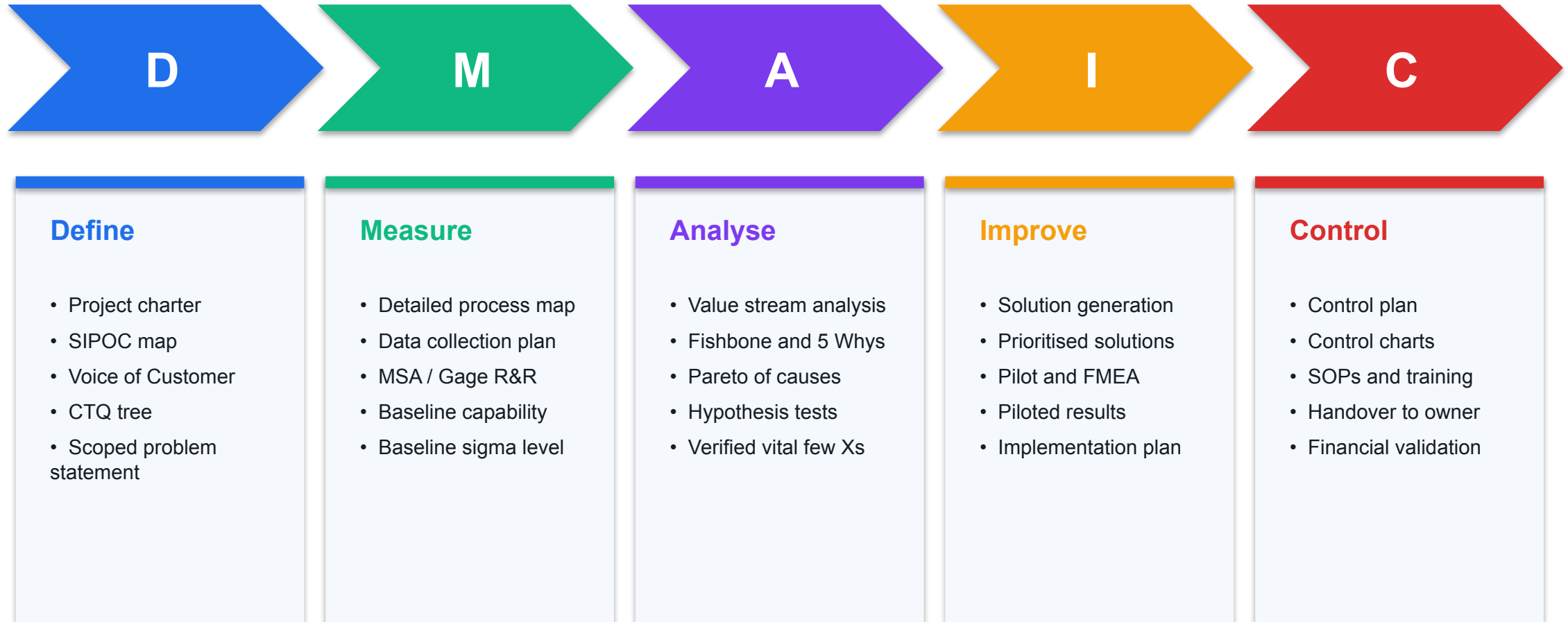
Benchmark

The best anyone achieves - internal sister site or external competitor. Southport DC: 98 per cent.

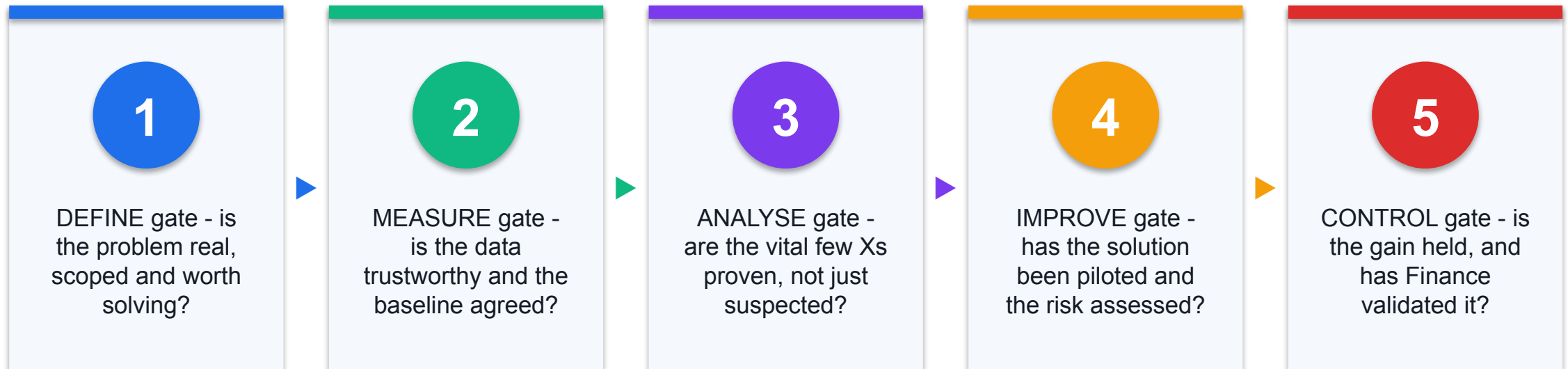
Using Them to Set the Goal

- Baseline tells you where you start - it must come from data, never from memory or opinion.
- Entitlement proves the target is achievable, because the process has already hit it at least once.
- The gap from baseline to entitlement is normally pure variation - exactly what DMAIC removes.
- Beyond entitlement usually needs capital or redesign, which is a different project and a different budget.
- A Green Belt goal that lands between baseline and entitlement is credible; beyond benchmark is fantasy.

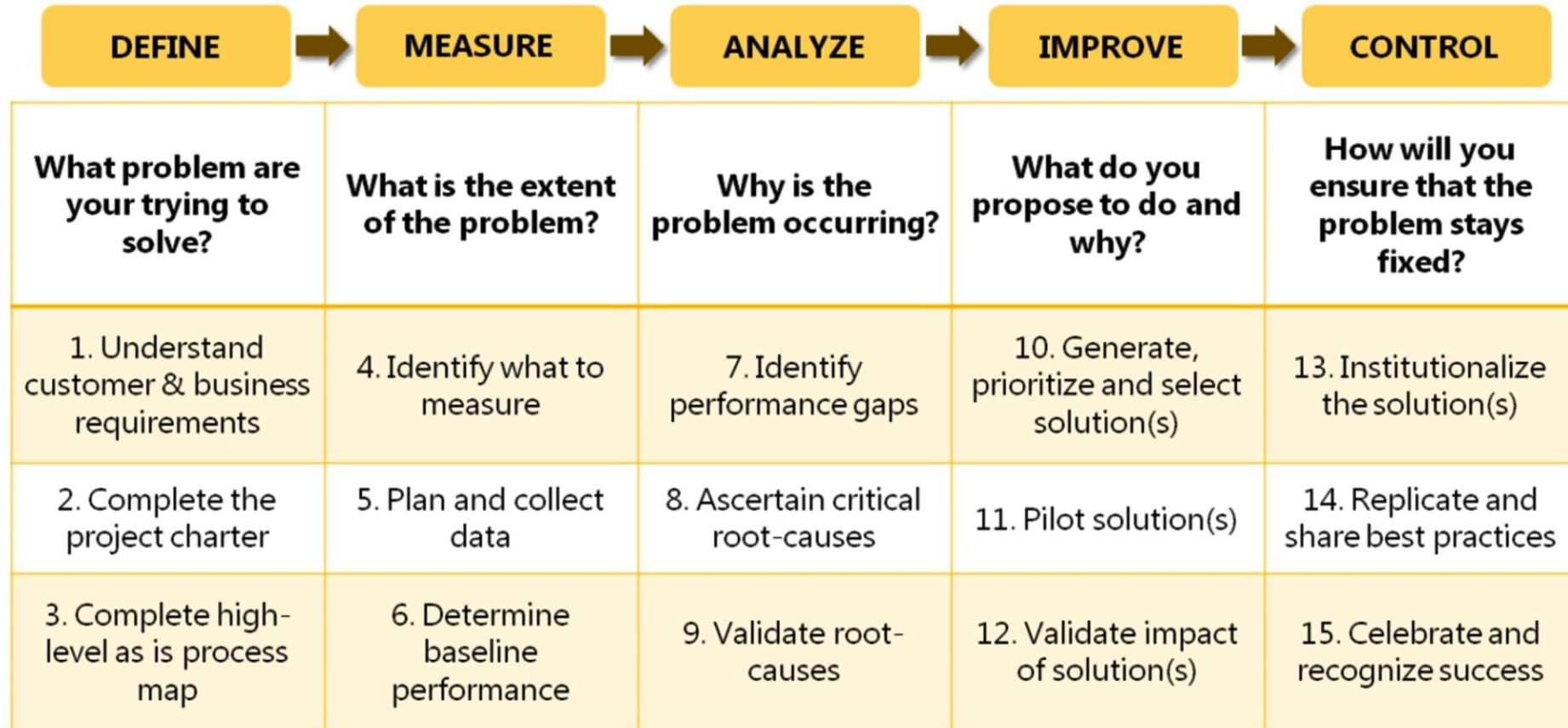
The DMAIC Roadmap and Its Deliverables



The Five Tollgate Reviews



The DMAIC Cycle



DMAIC is a closed loop - Control feeds the next Define, which is how continuous improvement is sustained.

DMAIC vs PDCA vs DFSS - When to Use Which

PDCA

Plan - Do - Check - Act

- For small, fast, local improvements
- Cause is largely understood already
- Days to weeks, minimal data
- No formal tollgates or charter
- The engine of daily kaizen

DMAIC

Define - Measure - Analyse - Improve - Control

- For an EXISTING process that underperforms
- Cause is unknown and must be proven
- 3 to 6 months, data and statistics
- Formal charter and five tollgates
- The Green Belt default

DFSS / DMADV

Define - Measure - Analyse - Design - Verify

- For a NEW process, product or service
- Or when DMAIC hits entitlement and stops
- Longest and most expensive route
- Requires design and capital authority
- Usually Black Belt or above

Checkpoint - Foundations

- Who ultimately defines quality, and why do the other two answers fail?
- State $Y = f(X)$ in your own words. Why can a manager not manage Y directly?
- What is the DPMO at 3 sigma and at 6 sigma, and why is six sigma 3.4 rather than 0.002?
- Name three things a Green Belt does that a Yellow Belt does not.
- Cause unknown, solution unknown - which method, and why not Just Do It?
- Define baseline, entitlement and benchmark using the Northwind numbers.

The Green Belt Role, Project Selection and Business Case

LAB 1

Establish what a Green Belt is accountable for versus a Yellow or Black Belt, then screen candidate projects against objective selection criteria and build the business case that wins sponsorship. A Green Belt leads the project and owns the analysis — this lab sets that expectation.

You'll build

A belt responsibility matrix, a scored project selection screen and a costed business case.

Tools & techniques: Belt role matrix, project selection criteria, COPQ estimate, business case

The Green Belt Role, Project Selection and Business Case

STEPS 1-4 OF 7

1

Compare the White, Yellow, Green, Black and Master Black Belt roles in a table — for each, record who leads, who analyses and who sponsors.

2

Write down the three things a Green Belt does that a Yellow Belt does not: leads a scoped project, runs the statistical analysis, and owns the tollgate reviews.

3

Open lab01-project-selection-candidates.xlsx. It lists four candidate improvement projects for this distribution centre with their volume, defect rate and cost inputs.

4

Score each candidate 1-5 against: measurable gap, data availability, manageable scope, sponsor support, and customer impact.

The Green Belt Role, Project Selection and Business Case

STEPS 5-7 OF 7

5

Reject any candidate that is a known solution in disguise ('install a new WMS') — a DMAIC project must start from a problem, not an answer.

6

Estimate the annualised COPQ for each candidate from the workbook: $\text{Annual Volume} \times \text{Defect Rate} \times (\text{Rework} + \text{Expedite} + \text{Credit})$ per unit.

7

Write a four-line business case: the gap, the annualised COPQ, the improvement target and the resource ask.

The Green Belt Role, Project Selection and Business Case

Check your work

Your selected project scores highest on the criteria table, contains no pre-selected solution, and your business case states an annualised dollar figure.

$Y = f(X)$, Sigma Level, DPMO and the DMAIC Roadmap

LAB 2

Every Six Sigma project rests on one model: the output Y is a function of the process inputs X s. Translate the Northwind problem into that model, then convert the baseline defect data into DPMO and a sigma level so improvement can be proven numerically later.

You'll build

A $Y = f(X)$ statement, a DPMO calculation and a baseline sigma level.

Tools & techniques: $Y = f(X)$ model, DPU, DPO, DPMO, sigma level, DMAIC tollgates

$Y = f(X)$, Sigma Level, DPMO and the DMAIC Roadmap

STEPS 1-4 OF 7

1

Write the project Y — the single output metric the customer feels. For Northwind: order fulfilment lead time, or percentage of orders shipped on time.

2

Brainstorm at least eight candidate Xs — the process inputs that could drive that Y (picking method, staffing, slotting, system downtime, order profile, shift, carrier cut-off).

3

Write the relationship formally as $Y = f(X_1, X_2, \dots, X_n)$ and state which Xs you can control and which you cannot.

4

Open lab02-baseline-defect-summary.xlsx: 4,200 orders shipped last month, 357 of them late. Calculate the defect rate as a proportion.

Y = f(X), Sigma Level, DPMO and the DMAIC Roadmap

STEPS 5-7 OF 7

5

Calculate DPMO using $DPMO = (\text{defects} / (\text{units} \times \text{opportunities per unit})) \times 1,000,000$. Treat each order as one opportunity for a late-delivery defect.

6

Look up the resulting DPMO on the sigma conversion table to read off the baseline sigma level.

7

Map the five DMAIC phases against your project and name the tollgate deliverable that ends each phase.

Y = f(X), Sigma Level, DPMO and the DMAIC Roadmap

Check your work

Your Y is a measurable customer-facing output, you have at least eight Xs, and your DPMO and sigma level are calculated from the 357/4,200 baseline.



DMAIC · DEFINE

Define — Scope the Problem

VOC · CTQ trees · Kano · Affinity · Charter · Problem statement · SIPOC · Stakeholders

Key Concepts — Define

1

Voice of the Customer

Capture what the customer actually says — reactive and proactive sources — before deciding anything.

2

Affinity diagram

Cluster raw VOC statements into natural themes so the signal emerges from the noise.

3

Kano analysis

Classify requirements as Must-Be, One-Dimensional or Delighter — they are not equally valuable.

4

CTQ tree

Translate each need into a specific, measurable Critical-to-Quality requirement with a target and limits.

5

Project charter

Problem, goal, scope, business case, team and timeline — the project's contract with the sponsor.

6

Problem statement

Process, time period, measurable gap and impact — and never a cause or a solution.

7

SIPOC

The macro as-is map: Suppliers, Inputs, Process, Outputs, Customers — agree the boundaries first.

8

Stakeholders & RACI

Name who is Responsible, Accountable, Consulted and Informed before the project starts.

What the Define phase must deliver

1

An approved Project Charter

Problem, goal, business case, scope, team, milestones and benefits - signed by the Champion.

2

Validated Voice of the Customer

Real verbatims from real customers, not the team's assumptions.

3

CTQs with operational definitions

Each critical-to-quality measure has a metric, target and spec limit.

4

A high-level process map (SIPOC)

Agreed start and stop points so everyone scopes the same process.

5

A stakeholder plan

Who has power, who has interest, who signs off, who resists.

6

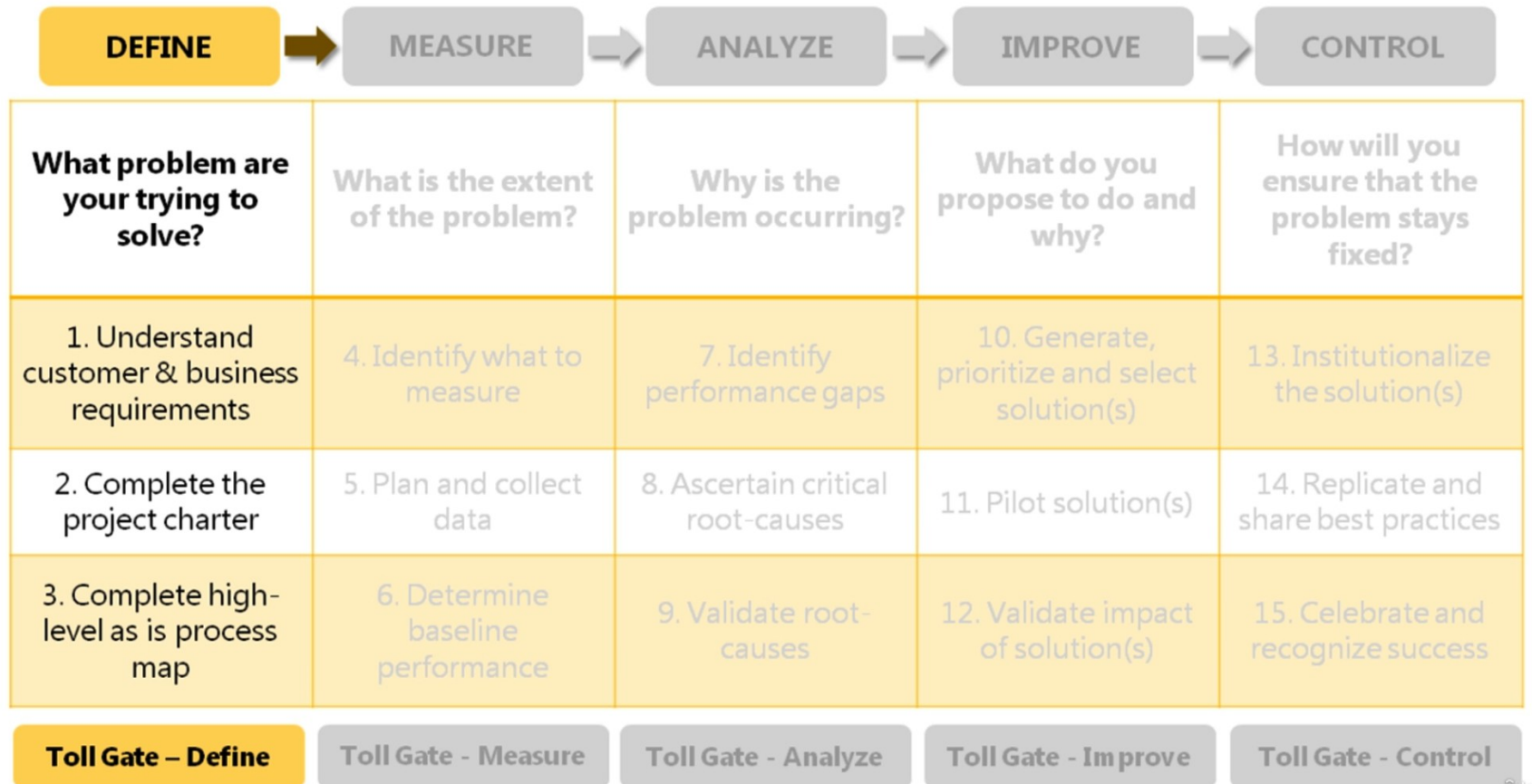
A passed tollgate review

Champion approval to spend measurement effort in the next phase.

The steps of the Define phase



Define phase - the standard sequence



Each step feeds the next: VOC drives CTQs, CTQs drive the goal, the goal drives the charter.

Northwind Retail Distribution Centre - our running case

- 12,000 sqm distribution centre fulfilling online retail orders across Singapore.
- Customers complain about late deliveries; negative feedback is spreading on social media.
- Baseline: 4,200 orders shipped last month, 357 of them late - 8.5 percent late shipments.
- Target: reduce late shipments from 8.5 percent to 3.0 percent.
- We will carry this one case through Define, Measure, Analyse, Improve and Control.

Two families of VOC sources

REACTIVE

Data that arrives whether you ask or not

- Complaints and contact-centre call logs
- Returns and refund requests
- Warranty and service claims
- Social media posts, reviews and ratings
- Cheap and continuous - but biased to the angry

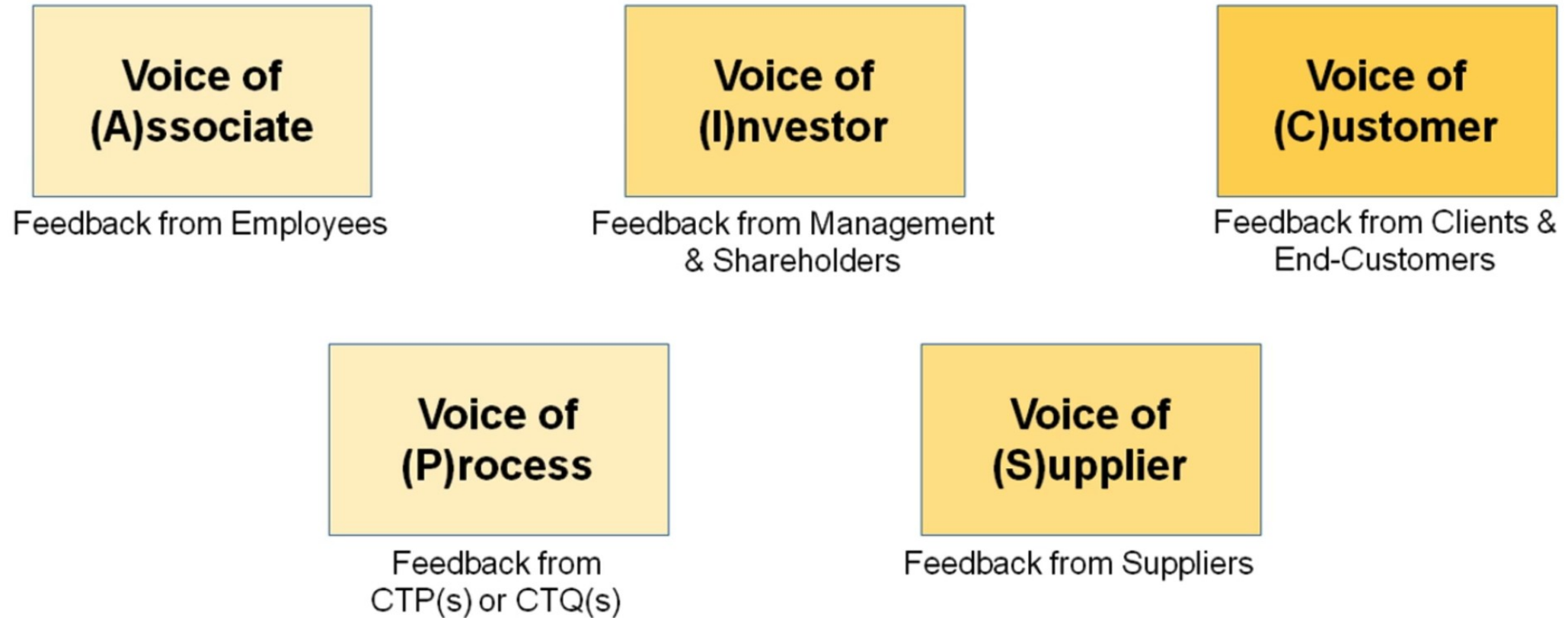
PROACTIVE

Data you must deliberately go and collect

- Structured surveys with a sampling plan
- One-to-one customer interviews
- Focus groups with 6-10 customers
- Gemba observation - watch the customer use it
- Costlier, but reaches the silent majority

DEFINE

The VOC source map



Use both families. Reactive data tells you what broke; proactive data tells you what matters.

Northwind - VOC verbatims from three channels

REACTIVE - calls and complaints

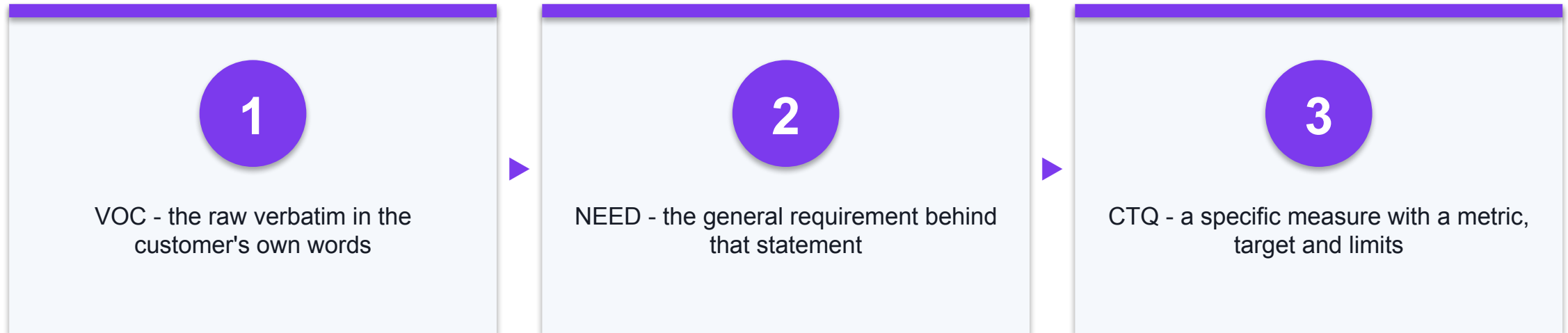
- 'It said two-day delivery and it arrived on day five.'
 - 'Nobody told me it was delayed - I had to chase.'
 - 'The tracking page still said packed for three days.'
- Source: 214 contact-centre calls, last 30 days

PROACTIVE - interviews and social listening

- 'I ordered for a birthday and it missed the date.'
 - 'Two of the four items came, the rest never did.'
 - 'Late again. Third time. Switching to a competitor.'
- Source: 61 social posts + 12 customer interviews

DEFINE

The three-level drill-down



Northwind - one verbatim drilled to a CTQ

VOC

What the customer said

- 'It said two-day delivery and
- it arrived on day five.'
- Emotional, vague, unmeasurable
- But it is the truth of the problem

NEED

What they actually require

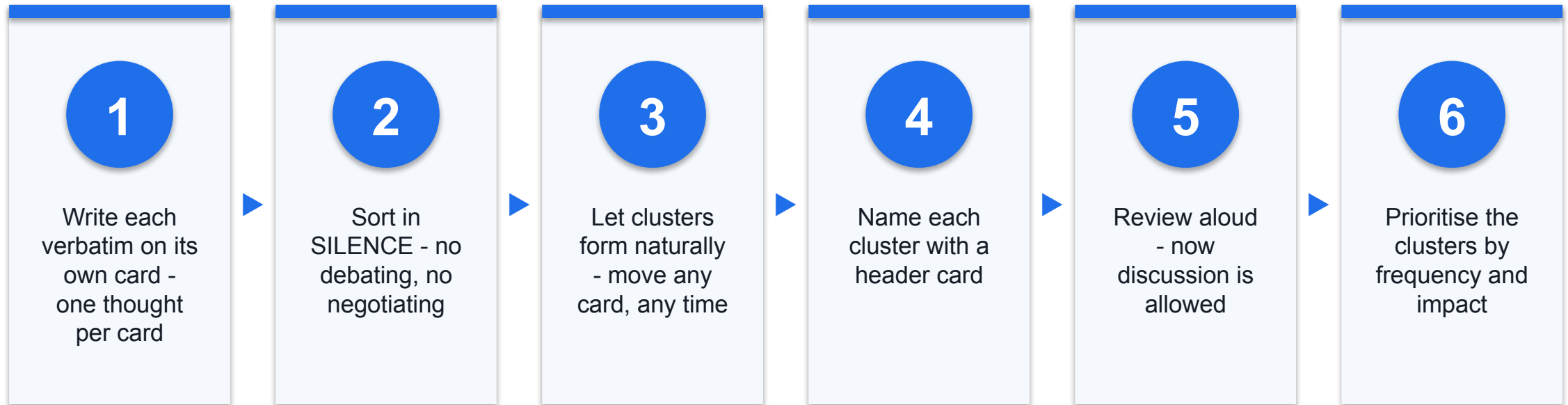
- My order must arrive by the date
- that was promised at checkout
- Still not measurable as written
- One need can spawn several CTQs

CTQ

What we will measure

- Metric: order dispatch lead time
- Target: 100 pct shipped same day
- USL: 24 hours order to despatch
- Defect: any order past 24 hours

The six steps of an affinity diagram



Why silence is the rule

1

Silence kills seniority

The loudest or most senior voice cannot dominate the grouping.

2

It is faster

Ten people sorting silently beat ten people arguing about one card.

3

It surfaces intuition

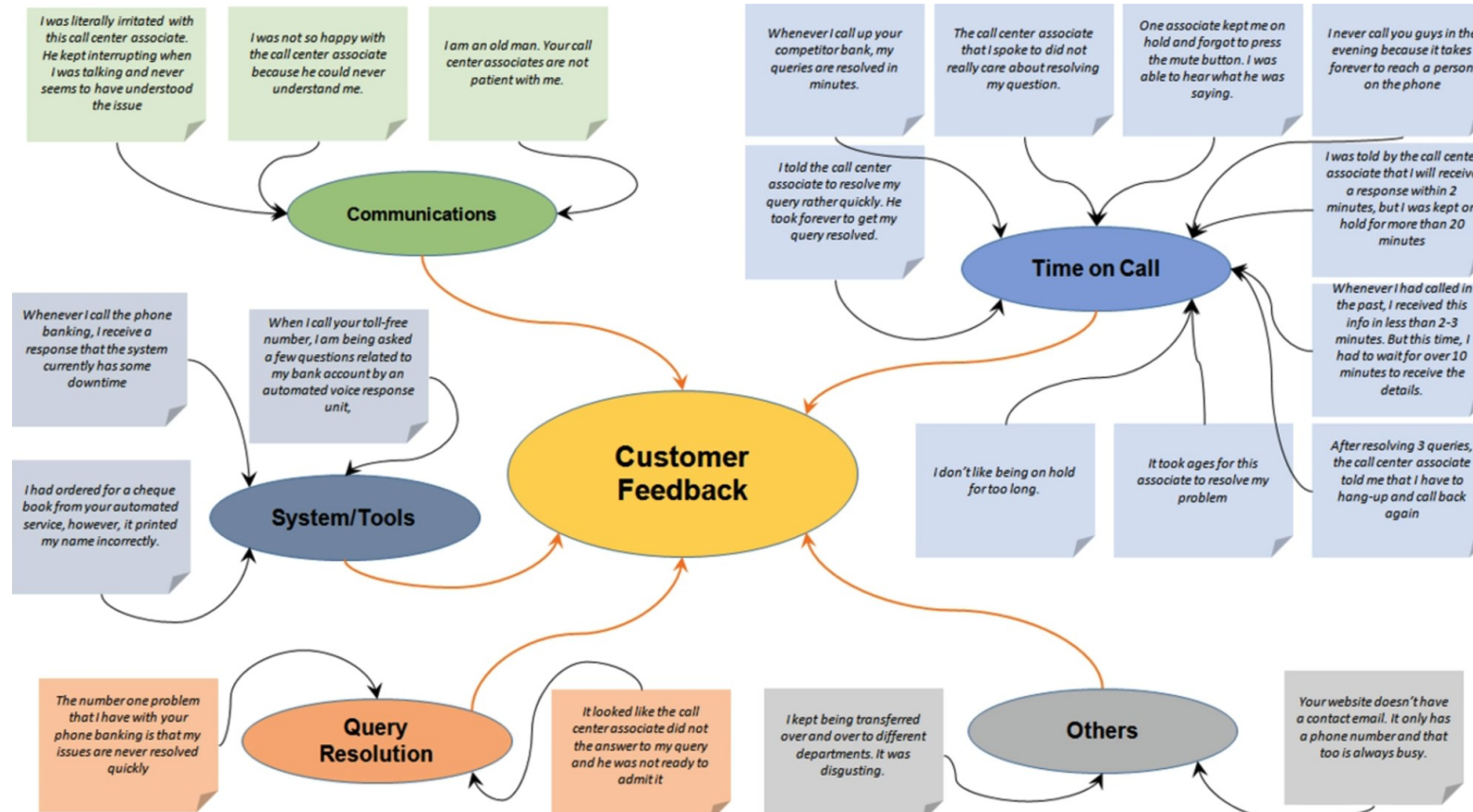
Pattern recognition works before it can be justified in words.

4

Duplicate moves are fine

If two people keep moving a card, that card belongs in a new cluster.

Affinity diagram in practice



Cards first, clusters second, names last. Never name the cluster before the cards decide it.

Northwind - the five clusters that emerged

1

Missed promise date

148 verbatims. The order arrived after the date shown at checkout.

2

No proactive notification

97 verbatims. Customer discovered the delay, we never told them.

3

Stale tracking information

64 verbatims. Tracking status not updated after packing.

4

Split and partial shipments

41 verbatims. Multi-line orders arriving in pieces.

5

Wrong or damaged items

23 verbatims. Out of scope - handled by a separate quality project.

The three primary Kano categories

MUST-BE

Basic / expected / dissatisfier

- Assumed, never articulated
- Absent: severe dissatisfaction
- Present: NO satisfaction at all
- Northwind: the parcel arrives intact
- You cannot win here - only lose

ONE-DIMENSIONAL

Performance / linear / spoken

- The more you give, the happier
- Customers ask for it explicitly
- This is where you compete
- Northwind: speed of delivery
- Satisfaction rises with performance

DELIGHTER

Attractive / excitement / unspoken

- Unexpected - never requested
- Absent: no dissatisfaction at all
- Present: disproportionate delight
- Northwind: live courier tracking map
- Decays into a Must-Be over time

DEFINE

The two categories most teams forget

1

INDIFFERENT

Performance changes but satisfaction does not move. Pure waste - stop investing in it.

2

REVERSE

More of the feature makes satisfaction fall. Over-packaging, over-notification, forced registration.

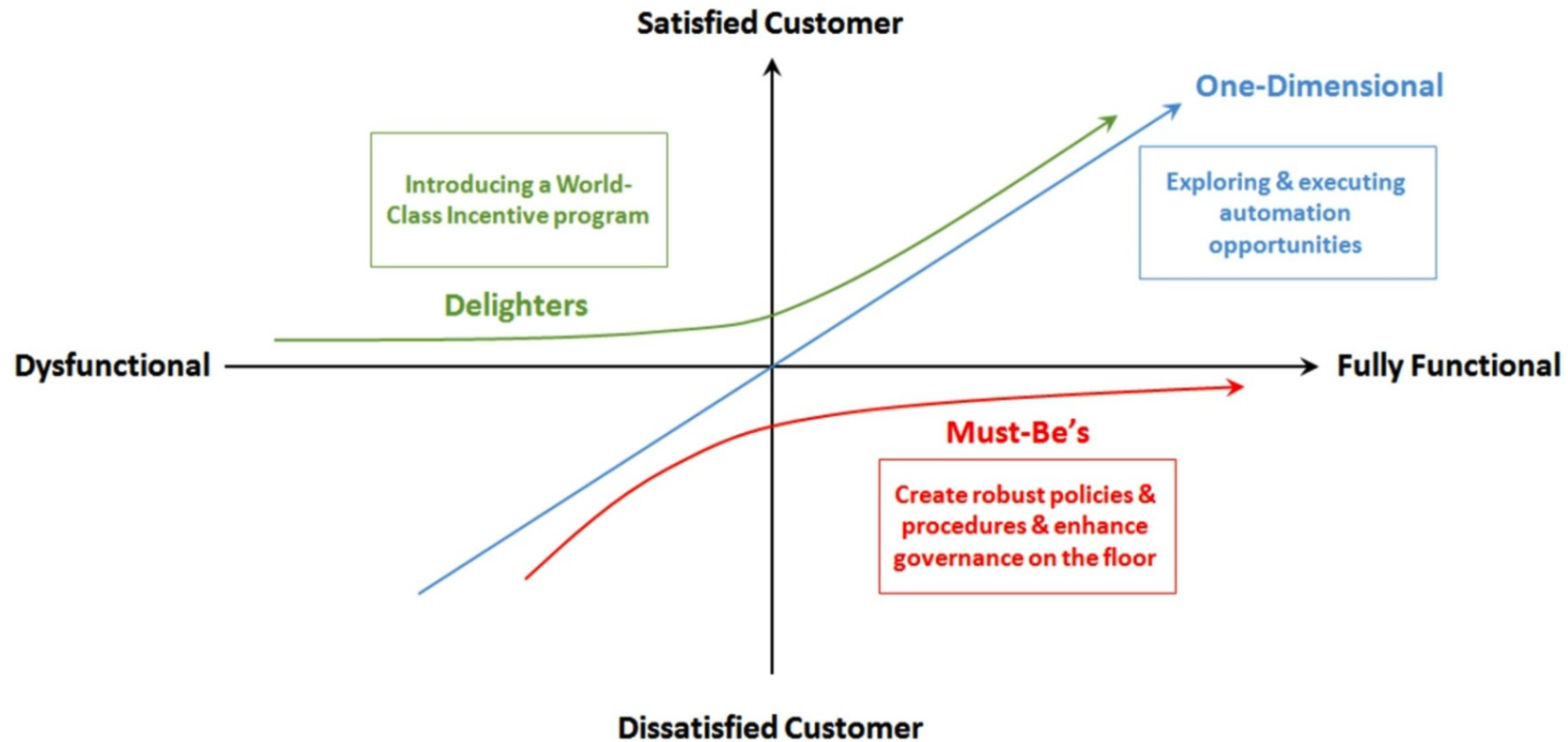
DEFINE

Delighters decay into Must-Bes.

Car air-conditioning was a delighter in 1965, a performance feature in 1985, and a must-be by 2005. Today its absence causes outrage and its presence earns nothing. Re-run Kano every 12 to 18 months or your CTQs quietly go stale.

DEFINE

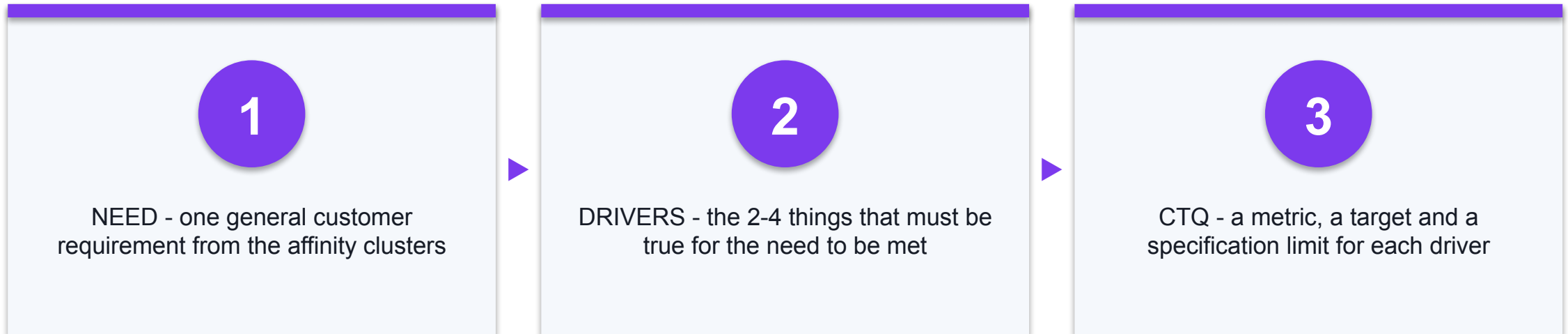
The Kano model



Vertical axis is satisfaction, horizontal is performance delivered. Only the middle curve passes through the origin.

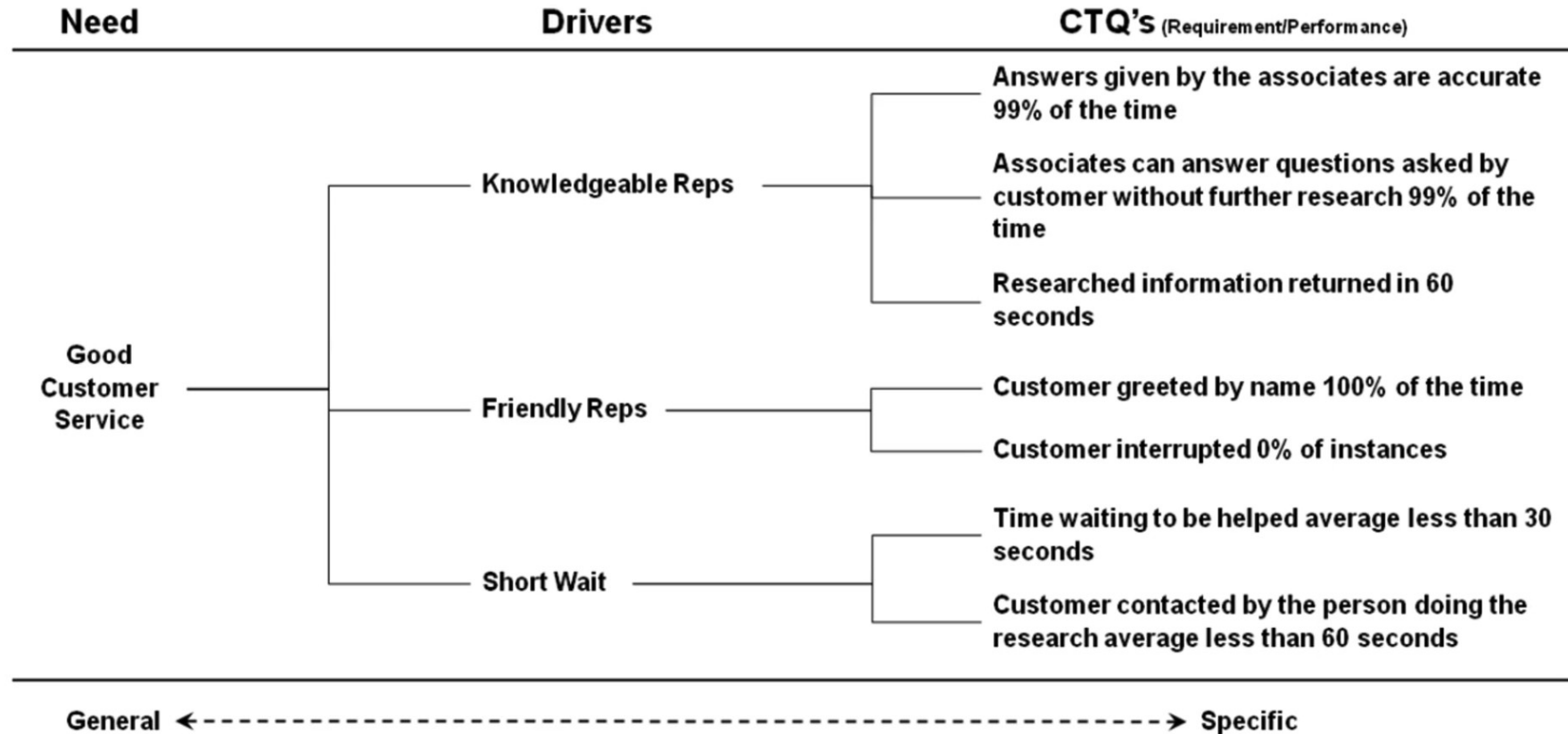
DEFINE

How a CTQ tree is built



DEFINE

CTQ tree structure



Read left to right. If a leaf cannot be measured this week, it is not yet a CTQ.

DEFINE

Northwind CTQ tree - need: my order arrives when promised

1

Driver 1 - Pick accuracy

CTQ: pick error rate. Target 0.2 pct. USL 0.5 pct of order lines.

2

Driver 2 - Despatch speed

CTQ: order-to-despatch hours. Target 12 h. USL 24 h from order receipt.

3

Driver 3 - Carrier collection

CTQ: on-time carrier pickup. Target 100 pct. LSL 98 pct of scheduled slots.

4

Driver 4 - Delay notification

CTQ: hours from delay detected to customer told. Target 1 h. USL 4 h.

Operational definitions - the discipline that saves the project

WHAT IS MEASURED

The characteristic, unambiguously

- Order-to-despatch elapsed hours
- Clock starts at payment confirmed
- Clock stops at carrier scan-out
- Excludes pre-orders and backorders

HOW IT IS MEASURED

Method, instrument and rounding

- Source: WMS timestamp table
- Reported to one decimal place
- Calendar hours, not working hours
- Extracted daily at 06:00

WHAT COUNTS AS A DEFECT

The decision rule

- Any order exceeding 24.0 hours
- Partial shipment: each line judged
- Cancelled orders are excluded
- Two people must agree on any edge case

The seven sections of a project charter

1

1. Problem statement

What is wrong, in which process, over what period, how big the gap, what it costs.

2

2. Goal statement

The same metric, with a baseline, a target and a completion date.

3

3. Business case

Why now, why this project, what happens if we do nothing.

4

4. Scope

In-scope and out-of-scope, plus the process start and stop points.

5

5. Team and roles

Champion, Sponsor, Green Belt, Black Belt mentor, process owner, members.

6

6. Milestones

Tollgate dates for D, M, A, I and C - with named review owners.

7

7. Benefits

Hard savings, soft savings and the customer-facing benefit.

Project charter template

Business Case: Write the business case here	Problem Statement: Write the problem statement here
Goal Statement: Write the goal statement here	Project Scope: Write the project scope here
Project Team: Write the project team details here	Project Milestones: Write the project milestones here

A charter is a living document - it is revised at every tollgate, never written once and filed.

Charter rules that separate Green Belts from amateurs

1

One page, always

If it needs three pages, the scope is too big - split the project.

2

Signed by the Champion

An unsigned charter has no resources and no authority behind it.

3

Numbers, never adjectives

'Significantly improve' is not a goal. '8.5 pct to 3.0 pct by 31 Oct' is.

4

Revised at each tollgate

Measure often changes the baseline. Update the charter, do not hide it.

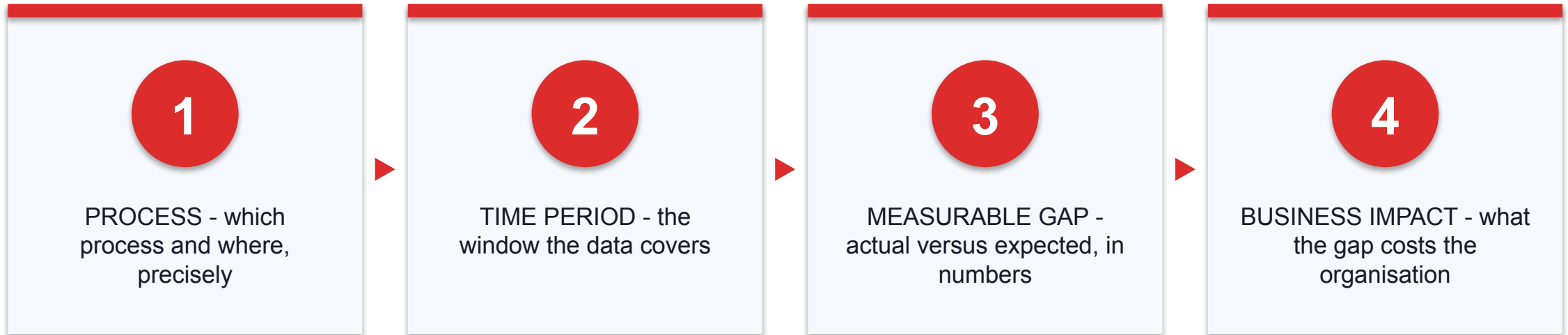
DEFINE

A problem statement has four components.

Process, time period, measurable gap, business impact. Anything else you add is either a cause or a solution - and both are forbidden at this stage.

DEFINE

The four required components



DEFINE

The two forbidden things

NO CAUSE

Do not state why it happens

- 'because the WMS is slow' is a hypothesis
- Causes are proved in Analyse, not asserted
- Naming a cause pre-closes the investigation
- It also assigns blame before evidence

NO SOLUTION

Do not state what to do about it

- 'we need a new WMS' is a solution
- Solutions are selected in Improve
- A solution in the problem statement
- turns a DMAIC project into a purchase order

DEFINE

Northwind - BAD versus GOOD problem statements

BAD

Vague, causal and solution-loaded

- 'Deliveries are always late because the
- warehouse system is slow and understaffed,
- so we need to buy new scanners.'
- No process, no period, no number, no cost
- States a cause AND a solution

GOOD

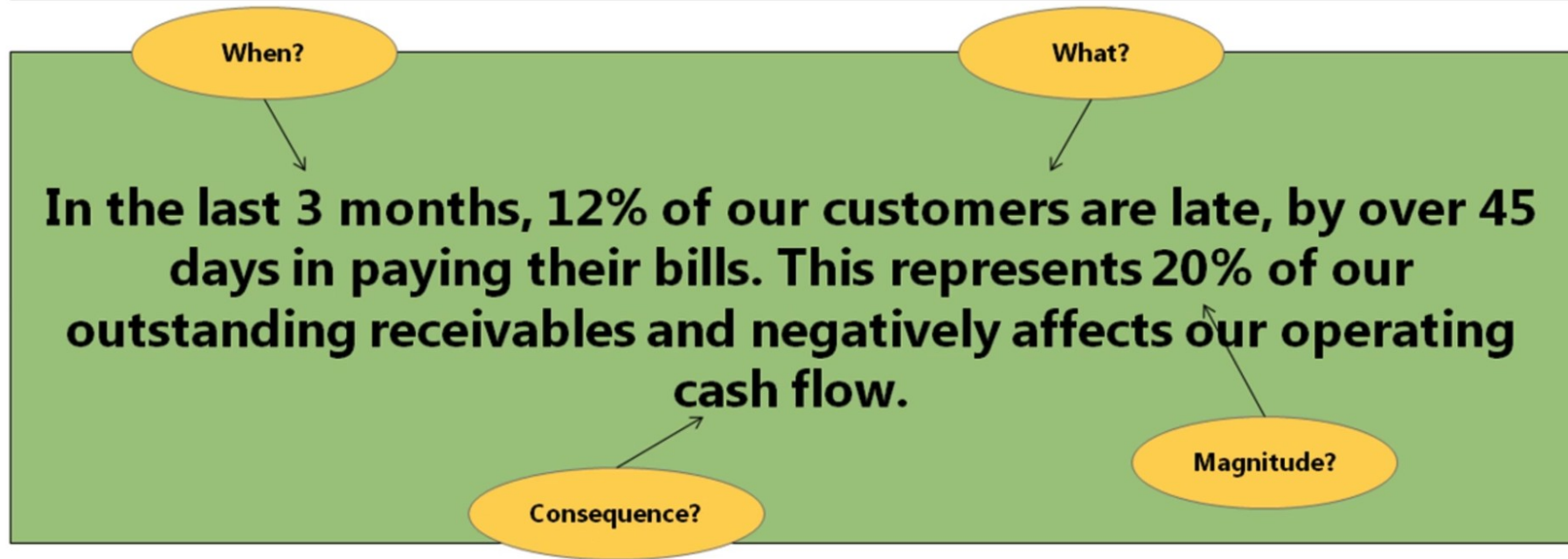
Four components, zero cause, zero fix

- 'In the Northwind DC outbound process,
- in the month of September, 357 of 4,200
- orders shipped late - 8.5 pct against a
- 3.0 pct standard - driving 214 complaint
- calls and an estimated \$186,000 annual COPQ.'

DEFINE

Problem statements - good and bad

"Our customers are angry with us and thus delay in paying their bills."



Read every statement back and ask: is there a why, or a what-to-do, hiding in here?

The four elements of a goal statement

1

METRIC

The identical measure used in the problem statement - percentage of orders shipped late.

2

BASELINE

Where it is today, from real data - 8.5 percent, September, 4,200 orders.

3

TARGET

Where it must get to - 3.0 percent, a 65 percent reduction in the defect rate.

4

DATE

When it must be achieved and verified - 31 October, confirmed by Control-phase data.

DEFINE

Northwind goal statement

Reduce late outbound shipments in the Northwind DC from 8.5 percent to 3.0 percent of orders by 31 October, sustained for three consecutive months, without increasing outbound cost per order.

Northwind - in scope and out of scope

IN SCOPE

- Outbound order fulfilment in the Singapore DC
- Order receipt through carrier scan-out
- Standard retail e-commerce orders
- Pick, pack, label and stage activities
- All three outbound shifts, Mon to Sat
- WMS despatch data, Jan to Sep

OUT OF SCOPE

- Inbound receiving and put-away
- Carrier transit time after scan-out
- B2B pallet and wholesale orders
- Returns, refunds and reverse logistics
- Product quality and packaging design
- The overseas fulfilment centres

DEFINE

Process start and stop points - the boundary that matters most

1

START POINT

Payment confirmed and order released to the warehouse management system.

2

STOP POINT

Carrier scan-out at the outbound dock door - custody transfers to the carrier.

3

Why fix it in Define

Different boundaries produce different baselines and different root causes.

4

Test the boundary

Everyone on the team must state the same start and stop point, unprompted.

Scope creep - how it starts and how to stop it

1

'While we are in there...'

A stakeholder adds a pet problem mid-project. Log it in the parking lot, do not adopt it.

2

Root causes outside the boundary

Legitimate finding - raise a separate project, do not silently expand this one.

3

Solution shopping

Improve ideas that need capex outside scope belong to a follow-on project.

4

The defence

Any scope change requires a re-signed charter and a revised milestone date.

What a Green Belt business case must contain

1

The current cost of the problem

COPQ today, calculated from real volumes and verified unit costs.

2

The size of the opportunity

The portion of that COPQ the goal statement actually recovers.

3

Why now

A regulatory date, a contract renewal, a churn trend or a capacity ceiling.

4

The cost of doing nothing

Project the trend forward twelve months and state the number.

5

Finance validation

A named Finance partner has agreed the assumptions in writing.

6

Strategic linkage

Which corporate objective this project moves, and by how much.

DEFINE

Northwind - building the COPQ number

Late shipment rate

$$357 / 4200 = 8.5 \text{ pct}$$

September actual against a 3.0 pct internal standard

Excess late orders

$$8.5 \text{ pct} - 3.0 \text{ pct} = 5.5 \text{ pct}$$

5.5 pct x 4,200 = 231 avoidable late orders per month

Monthly COPQ

$$231 \times \$67 = \$15,477$$

\$67 = expedite freight + service call + goodwill credit per order

Annualised COPQ

$$\$15,477 \times 12 = \$185,724$$

Rounded to \$186,000 - the figure in the problem statement

Every figure here is traceable to WMS volumes and Finance-agreed unit costs - never to an estimate on a whiteboard.

Hard savings versus soft savings

HARD SAVINGS

Finance will book them

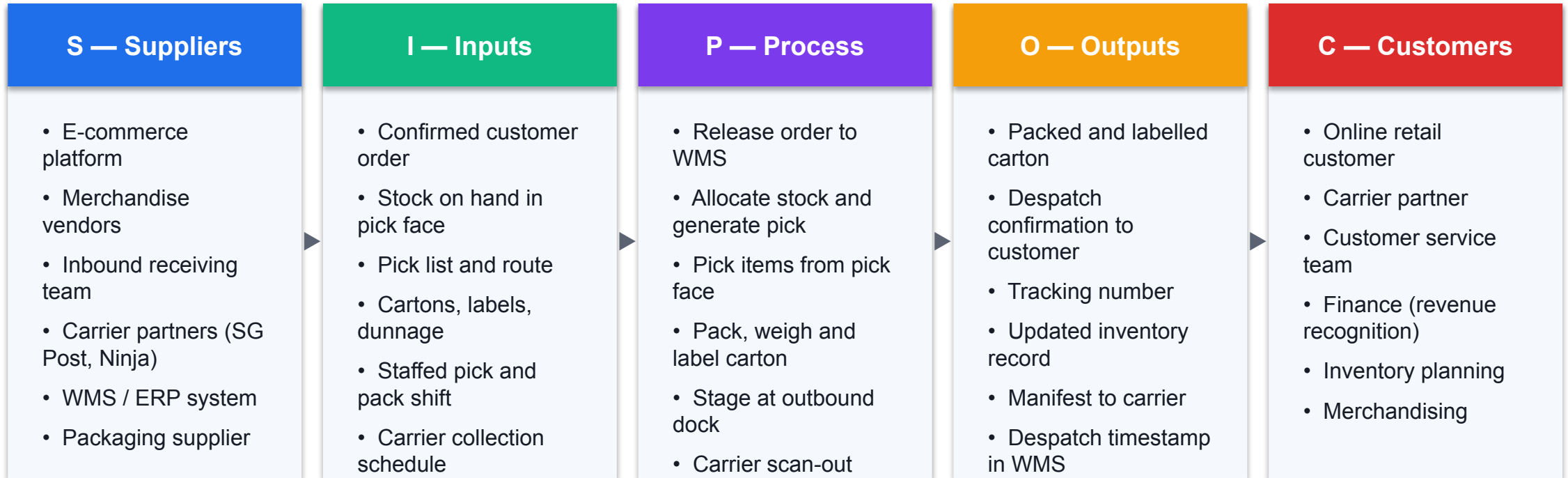
- Removed expedite freight charges
- Reduced overtime hours in outbound
- Avoided goodwill credits and refunds
- Show up on the P&L within 12 months
- Count these in the project benefit

SOFT SAVINGS

Real, but not bookable

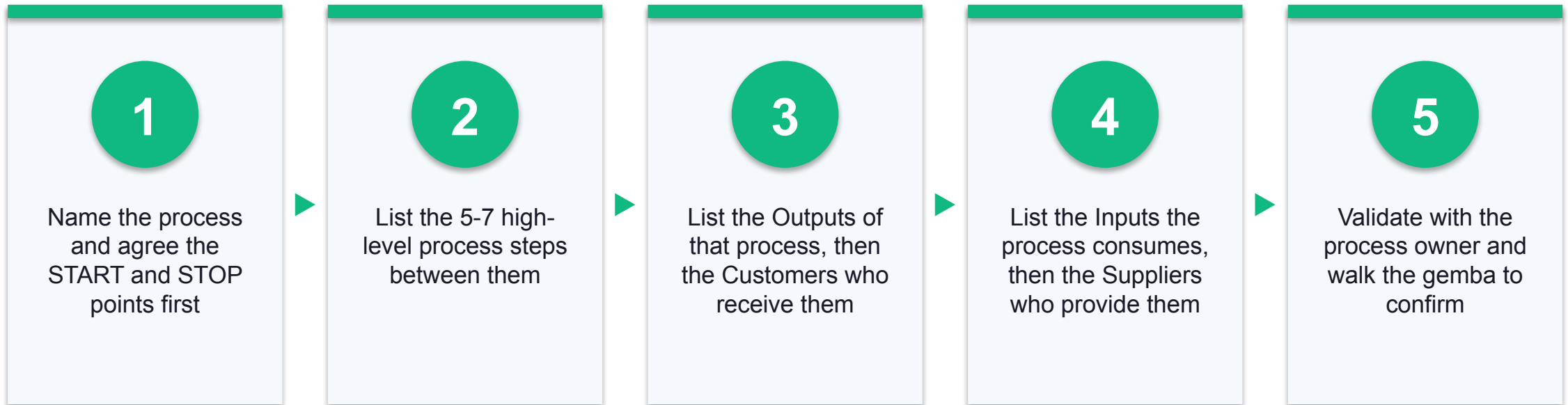
- Hours freed but not headcount removed
- Improved customer satisfaction score
- Reduced complaint-handling effort
- Lower risk of contract penalties
- Report them separately, never inflate

Northwind DC - outbound order fulfilment SIPOC



Read left to right: suppliers provide the inputs, the process transforms them, and the outputs go to the customers.

How to build a SIPOC - the working sequence



SIPOC rules Green Belts get wrong

1

Build P first, not S

Fix the process steps and boundary before anything else - the rest hangs off them.

2

5 to 7 steps only

SIPOC is a boundary tool, not a process map. Detail comes in the Measure phase.

3

Outputs before inputs

Work backwards from what the customer receives - it exposes missing steps.

4

Customers can be internal

Finance and Customer Service are customers of this process too.

5

Verbs for the process column

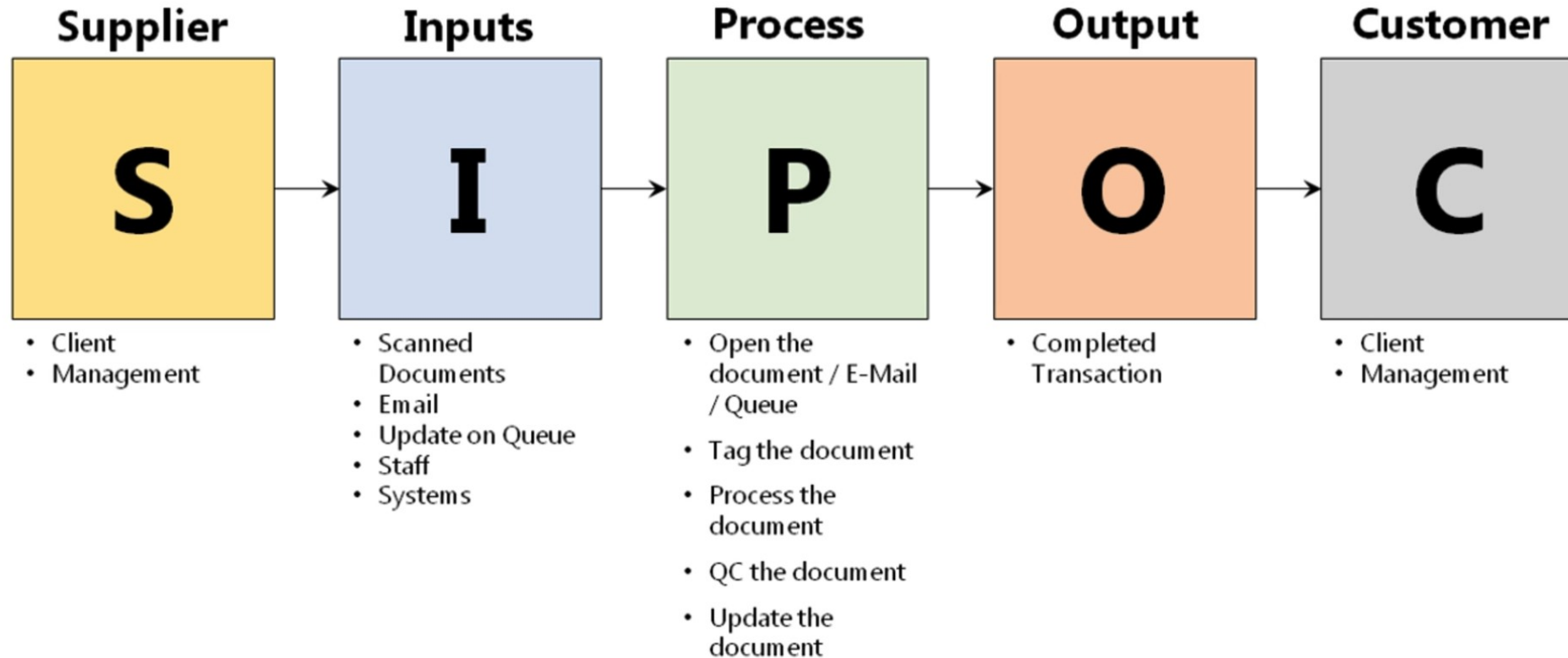
Every P entry starts with a verb: release, allocate, pick, pack, stage.

6

Walk it, do not imagine it

A SIPOC built in a meeting room is a hypothesis until you walk the floor.

SIPOC - worked example



The SIPOC is the contract on scope: if it is not in the P column, it is not in this project.

Build it in the browser - SIPOC & Process Map Builder

1

alfredang.github.io/sipoc

The tool you will use in Lab 6 and Lab 7. No install, no licence.

2

Boundaries first

It makes you name the process, the trigger and the end state before anything else.

3

SIPOC + pain points

Build all five columns, then tag at least three pain points on the steps.

4

Swimlane, generated

Assign an actor to each step and it draws the swimlane and the handoff table.

5

Check my SIPOC

Validates your diagram against the lab criteria before you move on.

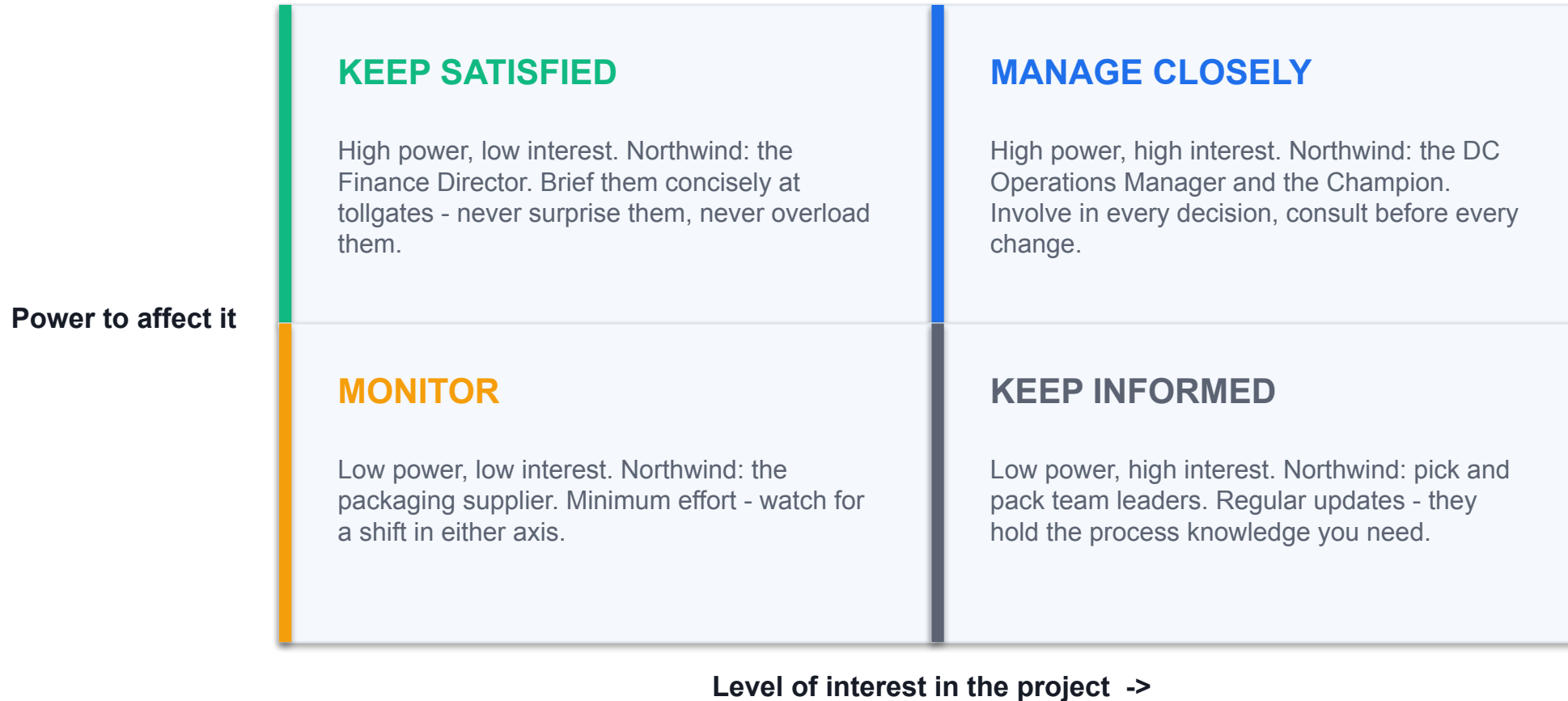
6

Export and share

Export to PNG or PDF for your project pack, or share the session with your team.

DEFINE

Power / interest grid



DEFINE

Running the stakeholder analysis

1

List everyone affected

Anyone who touches the process, funds it, measures it or feels its output.

2

Score power and interest

High or low on each axis - be honest, not political.

3

Record current versus needed

Where is each stakeholder now on resist-neutral-support, and where must they be?

4

Assign an owner per person

Every named stakeholder has one team member responsible for the relationship.

5

Re-map at every tollgate

Reorganisations and results move people between quadrants.

DEFINE

RACI: exactly one A per row.

Responsible does the work. Accountable owns the outcome and there is never more than one. Consulted gives input before the decision. Informed is told after it.

The four RACI letters

R - RESPONSIBLE

Performs the task

- Can be several people per activity
- Northwind: Green Belt runs the analysis
- Doing the work, not owning the result

A - ACCOUNTABLE

Owens the outcome

- EXACTLY ONE per row, no exceptions
- Northwind: DC Ops Manager owns delivery
- Two As means nobody is accountable

C - CONSULTED

Two-way, before

- Their input shapes the decision
- Northwind: Finance on the COPQ model
- Consult too many and you stall

I - INFORMED

One-way, after

- Told the outcome, no input expected
- Northwind: Customer Service team
- Cheap - so do not skip it

Northwind - the project RACI in practice

ACCOUNTABLE BY ACTIVITY

- Charter approval - A: Champion
- Data collection plan - A: Green Belt
- Baseline validation - A: Green Belt
- Root cause confirmation - A: Green Belt
- Solution sign-off - A: DC Ops Manager
- Control plan handover - A: Process Owner

RACI FAILURE MODES

- Common failure: two names in the A column
- Common failure: the Champion listed as R
- Common failure: everyone marked C
- Common failure: no I for the shop floor
- Fix: one A per row, then read it aloud
- Fix: the A must be able to say 'no'

Why people resist a Six Sigma project

1

Fear of job loss

Efficiency has historically meant headcount cuts. Say what will happen to roles, early.

2

Loss of expertise status

Standard work threatens the person who was the only one who knew the workaround.

3

Change fatigue

The fourth initiative this year. Show what you are stopping, not only starting.

4

Not invited, only informed

People defend what they helped build. Involve the floor in the fishbone.

5

No visible sponsorship

If the Champion never appears at the gemba, the team reads it as optional.

6

Measurement feels like policing

Explain that you are measuring the process, never the individual.

Practical change tactics for a Green Belt

1

Communicate the why first

People accept a change they understand the reason for - twice as fast.

2

Make the Champion visible

One gemba walk by the Champion outweighs ten emails from you.

3

Pilot small, publicise early

One shift, two weeks, then show the run chart to everyone.

4

Give credit to the floor

Name the operators who found the root cause, in front of management.

5

Address the loudest sceptic

Convert them and the rest follow. Ignore them and they run the project.

6

Plan the handover from day one

Control is designed in Define, not improvised at the end.

DEFINE

Define tollgate checklist

1

Charter signed

All seven sections complete, approved by the Champion and the process owner.

2

VOC evidence attached

Real verbatims from named sources, with volumes and dates.

3

CTQs operationally defined

Metric, target, spec limit and a defect rule two people agree on.

4

Baseline is from real data

8.5 pct comes from 4,200 WMS records, not from a manager's estimate.

5

SIPOC agreed and walked

Start and stop points confirmed at the gemba by the process owner.

6

Stakeholders mapped, RACI set

One A per row, engagement plan owned by a named team member.

7

Business case validated

COPQ assumptions agreed in writing by a named Finance partner.

8

Risks and resources logged

Team time released, risks named, escalation route agreed.

Three legitimate tollgate outcomes

PROCEED

Define is complete

- Charter signed, CTQs defined
- Baseline data validated
- Move to the Measure phase

REWORK

Something is not yet solid

- Scope too broad, or baseline unverified
- Fix the named gap, return in 1-2 weeks
- Far cheaper than discovering it in Analyse

STOP

The project should not continue

- Solution already known - just implement it
- Benefit does not justify the effort
- Stopping early is a success, not a failure

Define phase - checkpoint

- Name the four components of a problem statement and the two things it must never contain.
- Take one Northwind verbatim and drill it down to a CTQ with a metric and a limit.
- Why does a Delighter become a Must-Be, and what must you do about it?
- Which SIPOC column do you build first, and why does that order matter?
- How many people can be Accountable in one RACI row, and what breaks if there are two?
- State the Northwind goal statement with metric, baseline, target and date.

Hands-On Labs — Define

1

Lab 3 — Voice of the Customer, Affinity Diagram and Ka

A VOC log, an affinity diagram of themes and a Kano classification tab

2

Lab 4 — CTQ Tree — Translating Customer Needs into Mea

A three-level CTQ tree with metric, target, USL and LSL for each requi

3

Lab 5 — Project Charter, Problem Statement, Goal State

A complete one-page project charter with all seven sections signed off

4

Lab 6 — SIPOC, Stakeholder Analysis and RACI

A validated SIPOC (exported from the builder), a stakeholder power/int

Voice of the Customer, Affinity Diagram and Kano Analysis

LAB 3

Green Belts do not guess what the customer wants — they collect it, cluster it and classify it. Gather raw VOC, group it into themes with an affinity diagram, then use Kano analysis to separate the requirements that merely prevent dissatisfaction from the ones that actually delight.

You'll build

A VOC log, an affinity diagram of themes and a Kano classification table.

Tools & techniques: VOC sources, affinity diagram, Kano model, Must-Be / One-Dimensional / Delighter

Voice of the Customer, Affinity Diagram and Kano Analysis

STEPS 1-4 OF 7

1

List your VOC sources and split them into reactive (complaints, returns, social media) and proactive (surveys, interviews, gemba walks).

2

Record at least twelve verbatim customer statements for the Northwind order-fulfilment process — use the customer's own words, not your summary.

3

Write each verbatim on a separate sticky note or row so it can be moved independently.

4

Build the affinity diagram: cluster the verbatims into natural themes in silence first, then name each cluster.

Voice of the Customer, Affinity Diagram and Kano Analysis

STEPS 5-7 OF 7

5

Classify each theme against the Kano model — Must-Be (expected, causes dissatisfaction if absent), One-Dimensional (more is better), or Delighter (unexpected).

6

Mark which Kano category each theme belongs to and note that Must-Be requirements never generate satisfaction, only its absence.

7

Identify the one theme that, if fixed, would most reduce complaints — this becomes your project focus.

Voice of the Customer, Affinity Diagram and Kano Analysis

Check your work

You have at least twelve verbatims, every verbatim sits in a named affinity cluster, and every cluster carries a Kano classification.

CTQ Tree — Translating Customer Needs into Measurable Requirements

LAB 4

A customer need is not measurable; a CTQ is. Drive each VOC theme down through need to driver to a Critical-to-Quality requirement that carries a metric, a target and specification limits — because the Measure phase can only baseline what has been defined numerically.

You'll build

A three-level CTQ tree with metric, target, USL and LSL for each requirement.

Tools & techniques: CTQ tree, operational definitions, specification limits, performance metrics

CTQ Tree — Translating Customer Needs into Measurable Requirements

STEPS 1-4 OF 7

1

Take your top VOC theme (for example 'my order arrives late') and write it at the root of the tree.

2

Level 2 — break the need into drivers: what specifically must be true for that need to be met (picked on time, packed correctly, handed to carrier before cut-off).

3

Level 3 — convert each driver into a CTQ with a unit of measure (order-to-ship hours, picking accuracy percentage, carrier cut-off compliance rate).

4

For each CTQ write an operational definition: exactly how it is measured, by whom, from which system field, and when the clock starts and stops.

CTQ Tree — Translating Customer Needs into Measurable Requirements

STEPS 5-7 OF 7

5

Set the target and the specification limits (USL/LSL) for each CTQ from the customer promise, not from current performance.

6

Check each CTQ against the test: could two different people measure this and get the same number? If not, tighten the operational definition.

7

Select the one CTQ that becomes your project Y and confirm it matches the Y you wrote in Lab 2.

CTQ Tree — Translating Customer Needs into Measurable Requirements

Check your work

Every CTQ has a unit of measure, a target, specification limits and an operational definition unambiguous enough for two people to apply identically.

Project Charter, Problem Statement, Goal Statement and Scope

LAB 5

The charter is the project's contract with its sponsor. Write a problem statement that quantifies the gap without naming a cause or a solution, a goal statement that is measurable and time-bound, and an explicit in-scope/out-of-scope table that prevents scope creep later.

You'll build

A complete one-page project charter with all seven sections signed off.

Tools & techniques: Project charter, problem statement, goal statement, scope, business case, milestones

Project Charter, Problem Statement, Goal Statement and Scope

STEPS 1-4 OF 8

1

Write the problem statement with all four components: the process, the time period, the measurable gap and the business impact.

2

Check the problem statement names NO cause and NO solution — if it contains 'because' or 'by implementing', rewrite it.

3

Write the goal statement using the metric, baseline, target and date: 'reduce late shipments from 8.5% to 3.0% by 31 December'.

4

Build the in-scope / out-of-scope table — name the process start point, the stop point, and at least three things explicitly excluded.

Project Charter, Problem Statement, Goal Statement and Scope

STEPS 5-8 OF 8

5

State the business case: the annualised COPQ from Lab 1 and the benefit expected if the goal is achieved.

6

List the team: sponsor, process owner, Green Belt, Black Belt mentor and subject matter experts, with named individuals.

7

Add the DMAIC milestone schedule with a tollgate review date for each of the five phases.

8

Review the charter against the sponsor's priorities and confirm they would sign it.

Project Charter, Problem Statement, Goal Statement and Scope

Check your work

Your problem statement contains process, period, measurable gap and impact but no cause or solution; your goal statement contains metric, baseline, target and date.

SIPOC, Stakeholder Analysis and RACI

LAB 6

Before mapping the process in detail, agree its boundaries and its customers with SIPOC. Then identify every stakeholder the project touches and fix accountability with a RACI — because Green Belt projects fail on resistance far more often than on analysis.

You'll build

A validated SIPOC (exported from the builder), a stakeholder power/interest grid and a RACI matrix.

Tools & techniques: SIPOC & Process Map Builder, process boundaries, pain points, stakeholder analysis, power/interest grid, RACI

SIPOC, Stakeholder Analysis and RACI

STEPS 1-4 OF 11

1

Open the SIPOC & Process Map Builder — it walks you through the boundaries, the SIPOC and the pain points in order.

<https://alfredang.github.io/sipoc/>

2

Agree the process boundaries first — write the explicit start trigger and stop event for the Northwind order-fulfilment process. Boundary disputes are where most SIPOCs fail.

3

List the Process steps: five to seven high-level steps only. Resist the urge to detail — that comes in Lab 7.

4

Work outward: list the Outputs of the process and the Customers who receive each output.

SIPOC, Stakeholder Analysis and RACI

STEPS 5-8 OF 11

5

List the Inputs each step consumes and the Suppliers who provide them, internal and external.

6

Tag at least three pain points on the process steps — the delays, rework loops and unclear handoffs you already suspect.

7

Run 'Check my SIPOC' in the tool to validate your diagram against the lab criteria, then export it as PNG or PDF for your project pack.

8

Validate the SIPOC with someone who does the work — SIPOCs built in a meeting room are usually wrong.

SIPOC, Stakeholder Analysis and RACI

STEPS 9-11 OF 11

9

Identify every stakeholder and plot them on a power/interest grid: manage closely, keep satisfied, keep informed, or monitor.

10

Build the RACI for the project deliverables — exactly one Accountable per row, no more.

11

For each high-power stakeholder, note their likely objection and your engagement approach.

SIPOC, Stakeholder Analysis and RACI

Check your work

'Check my SIPOC' passes, all five columns are populated with explicit boundaries and at least three pain points tagged, and your RACI has exactly one Accountable per deliverable.

Recap — Define

- Voice of the Customer — Capture what the customer actually says — reactive and proactive sources — before deciding anything.
- Affinity diagram — Cluster raw VOC statements into natural themes so the signal emerges from the noise.
- Kano analysis — Classify requirements as Must-Be, One-Dimensional or Delighter — they are not equally valuable.
- CTQ tree — Translate each need into a specific, measurable Critical-to-Quality requirement with a target and limits.
- Project charter — Problem, goal, scope, business case, team and timeline — the project's contract with the sponsor.
- Problem statement — Process, time period, measurable gap and impact — and never a cause or a solution.

DMAIC · MEASURE

Measure — Quantify Performance

Process mapping · VSM · Takt · Data types · Sampling · Sample size · MSA/Gage R&R · Yield · DPMO ·
Baseline capability

Key Concepts — Measure (1)

1

Process mapping

Flowcharts and swimlanes expose the handoffs, delays and rework loops where defects are born.

2

Value stream mapping

Map material AND information flow, then compare value-added time against total lead time.

3

Lean metrics

WIP, lead time, cycle time, throughput and takt time — takt = available time / customer demand.

4

Types of data

Continuous vs discrete; nominal vs ordinal. The data type dictates which statistical tool is legal.

5

Sampling

Simple random, stratified, systematic and cluster — a biased sample invalidates every later conclusion.

6

Sample size

$n = (1.96s/d)^2$ for continuous data — how much data is enough, decided before you collect it.

7

MSA / Gage R&R

Before trusting the data, prove the measurement system: accuracy, repeatability, reproducibility.

8

Yield, DPU, DPO, DPMO

Classic yield hides rework; first pass and rolled throughput yield expose the hidden factory.

Key Concepts — Measure (2)

1

Baseline capability

Convert the baseline into a sigma level and Cp/Cpk so improvement can be proven later.

MEASURE

Measure replaces opinion with data

Define told us what hurts. Measure tells us how much, how often and where - in numbers a finance director would sign off on.

What the Measure phase delivers

1

A validated process map

How the work actually flows today, including every handoff and rework loop

2

A data collection plan

What is measured, by whom, how often, with what operational definition

3

A trustworthy measurement system

MSA proves the gage is not the source of the variation

4

A quantified baseline

DPMO, yield, sigma level and process capability before any change

5

Refined problem statement

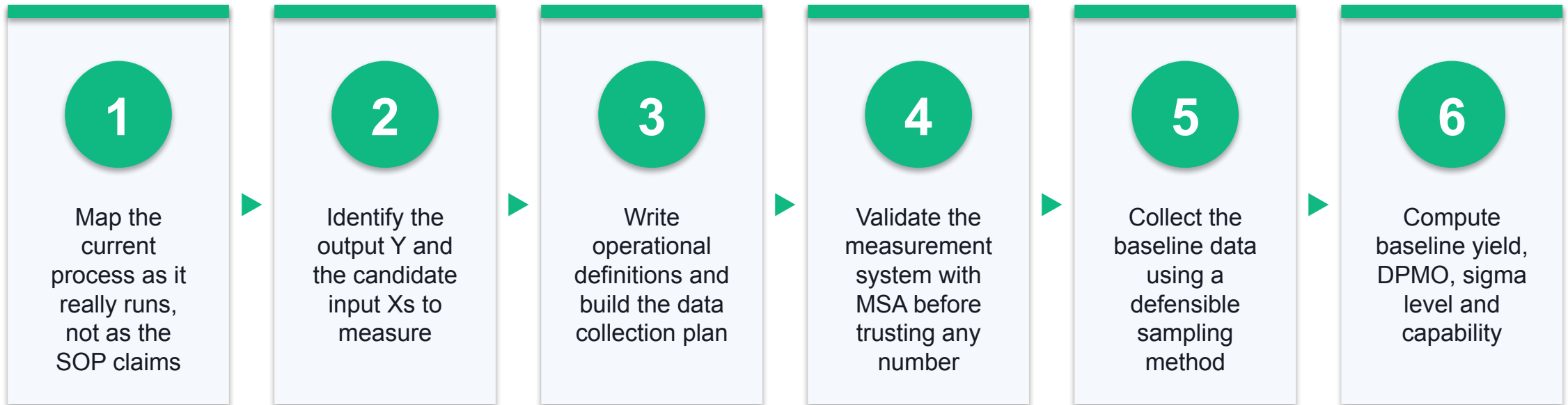
The Define scope narrowed to where the data says the pain sits

6

Evidence for the tollgate

Numbers the sponsor can act on, not anecdotes

Steps of the Measure phase

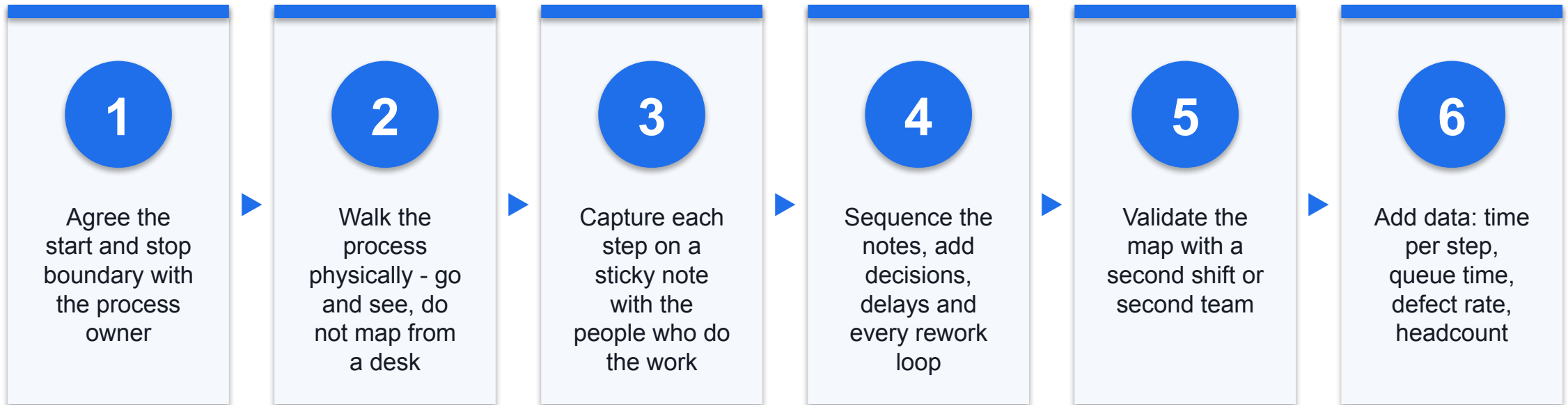


Standard process mapping symbols

SYMBOL	DESCRIPTION & USAGE
 Connector	• Connects any two steps and shows the path or direction of the process
 Delay	• Identifies when the process comes to a temporary halt • Also identifies what has to happen before the process resumes
 Document	• Indicates that a written document is prepared or used at that step of the process; the name of the document appears in the symbol

Oval = start/stop. Rectangle = activity. Diamond = decision. D-shape = delay. Document shape = paperwork produced.

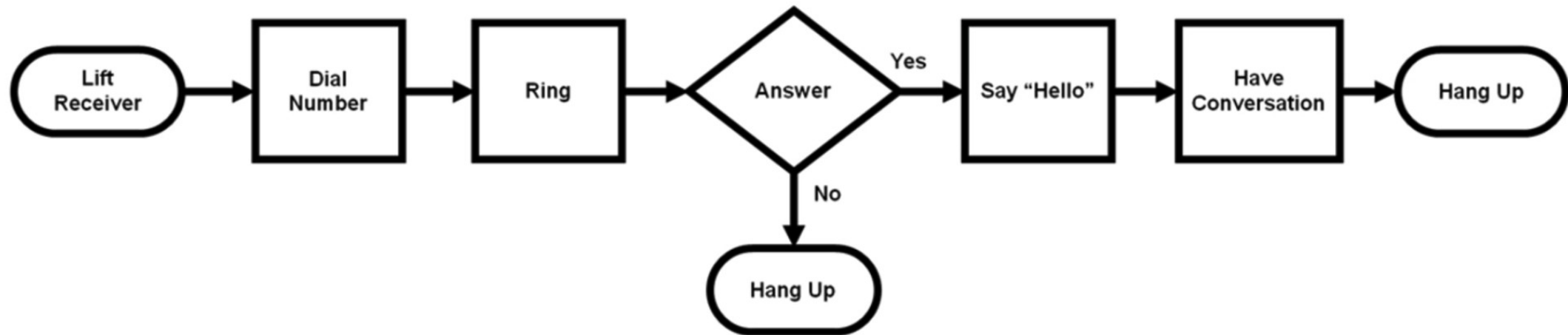
Steps to create a process map



Key questions to evaluate a process map

- Is any step done twice, or done differently by different people on different shifts?
- Where does work stop and wait, and how long does it sit there on average?
- Which steps exist only to check, inspect or correct someone else's earlier work?
- Where does the work change hands, and what is lost or delayed at each handoff?
- Which steps would the customer actually be willing to pay for?
- Which decisions are made without a written rule, so each person decides differently?

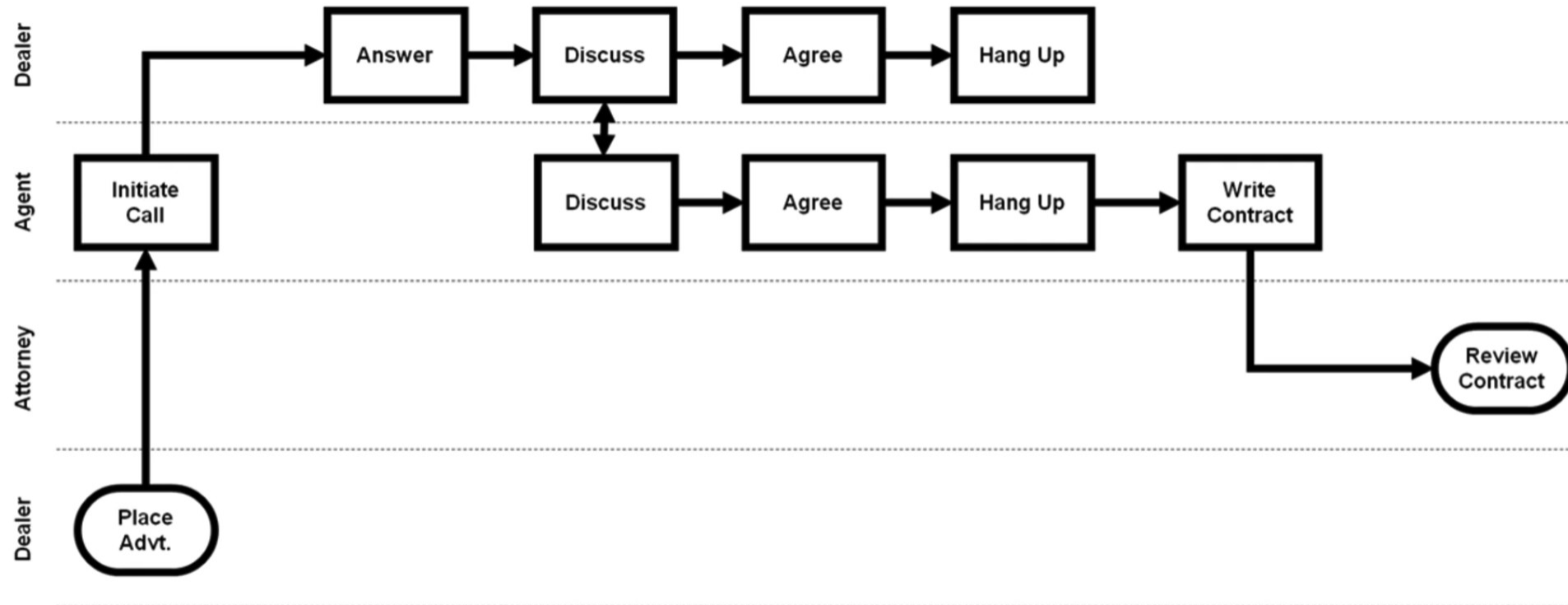
A worked process flowchart



Notice the rework loop: every backward arrow is a hidden factory consuming capacity nobody has budgeted for.

MEASURE

Swimlane (deployment) process map



Each horizontal lane is one role or department. Count the crossings - that count predicts your defect rate better than any survey.

How to read a swimlane map

1

One lane per actor

Role, team or system - never a person's name. Lanes survive staff turnover.

2

Count the crossings

Every lane crossing is a handoff: a queue, a re-entry, a chance to lose context.

3

Look for ping-pong

Work bouncing between two lanes signals unclear ownership or missing authority.

4

Find the orphan steps

Steps nobody claims are where things silently stall over a weekend.

5

Mark the decision lanes

Approvals concentrated in one lane create a bottleneck by design.

6

Time each crossing

Handoff wait time is usually larger than all touch time combined.

Northwind: swimlanes in order fulfilment

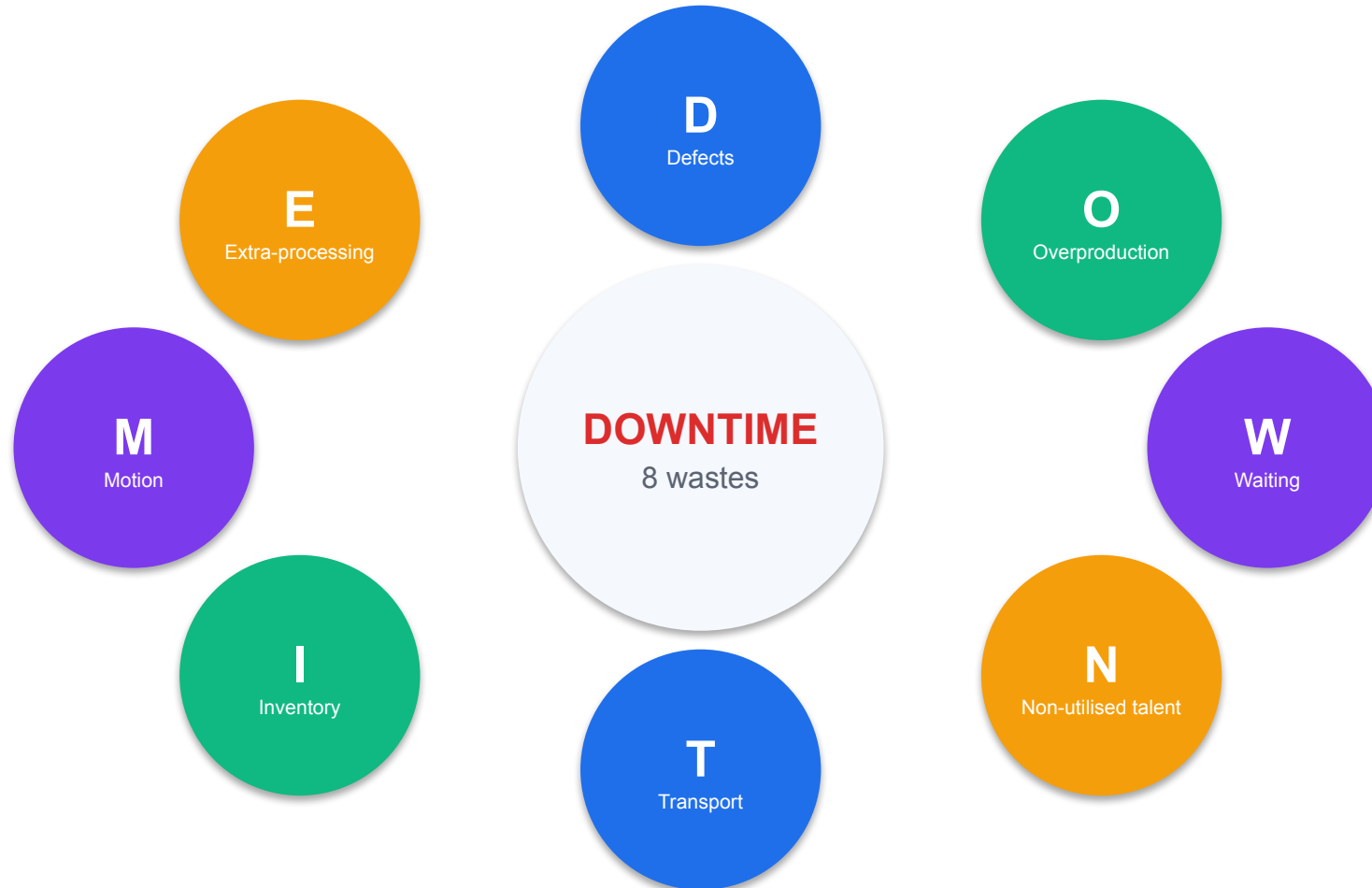
THE FLOW

- Customer Service takes the order and validates credit
 - Handoff 1 to Warehouse Planning for wave allocation
- Warehouse Planning releases the pick wave twice daily
 - Handoff 2 to Pickers on the floor
- Pickers pick, Packers pack and label
 - Handoff 3 to Dispatch for carrier manifest

WHAT THE MAP REVEALED

- Six lane crossings per order
 - Average queue at each crossing: 40 to 220 minutes
- Touch time across all steps: about 34 minutes
 - Everything else is waiting
- 357 of 4,200 orders late each month - 8.5 percent
 - Most late orders cross a crossing on a Friday

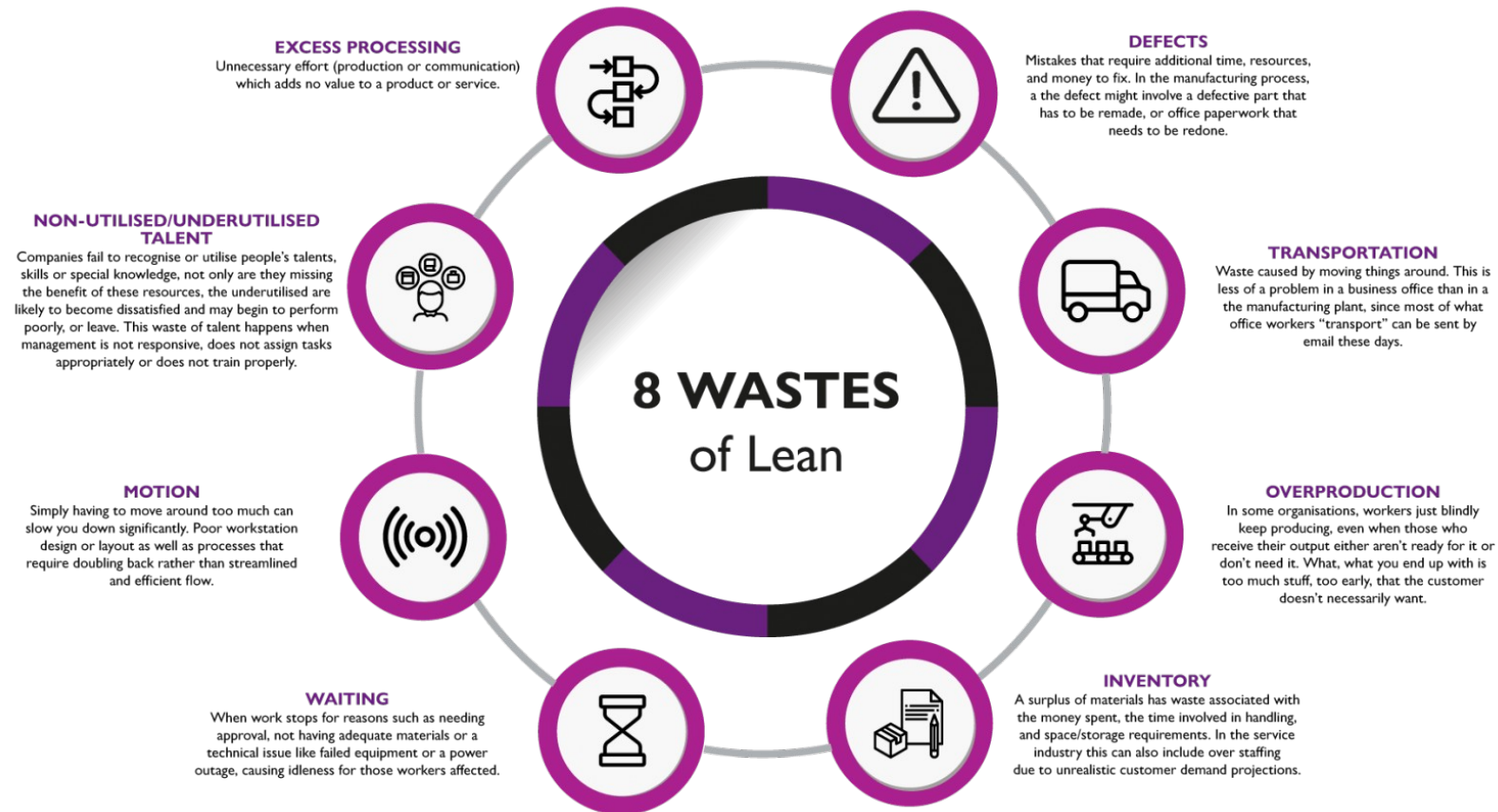
The 8 wastes of Lean - DOWNTIME



Waste consumes capacity, cash and lead time while adding nothing the customer values.

MEASURE

The eight wastes in detail



Learn DOWNTIME as a walking checklist - one pass of the floor should produce at least one example of each.

MEASURE

DOWNTIME at Northwind

1

D - Defects

Mis-picks returned by the customer; wrong labels forcing a full re-pack.

2

O - Overproduction

Printing pick lists for the whole day at 06:00 when only half can be picked by noon.

3

W - Waiting

Packed cartons sitting on the dispatch dock waiting for the 16:00 carrier collection.

4

N - Non-utilised talent

Experienced pickers who know the slotting problem are never asked about it.

5

T - Transport

Fast-moving SKUs slotted at the far aisle, so every wave walks the full length.

6

I - Inventory

Six weeks of slow-moving stock blocking the prime pick face near dispatch.

7

M - Motion

Packers reaching across the bench for tape and labels 300 times a shift.

8

E - Extra-processing

Three separate manual checks of the same pick before it reaches dispatch.

Value-added analysis - the three categories

Value-Added

Customer would pay for it - VA

- Physically transforms the product or service
- Done right the first time
- The customer explicitly wants it
- Northwind: picking, packing, loading
- Typically 5 to 20 percent of steps

Business-Value-Added

Required, but not customer value - BVA

- Needed for legal, tax or regulatory reasons
- Required to keep the business running
- Customer would not pay extra for it
- Northwind: customs paperwork, credit check
- Minimise it - you cannot delete it

Non-Value-Added

Pure waste - NVA, delete it

- Inspection, rework, transport, waiting
- Adds cost and lead time, adds no value
- Northwind: triple checks, dock queue
- Typically 60 to 90 percent of lead time
- This is your improvement opportunity

The three layers of a VSM

1

Material flow

Bottom half, left to right: how the physical work or the order moves through the process.

2

Information flow

Top half, right to left: forecasts, schedules and signals that trigger the work.

3

The timeline ladder

Underneath: process time on the lower step, waiting time on the upper step.

4

Data boxes

Under each process: cycle time, changeover, uptime, headcount, batch size.

5

Inventory triangles

Between processes: how much work sits in queue and for how long.

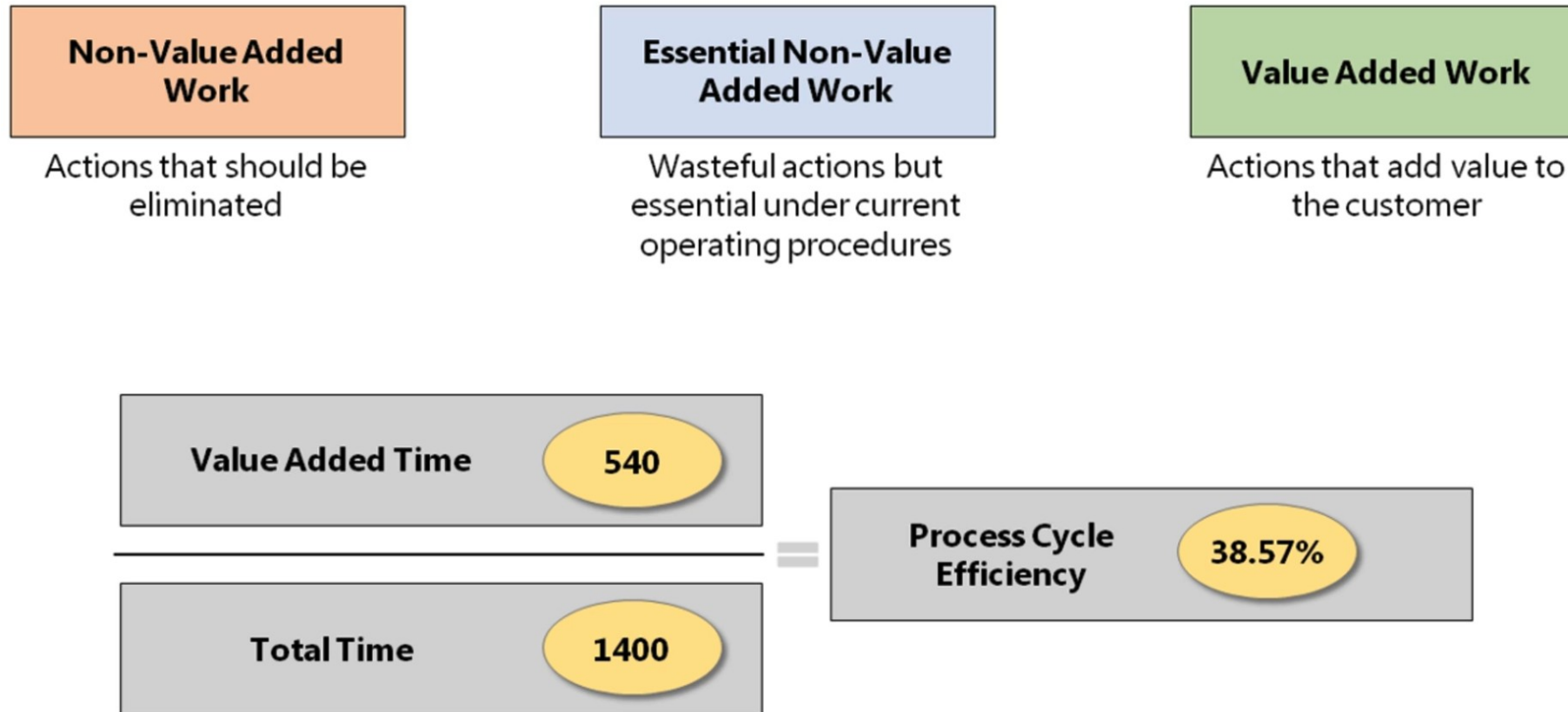
6

The summary line

Total lead time versus total value-added time - the efficiency headline.

MEASURE

A complete value stream map



Read the information flow right to left across the top, the material flow left to right along the bottom, and the ladder for the timeline.

Standard VSM icons



Customer /
Supplier Icon



Process
Flow



Kaizen
Burst



Operator



Timeline



Go & See



Electronic
Flow



Production
Kanban



Signal
Kanban



Inventory



Shipments



Push Arrow

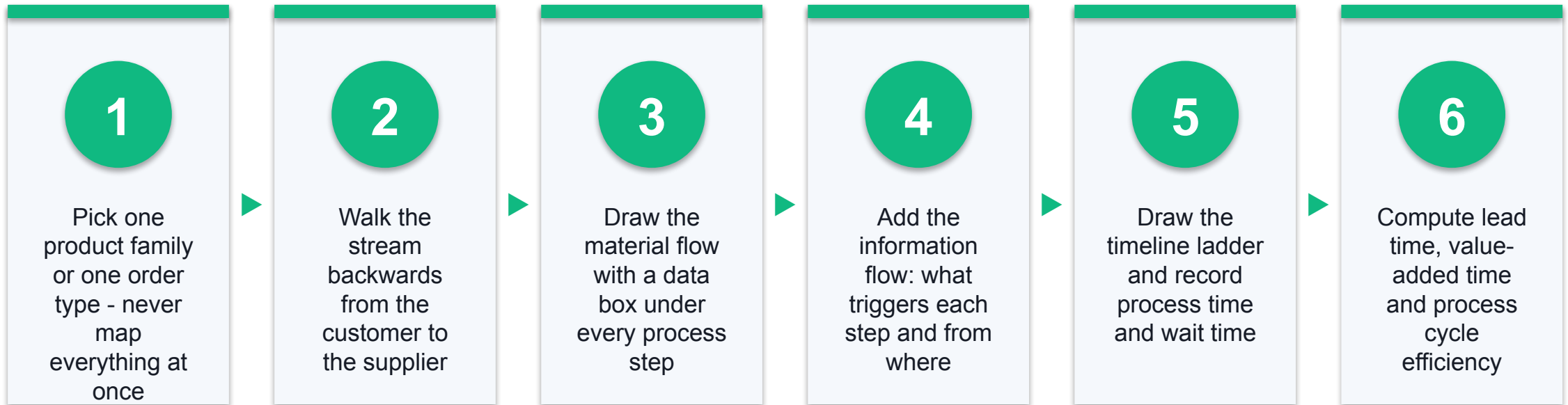


FIFO

Use the standard icon set so any Lean practitioner anywhere can read your map without a legend.

MEASURE

Steps to create a value stream map



MEASURE

Process cycle efficiency

Process Cycle Efficiency

$$\text{PCE} = \text{VA time} / \text{Lead time}$$

Northwind: 34 min VA against 4 days (5,760 min) lead time.

Northwind PCE

$$34 / 5760 = 0.0059$$

0.59 percent - well under the 10 percent typical of service processes.

Improvement target

Halve lead time, PCE doubles

Attack the queues, not the touch time - the queues own the lead time.

Most service and transactional processes run under 10 percent PCE. World class is above 25 percent.

The core Lean metrics

1

Work in Process (WIP)

How many units are inside the process right now, including everything queuing.

2

Lead time

Elapsed time from customer request to customer receipt, waiting included.

3

Cycle time

Time to complete one unit at a step, or the interval between completed units.

4

Throughput

Units completed per unit of time - the actual output rate of the process.

5

Takt time

The customer demand rhythm the process must match to avoid backlog or idle time.

6

Little's Law

Lead time = WIP / Throughput. Cut WIP and lead time falls with no extra staff.

Takt time explained

$$T = \frac{T_a}{T_d}$$

where, T stands for Takt Time

T_a stands for Net Available Time

T_d is the total customer demand (units produced per day)

**German word for
"Drum beat"**

**Establishes the pace of
the process needed to
meet customer demand**

**Shows where the flow
is uneven**

Takt = available working time divided by customer demand for that same period.

MEASURE

Takt time - worked example

Available time

$$58 \times 7.5 \times 3600 \text{ sec}$$

58 staff productive for 7.5 hours per day =
1,566,000 seconds available.

Takt time

$$\text{Takt} = \text{Avail time} / \text{Demand}$$

1,566,000 / 4,500 transactions = 348 seconds
per transaction.

Interpretation

One unit every 348 sec

Any step whose cycle time exceeds 348 sec
is a bottleneck and must be rebalanced.

Use only productive time - exclude breaks, briefings and planned downtime from the available time.

Takt vs cycle time vs lead time

Takt time

Set by the customer, not by you

- Available time divided by demand
- Changes only when demand changes
- The target rhythm for every step
- Northwind: 348 seconds per transaction

Cycle time

Set by the process design

- Time for one step to finish one unit
- Compare each step against takt
- Steps above takt are bottlenecks
- Balance the line towards takt

Lead time

What the customer experiences

- Request to receipt, waiting included
- Dominated by queues, not touch time
- Little's Law: WIP / throughput
- Northwind: about 4 days

Types of data

Qualitative

Attribute / categorical data

- Describes a category or an attribute
- Counted, not measured
- Pass/fail, late/on-time, defect type
- Needs large samples for precision
- Northwind: order late yes or no

Quantitative

Variable / measurement data

- A number on a real measurement scale
- Measured on a continuum
- Cycle time, weight, distance, cost
- Far more information per data point
- Northwind: hours from order to despatch

Discrete, continuous, nominal, ordinal

1

Discrete data

Counts of whole things - 3 defects, 17 late orders. No meaningful value between 3 and 4.

2

Continuous data

Any value on a scale - 4.37 hours, 12.6 kg. Infinitely divisible, richest information.

3

Nominal data

Named categories with no order - carrier A, B, C; defect type; region.

4

Ordinal data

Ordered categories with unequal gaps - satisfaction 1 to 5, priority high/medium/low.

5

Why it matters

t-tests and capability need continuous data. Proportions and chi-square need attribute.

6

Green Belt rule

Prefer continuous wherever you can get it - you need far fewer samples for the same power.

The data collection plan - six columns

1

What

The exact measure: 'hours from order cut-off to carrier scan', not 'delivery performance'.

2

Who

The named person or system responsible for capturing each observation.

3

When

Frequency and timing - every order, hourly sample, one shift in three.

4

How

The instrument, the system report, the check sheet, the exact recording method.

5

Sample size

Calculated, not guessed - see the sample size formula in the next section.

6

Stratification

The factors recorded alongside each point: shift, carrier, SKU class, day of week.

Operational definitions - two people, one number

1

The test

Two trained people measure the same thing independently and get the same number.

2

Define the measure

'Late' means the carrier scan timestamp is after 18:00 on the promised date.

3

Define the method

Which system field, which report, which rounding rule, which time zone.

4

Define the boundary

What counts and what is excluded - cancellations, samples, internal transfers.

5

Define the decision

The exact criterion that separates conforming from non-conforming.

6

Test it before use

Run a pilot with two people. If they disagree, the definition is not yet done.

Sampling techniques

Simple Random Sampling

Stratified Random Sampling

Systematic Sampling

Cluster Sampling

Only random-based methods support statistical inference. Convenience and judgment sampling do not.

The four random sampling techniques

1

Simple random

Every unit has an equal chance of selection. Use a random number generator, not intuition.

2

Stratified random

Split the population into homogeneous strata, then sample randomly within each stratum.

3

Systematic

Take every Nth unit after a random start. Beware hidden cycles matching your interval.

4

Cluster

Randomly select whole groups (a depot, a shift) and measure everything within them.

Random vs non-random sampling

Random methods

Valid for statistical analysis

- Simple random, stratified, systematic, cluster
- Selection probability is known
- Sampling error can be quantified
- Confidence intervals are meaningful
- Conclusions generalise to the population

Non-random methods

NOT valid for statistical analysis

- Convenience sampling - whatever is nearby
- Judgment sampling - what looks typical
- Selection probability is unknown
- Bias cannot be measured or corrected
- Use only for scoping, never for inference

Sample size for continuous data

$$n = \left[\frac{1.96 s}{d} \right]^2$$

n = sample size
s = standard deviation
d = margin of error
1.96 = 95% Confidence

n = (1.96 s / d) squared, where s is the standard deviation and d is the acceptable margin of error.

Sample size - continuous data

Formula

$$n = (1.96 s / d)^2$$

n = sample size, s = standard deviation, d = margin of error, 1.96 = 95 percent confidence.

Worked example

$$n = (1.96 \times 10 / 5)^2$$

Estimate mean cycle time within 5 days, preliminary s = 10 days.

Result

$$3.92^2 = 15.4 \rightarrow n = 16$$

Sixteen observations. Always round the calculated sample size UP.

Halving the margin of error d quadruples the required sample size - precision is expensive.

Sample size for discrete data

$$n = \left[\frac{1.96}{d} \right]^2 [p (1 - p)]$$

n = sample size
d = margin of error
p = proportion defective
1.96 = 95% Confidence

Attribute data needs far larger samples than continuous data for the same confidence - another reason to prefer continuous measures.

Sample size in practice

1

Where does s come from

A pilot sample of 20 to 30 points, or historical data from the same process.

2

Choosing d

Set by the decision, not by convenience: how wrong can you afford the estimate to be.

3

Why 1.96

The z value for 95 percent confidence - the Six Sigma community default.

4

Always round up

15.4 becomes 16. Rounding down silently degrades your confidence level.

5

More is not always better

Beyond the calculated n you spend money to shrink an already adequate interval.

6

Attribute data costs more

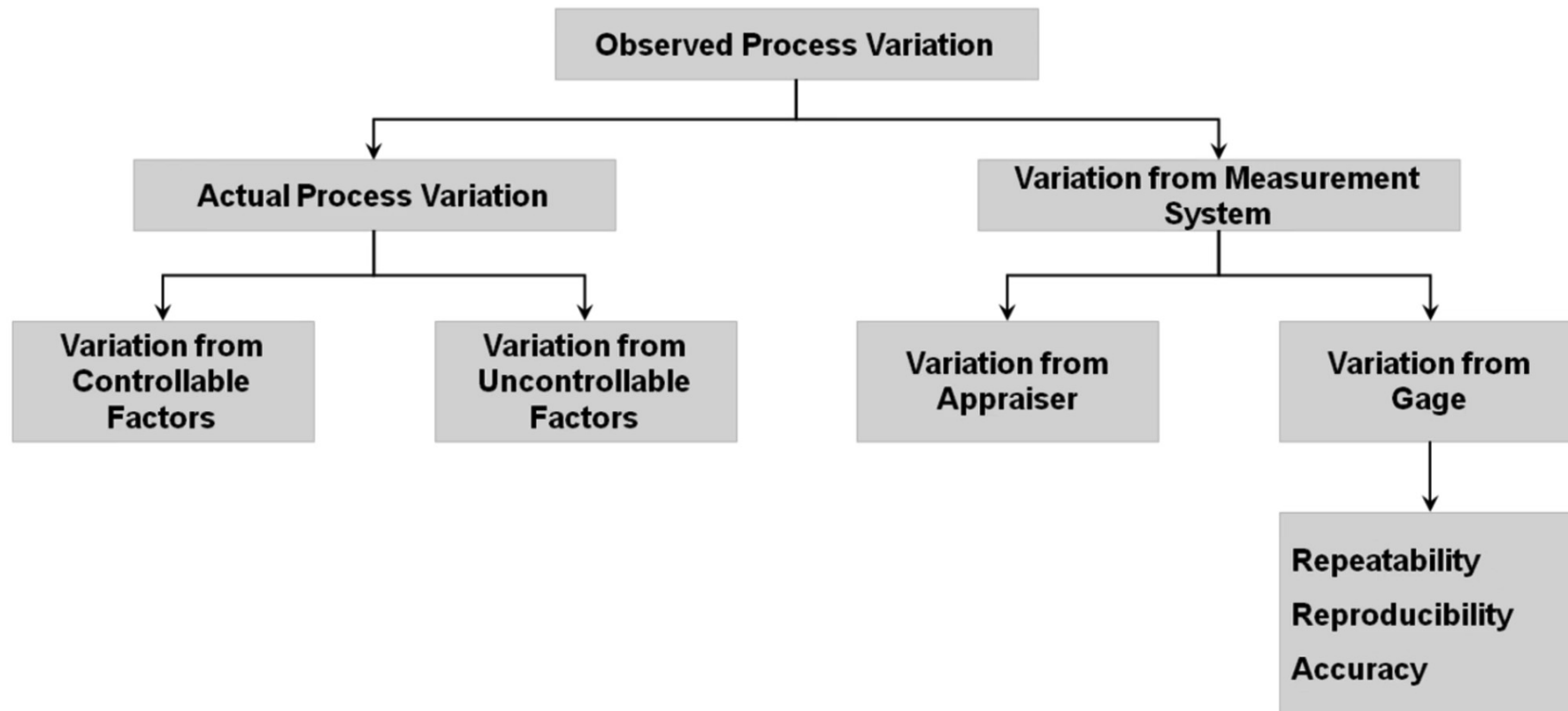
Estimating a proportion precisely may need hundreds of units, not sixteen.

MEASURE

If the gage varies, you chase phantoms

Total observed variation = actual process variation + measurement system variation.
MSA proves how much of what you see is real before you spend a cent fixing it.

Sources of variation - the MSA tree



Observed variation splits into true process variation and measurement system variation, which splits again into repeatability and reproducibility.

The variation equation

Total observed

$$\text{Var}(\text{total}) = \text{Var}(\text{proc}) + \text{Var}(\text{MS})$$

Variances add, standard deviations do not.
Always work in variance terms.

Measurement system

$$\text{Var}(\text{MS}) = \text{Var}(\text{rep}) + \text{Var}(\text{repro})$$

Repeatability is the equipment; reproducibility is the appraiser.

The goal

$$\text{Var}(\text{MS}) \text{ small vs } \text{Var}(\text{total})$$

Aim for the measurement system to consume under 10 percent of total variation.

Resolution - the ten-bucket rule

1

The rule

The device increment must be about one tenth of the tolerance or the variation of interest.

2

Bad example

Scale reads to 1 kg but the spec band is 2 kg - you have only 2 buckets, not 10.

3

Good example

Timestamp to the second when detecting improvements measured in minutes.

4

Symptom of failure

Data clumps onto only two or three distinct values across the whole sample.

5

Northwind case

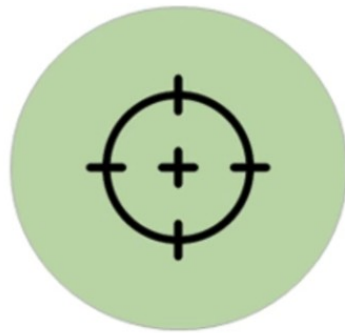
Despatch recorded to the nearest day cannot detect a 4-hour improvement.

6

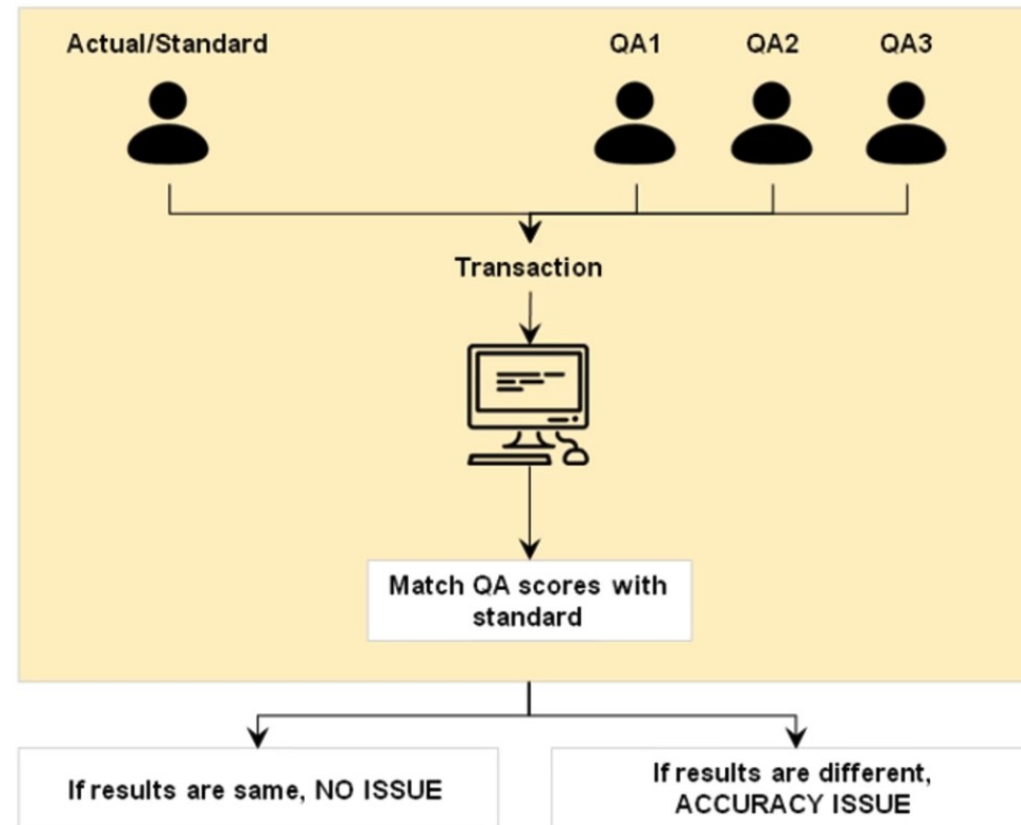
The fix

Change the recording increment before doing anything else - it is often free.

Accuracy, linearity and stability



Accuracy



Accuracy is the offset from truth. Linearity is whether that offset changes across the range. Stability is whether it drifts over time.

Accuracy, linearity, stability

Accuracy (Bias)

Offset from the true value

- Difference between the average reading and truth
- Requires a known reference standard
- Constant bias is correctable by calibration
- Example: scale reads 0.4 kg high every time

Linearity

Does the bias change across the range

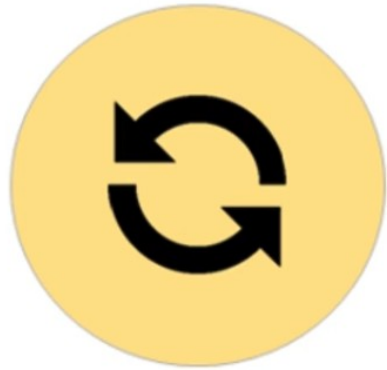
- Bias measured at low, mid and high values
- Accurate at 10 kg but 3 percent high at 100 kg
- Detected by measuring several known standards
- A linearity failure cannot be fixed by one offset

Stability

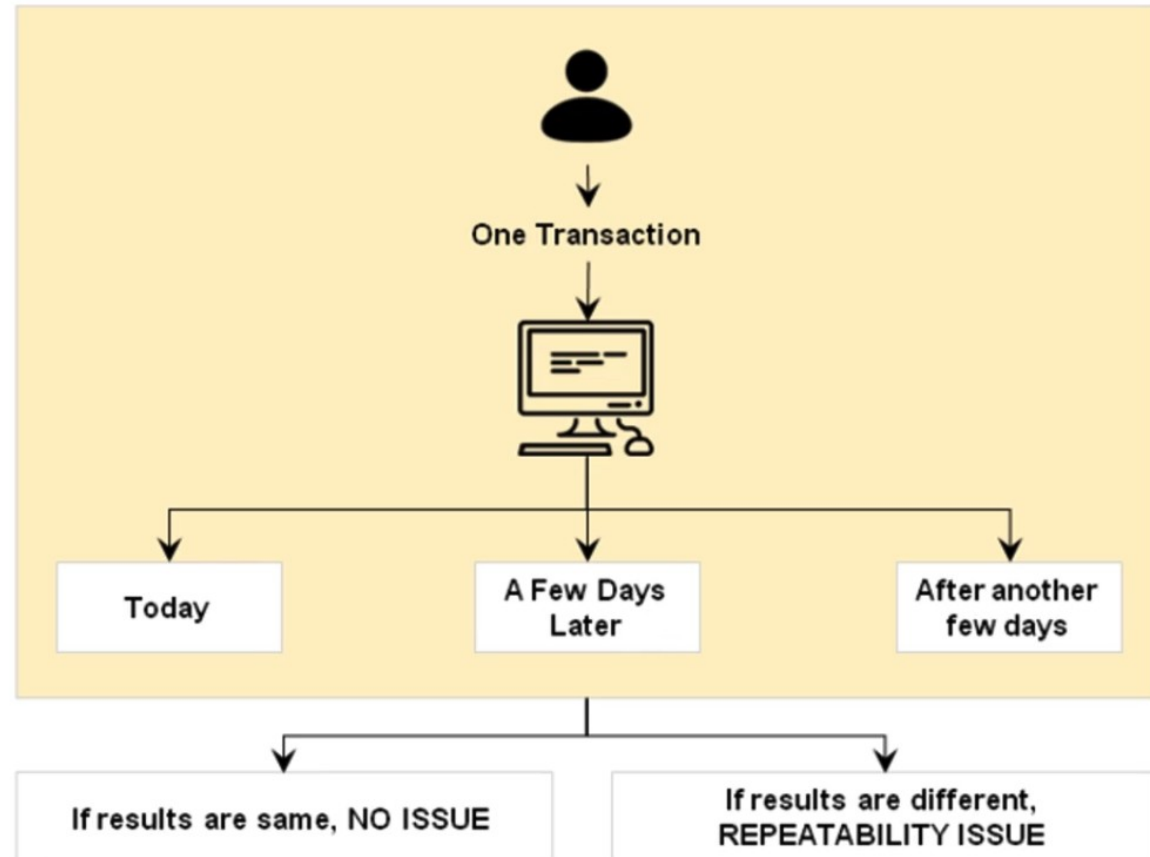
Does the bias drift over time

- Same standard measured repeatedly over weeks
- Plotted on a control chart to detect drift
- Drives the calibration interval
- Example: gage drifts after 200 uses

Repeatability and reproducibility



Repeatability



Gage R&R quantifies both components so you know whether to fix the equipment or retrain and re-standardise the people.

Repeatability vs Reproducibility

Repeatability

Equipment variation - EV

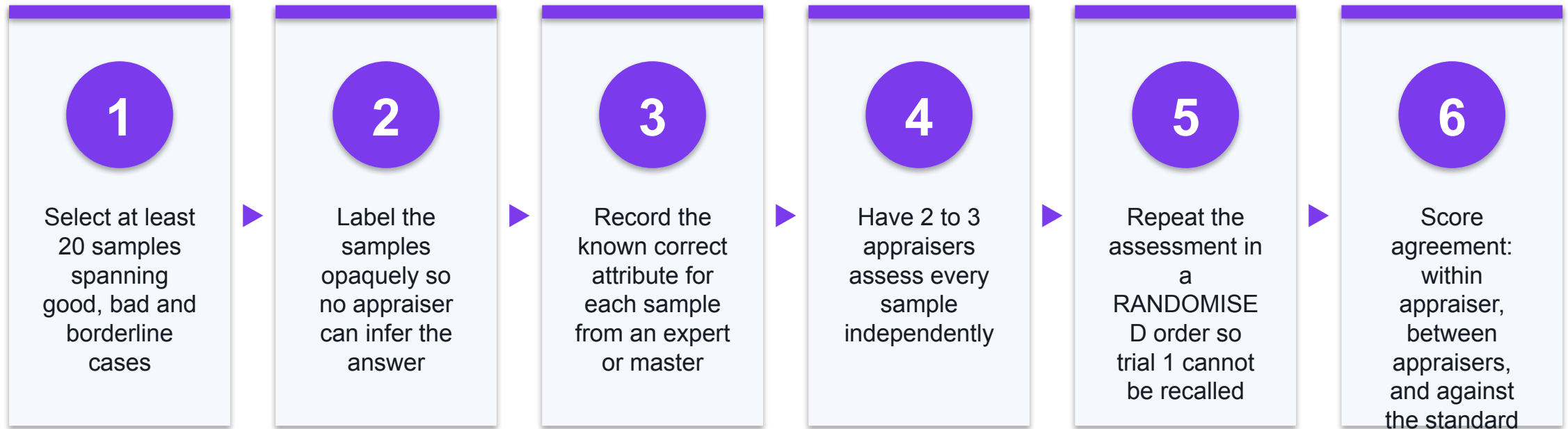
- SAME appraiser, SAME item, repeated measures
- Variation inherent in the device or method
- Poor result means the instrument is the problem
- Fix: better gage, better fixture, finer resolution
- Northwind: same clerk, same order, two readings

Reproducibility

Appraiser variation - AV

- DIFFERENT appraisers, SAME item, same conditions
- Variation caused by who is doing the measuring
- Poor result means the definition or training failed
- Fix: operational definitions, training, standard work
- Northwind: three clerks grading the same 'late' order

Attribute Gage R&R - the procedure



Attribute Gage R&R - rules that make it valid

1

At least 20 samples

Fewer than 20 and a single disagreement swings the percentage wildly.

2

Include the borderline

Easy pass and easy fail items flatter the system. Edge cases are the real test.

3

Opaque labelling

Random codes, no sequence, no visible grouping by known condition.

4

Known standard required

Without an expert-agreed truth you can measure agreement but not correctness.

5

Randomise every trial

Otherwise appraisers repeat their memory of trial 1 instead of re-assessing.

6

Blind the appraisers

Never let them see each other's calls or their own previous scores.

MEASURE

Variable Gage R&R - acceptance criteria

1

% Study Variation

0 to 10 pass | 10 to 30 caution | 30 and above fail

2

% Tolerance

0 to 10 pass | 10 to 30 caution | 30 and above fail

3

% Contribution

0 to 1 pass | 1 to 9 caution | 10 and above fail

4

Distinct Categories

10 or more pass | 6 to 10 caution | 1 to 5 fail

Reading the Gage R&R acceptance table

- Percent Study Variation compares measurement error against the total observed variation.
- Percent Tolerance compares measurement error against the customer specification width.
- Percent Contribution is variance-based, so its thresholds are the squares of the others.
- Number of Distinct Categories is how many separate levels the system can actually tell apart.
- DECISION RULE: one caution with all other metrics passing is acceptable for use.
- Anything landing in the fail zone means repair or replace the measurement system first.

What to do with a failing Gage R&R

Repeatability fails

The equipment is the problem

- Improve gage resolution or condition
- Add a fixture to standardise positioning
- Service, calibrate or replace the device
- Automate the reading where possible

Reproducibility fails

The people or the method are the problem

- Rewrite the operational definition
- Retrain all appraisers to one standard
- Add reference samples and visual aids
- Create standard work for the measurement

Both fail

Stop the project data collection

- Do not collect a baseline with this system
- Fix the measurement system as a mini-project
- Re-run the Gage R&R to confirm the fix
- Only then proceed to baseline collection

DUDO analysis

- A single parameter updated incorrectly is considered an error
- If more than 3 parameters updated incorrectly, the form is considered defective

	Scenario 1	Scenario 2
• No. of Units =	1	1
• Opportunities of Errors =	8	8
• Defects =	2	5
• Defective =	No	Yes

"A unit can have multiple defects, but it can only either be defective or non-defective"

©dax

Defect, Unit, Defect Opportunity, Observed defects - agree all four before you compute a single metric.

DUDO - the four definitions

1

D - Defect

Any failure to meet a customer requirement - one CTQ not satisfied on one unit.

2

U - Unit

The thing being produced or delivered that the customer receives. One order at Northwind.

3

D - Defect Opportunity

Every distinct way a single unit could fail. A form with 10 fields has 10.

4

O - Observed defects

The actual count of defects found in the sample you inspected.

5

Why opportunities matter

DPMO normalises complexity so a 10-field form compares fairly with a 60-field one.

6

Be conservative

Inflating the opportunity count artificially flatters your sigma level. Auditors check this.

DUDO applied to Northwind

DEFINITIONS

- Unit: one customer order
- Defect: order not despatched by the promised cut-off
- Opportunities: 1 per order (on-time or not)
- Observed: 357 late in the month

THE BASELINE

- Baseline volume: 4,200 orders per month
- Defect rate: $357 / 4,200 = 8.5$ percent
- First pass yield: 91.5 percent
- This is the number the whole project moves

Three ways to measure yield

Classic Yield

Units out / units in

- Counts anything that eventually shipped
- Rework and retest are invisible
- Always the most flattering number
- Hides all the cost of poor quality

First Pass Yield

Right first time at ONE step

- Units through with no rework or repair
- Computed step by step
- Exposes rework at each individual step
- The honest per-step measure

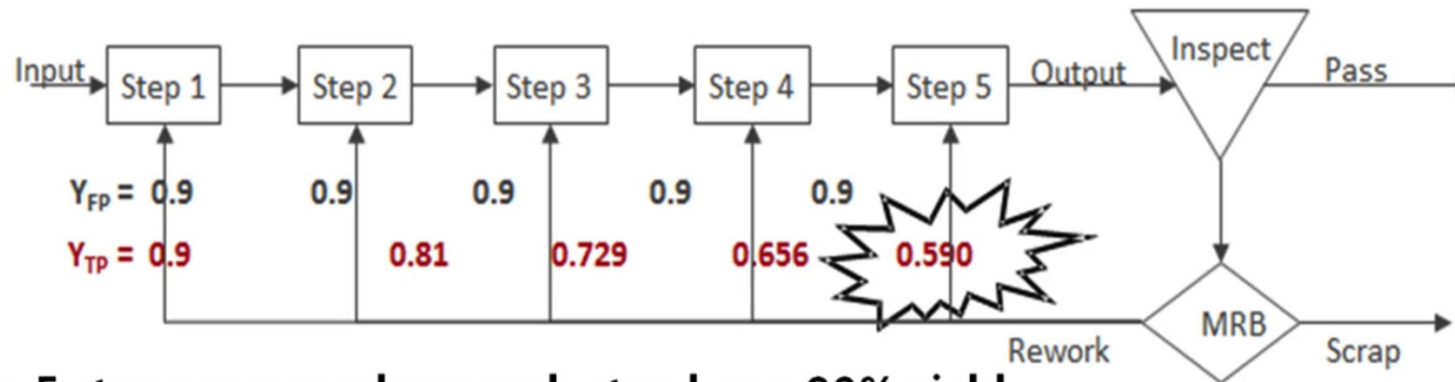
Rolled Throughput Yield

Right first time ACROSS all steps

- $RTY = FPY1 \times FPY2 \times \dots \times FPYn$
- Probability a unit passes every step cleanly
- Falls fast as step count rises
- The true customer experience

MEASURE

Rolled throughput yield



For a 5 step process where each step has a 90% yield ...

$$Y_{TP} = (Y_{FP1}) \times (Y_{FP2}) \times (Y_{FP3}) \times (Y_{FP4}) \times (Y_{FP5}) = 0.590 = 59\%$$

If each individual yield is 99%, for a 5 step process the Y_{TP} = 95%

If four steps have a 99% yield and one has a 90% yield, the Y_{TP} drops to 86%!

Poor quality accumulates rapidly!

Multiply the first pass yields of every step. Long processes with good-looking steps still deliver poor overall yield.

MEASURE

Rolled throughput yield - worked example

RTY formula

$$\text{RTY} = \text{FPY1} \times \text{FPY2} \times \dots \times \text{FPYn}$$

Multiply the first pass yield of every step in the process chain.

Five steps at 95%

$$\text{RTY} = 0.95^5$$

$$0.95 \times 0.95 \times 0.95 \times 0.95 \times 0.95 = 0.7738.$$

The hidden factory

77.4% vs 95% headline

Each step looks excellent, yet 22.6 percent of units are reworked somewhere.

The hidden factory is the capacity, cost and lead time consumed by rework that no budget line shows.

MEASURE

DPU, DPO and DPMO

DPU

$$\text{DPU} = \text{Defects} / \text{Units}$$

Average defects carried by each unit produced.

DPO

$$\text{DPO} = \text{Defects} / (\text{U} \times \text{O})$$

Defects per single opportunity, normalised for unit complexity.

DPMO

$$\text{DPMO} = \text{DPO} \times 1,000,000$$

Scaled to a million opportunities so any two processes can be compared.

MEASURE

DPMO - worked example (forms)

Opportunities

$$10 \text{ fields} \times 90 \text{ forms} = 900$$

Each form has 10 fields, so each form carries 10 defect opportunities.

Observed defects

2 errors found

Two incorrectly completed fields across the entire 90-form sample.

DPMO

$$(2 / 900) \times 1,000,000$$

= 2,222 DPMO. Note this is far better than 2 defects in 90 forms sounds.

The opportunity count changes the answer dramatically - define it once, document it, never change it mid-project.

MEASURE

DPMO - Northwind baseline

Units and defects

4,200 orders, 357 late

One opportunity per order: the order is either on time or it is not.

DPMO

$(357 / 4200) \times 1,000,000$

= 85,000 DPMO. This is the number the project must move.

Sigma level

85,000 DPMO -> ~2.9 sigma

Between 3 sigma (66,800) and 2 sigma (308,000) - closer to 3.

Northwind operates at roughly 2.9 sigma - typical for an unimproved transactional process.

MEASURE

The sigma scale - DPMO at each level



Each sigma level is roughly a ten-fold reduction in defects - the gains get harder as you climb.

Descriptive statistics - centre and spread

Measures of centre

Where the data sits

- Mean: the arithmetic average, uses every value
- Median: the middle value when sorted
- Mode: the most frequently occurring value
- Median resists outliers; the mean does not
- Report both when the data is skewed

Measures of spread

How much the data varies

- Range: maximum minus minimum, uses 2 points
- Standard deviation: average distance from the mean
- Variance: standard deviation squared - variances add
- Spread is the enemy - Six Sigma attacks variation
- Northwind despatch: mean 6.4 h, s 3.1 h

Reading histogram shape

1

Bell shaped

Symmetric, single peak. Common cause variation only. Normal tools apply.

2

Right skewed

Long tail towards high values. Typical of times, delays and costs.

3

Left skewed

Long tail towards low values. Often a ceiling or a cap in the process.

4

Bi-modal - two peaks

TWO PROCESSES measured as one. Stratify by shift, carrier, line, then re-plot.

5

Truncated / cut off

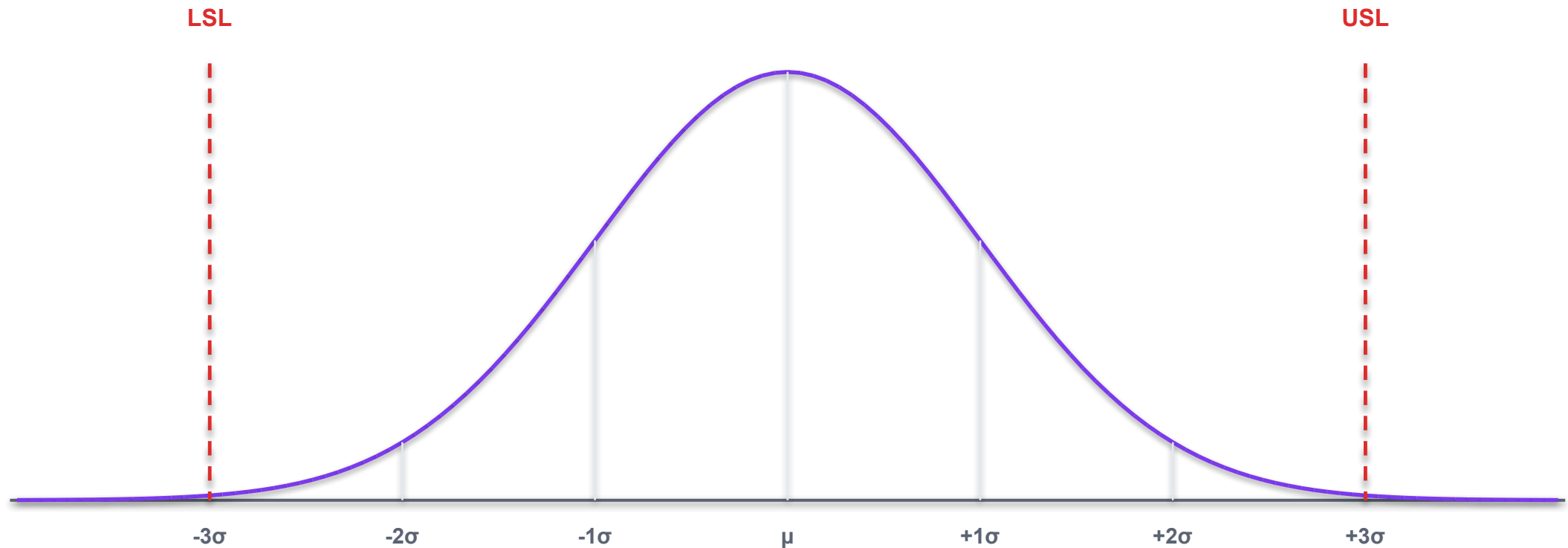
A cliff edge usually means data is being filtered, rounded or censored.

6

Isolated island

A small separate cluster - a distinct special cause worth investigating alone.

The normal distribution and the empirical rule



68.26 percent within ± 1 sigma, 95.46 percent within ± 2 sigma, 99.73 percent within ± 3 sigma.

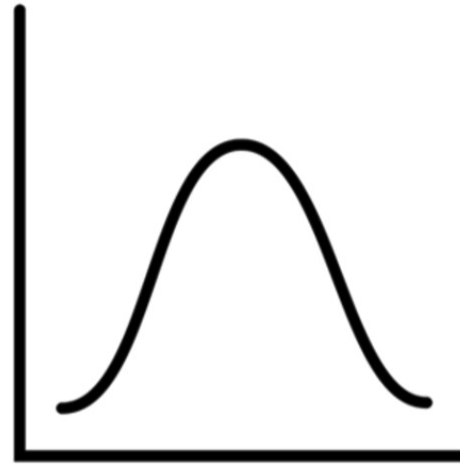
The normal distribution

Mean = Median = Mode

Curve is symmetrical on both sides

Area under the curve = 1

Tails are asymptotic



Symmetric about the mean, fully described by two parameters: the mean (location) and the standard deviation (spread).

The empirical rule in practice

1

+/- 1 sigma

68.26 percent of all observations fall inside one standard deviation of the mean.

2

+/- 2 sigma

95.46 percent of all observations fall inside two standard deviations.

3

+/- 3 sigma

99.73 percent - which is why 3 sigma control limits flag only rare events.

4

Beyond 3 sigma

About 0.27 percent, or 2,700 per million - too many for a Six Sigma process.

5

Northwind applied

Mean 6.4 h, s 3.1 h means about 95 percent despatch between 0.2 and 12.6 hours.

6

Check normality first

Use a histogram and a normal probability plot before applying these percentages.

Voice of the customer vs voice of the process

Voice of the Customer

Specification limits - USL and LSL

- Set by the customer, contract or regulation
- USL: upper specification limit
- LSL: lower specification limit
- Never calculated from your own process data
- Northwind: despatch within 8 hours of cut-off

Voice of the Process

Control limits and the distribution

- Calculated from what the process actually does
- Mean and standard deviation of real output
- Control limits are ± 3 sigma of the process
- Tells you what the process CAN deliver today
- Northwind: mean 6.4 h, s 3.1 h

The Cp formula

$$C_p = \frac{USL - LSL}{6s} = \frac{\text{Spec. Width}}{\text{Process Spread}}$$

$C_p \geq 1$ -----► Potentially capable

$C_p = (USL - LSL) / 6s$ - the ratio of the allowed spread to the actual process spread. C_p ignores centring.

MEASURE

Baseline capability - Cp

Cp

$$Cp = (USL - LSL) / 6s$$

Specification width divided by the natural process width of six standard deviations.

Interpretation

Cp = 1.0 means just fits

Cp below 1 cannot meet spec even when perfectly centred. Cp of 1.33 is a common minimum.

The limitation

Cp ignores the mean

A perfectly narrow process sitting off-centre still produces defects. Cpk fixes this - taught in Control.

Cp is potential capability. Cpk is actual capability and is covered fully in the Control phase.

Interpreting the baseline capability number

1

Cp below 1.00

The process spread is wider than the specification.
Defects are guaranteed.

2

Cp 1.00 to 1.33

Marginal. Any small drift in the mean immediately produces defects.

3

Cp 1.33 or above

The commonly accepted minimum for a controlled process.

4

Cp 2.00

The Six Sigma standard - spec width is twice the natural process width.

5

Northwind baseline

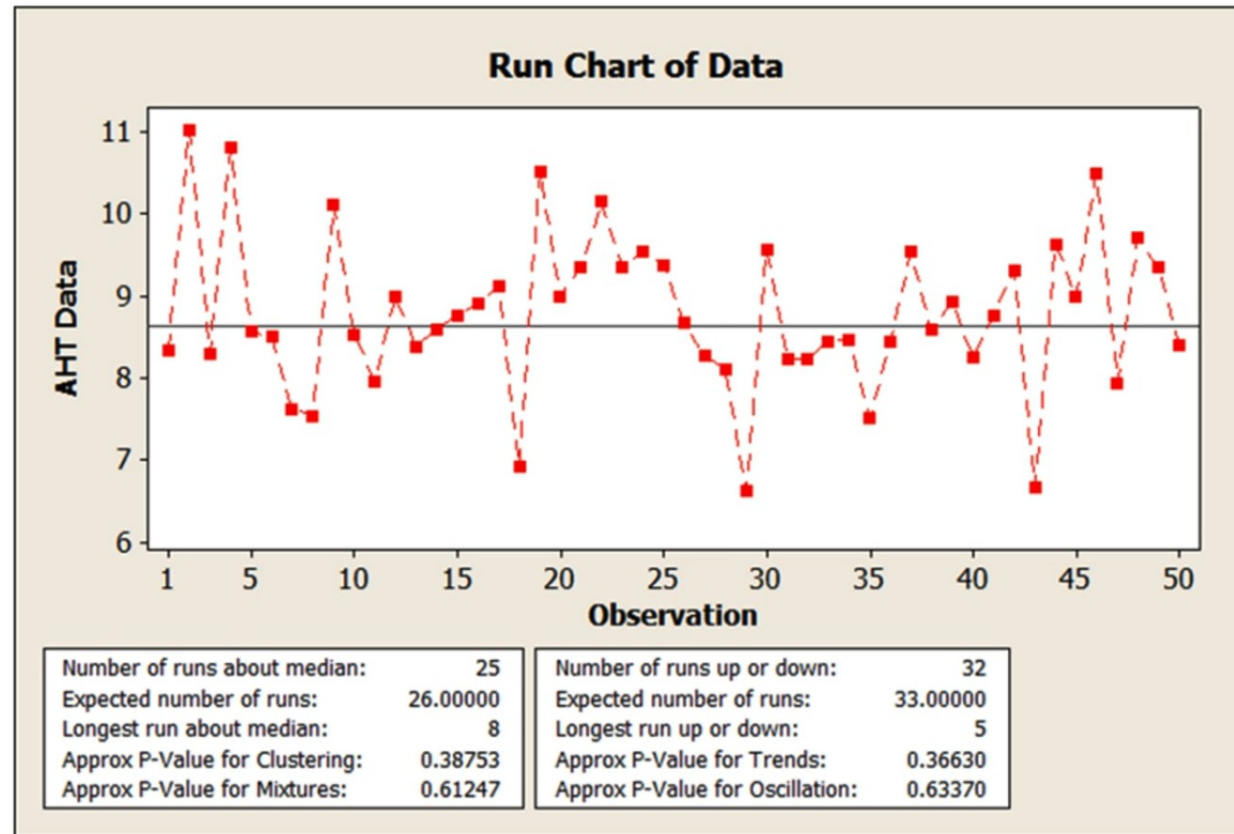
Wide spread against an 8-hour promise gives a Cp well below 1.

6

What Cp does not tell you

Nothing about centring. Always pair Cp with Cpk before concluding.

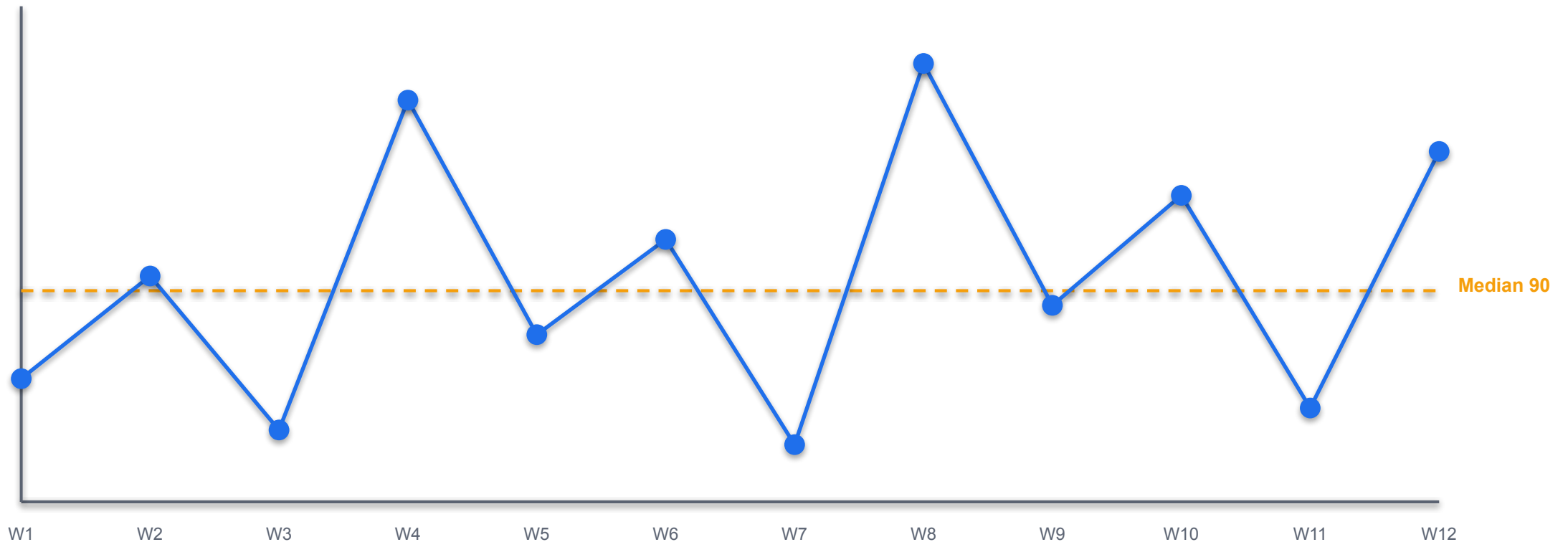
Baseline run chart



Plot the baseline over time before summarising it - a stable process and a drifting one can share the same mean and standard deviation.

MEASURE

Northwind baseline - late orders per week



Common cause variation around a stable median - the process is consistently bad, not occasionally bad. That means a system fix, not a people fix.

Common Measure phase mistakes

1

Skipping MSA

The single most common Green Belt failure - the whole baseline becomes unusable.

2

Mapping from a desk

The SOP map is fiction. Walk the process or you will map the wrong thing.

3

Guessing the sample size

Undersized samples give confident conclusions that do not replicate.

4

Ignoring bi-modality

Summarising two processes as one produces a mean that describes neither.

5

Inflating opportunities

Gaming the opportunity count to flatter DPMO destroys project credibility.

6

No stratification data

Forgetting to record shift and carrier means re-collecting everything later.

Measure phase - checkpoint

- Which is the bigger lever at Northwind: touch time or queue time, and how do you know?
- State the sample size formula for continuous data and explain each of its terms.
- Explain the difference between repeatability and reproducibility, with a fix for each.
- Five steps each at 95 percent first pass yield - what is the RTY, and what does the gap mean?
- Compute Northwind's DPMO from 4,200 orders and 357 late, and give the sigma level.
- Your histogram is bi-modal. What has almost certainly happened, and what do you do next?

WHAT YOU'LL DO

Hands-On Labs — Measure (1)

1

Lab 7 — Detailed Process Mapping and Swimlane Analysis

A detailed process map and a swimlane map with pain points and handoff

2

Lab 8 — Value Stream Mapping, Takt Time and the Eight

A current-state VSM with a timeline ladder, a takt time calculation and

3

Lab 9 — Data Types, Operational Definitions and the Data

A data type classification, operational definitions and a complete data

4

Lab 10 — Sampling Techniques and Sample Size Calculation

A justified sampling plan and calculated sample sizes for continuous data

5

Lab 11 — Measurement System Analysis and Gage R&R

An attribute Gage R&R study with repeatability, reproducibility and accuracy

6

Lab 12 — Yield, DPU, DPO, DPMO, RTY and the Hidden Factory

A metrics worksheet with yield, FPY, RTY, DPU, DPO and DPMO calculated

Hands-On Labs — Measure (2)

1

Lab 13 — Descriptive Statistics, Normality and Baseline

A descriptive statistics summary, a normality assessment and calculate

Detailed Process Mapping and Swimlane Analysis

LAB 7

The SIPOC gave the macro view; now build the detailed map that shows every actor, system and handoff. Handoffs are where delay and defects are created, so a swimlane map that puts each actor in their own lane makes the problem visible in a way a flowchart cannot.

You'll build

A detailed process map and a swimlane map with pain points and handoffs marked.

Tools & techniques: SIPOC & Process Map Builder, process symbols, flowchart, swimlane map, handoff analysis, pain points

Detailed Process Mapping and Swimlane Analysis

STEPS 1-4 OF 9

1

Review the standard process symbols: oval for start/stop, rectangle for activity, diamond for decision, D-shape for delay, and the document symbol.

2

Walk the process physically (go to gemba) and record every step in sequence as it actually happens — not as the SOP says it should.

3

Open lab07-process-steps-timing.xlsx — a gemba walk record of all 14 as-is steps with actor, system, elapsed time and handoff flag. Build the detailed process map from it.

4

Continue in the SIPOC & Process Map Builder — assign an actor to each step and it generates the swimlane and the handoff table for you.

<https://alfredang.github.io/sipoc/>

Detailed Process Mapping and Swimlane Analysis

STEPS 5-8 OF 9

5

Redraw the same flow as a swimlane map, giving each actor or department its own lane.

6

Count the handoffs in the workbook's Handoff column — every lane crossing is a queue and a risk of information loss.

7

Mark every decision diamond that creates a rework loop and note what percentage of work takes the rework path.

8

Tag pain points: delays, rework, unclear ownership, duplicate data entry and waiting for approval.

Detailed Process Mapping and Swimlane Analysis

STEPS 9-9 OF 9

9

Identify the three steps you suspect consume the most elapsed time — you will test that suspicion with data in Lab 9.

Detailed Process Mapping and Swimlane Analysis

Check your work

Every lane crossing on your swimlane map is marked as a handoff with a named owner on both sides, and every rework loop is labelled with its percentage.

Value Stream Mapping, Takt Time and the Eight Wastes

LAB 8

A value stream map shows material AND information flow, plus the timeline ladder that exposes how little of the total lead time actually adds value. Combine it with a takt time calculation and a waste walk to quantify the opportunity.

You'll build

A current-state VSM with a timeline ladder, a takt time calculation and a waste walk log.

Tools & techniques: VSM icons, material and information flow, timeline ladder, takt time, DOWNTIME wastes

Value Stream Mapping, Takt Time and the Eight Wastes

STEPS 1-4 OF 9

1

Review the standard VSM icons: process box, data box, inventory triangle, push arrow, supermarket, kanban and the kaizen burst.

2

Open lab08-vsm-timeline-demand.xlsx. Draw the process boxes left to right and add each one's data box: cycle time, changeover, uptime and operators.

3

Add the information flow across the top — how does each step know what to work on next?

4

Add inventory triangles between steps with the observed queue quantity, and convert each to days of supply.

Value Stream Mapping, Takt Time and the Eight Wastes

STEPS 5-8 OF 9

5

Draw the timeline ladder along the bottom: value-added time on the lower rungs, waiting time on the upper rungs.

6

Sum both rows and calculate process cycle efficiency = value-added time / total lead time using the workbook's Waiting Time column. Typical service processes run under 10%.

7

Calculate takt time = available working time / customer demand (1,350 min/day and 140 orders/day, given in the workbook notes), then compare each step's cycle time against takt.

8

Run a waste walk and tag every observation against the eight wastes (DOWNTIME): Defects, Overproduction, Waiting, Non-utilised talent, Transport, Inventory, Motion, Extra-processing.

Value Stream Mapping, Takt Time and the Eight Wastes

STEPS 9-9 OF 9

9

Mark kaizen bursts on the VSM where waste is concentrated — these become improvement candidates in the Improve phase.

Value Stream Mapping, Takt Time and the Eight Wastes

Check your work

Your VSM shows both material and information flow, your timeline ladder yields a process cycle efficiency percentage, and every step's cycle time is compared against takt.

Data Types, Operational Definitions and the Data Collection Plan

LAB 9

The data type determines which statistical tool is legal later — continuous data unlocks far more powerful analysis than discrete. Classify your data, write operational definitions that two people would apply identically, and plan the collection before touching the process.

You'll build

A data type classification, operational definitions and a complete data collection plan.

Tools & techniques: Continuous vs discrete, nominal vs ordinal, operational definitions, check sheets, data collection plan

Data Types, Operational Definitions and the Data Collection Plan

STEPS 1-4 OF 8

1

Classify data as qualitative or quantitative, then quantitative as discrete (countable) or continuous (measurable on a scale).

2

Classify qualitative data as nominal (no order — carrier name, defect type) or ordinal (ordered — priority, satisfaction rating).

3

Open lab09-data-collection-fields.xlsx and classify every available field as Continuous, Discrete, Nominal or Ordinal in the blank Data Type column.

4

Write the operational definition for each measure: what exactly is counted, when the clock starts and stops, which system field, and what is excluded.

Data Types, Operational Definitions and the Data Collection Plan

STEPS 5-8 OF 8

5

Test each definition by having two people apply it to the same five records — if they disagree, the definition is not yet operational.

6

Build the data collection plan table: measure, data type, operational definition, source, who collects, how often, sample size, and how it is recorded.

7

Design the check sheet or data capture form so it takes under 30 seconds to complete — complex forms do not get filled in.

8

Add a stratification plan: capture shift, product family, carrier and picker at the same time so the Analyze phase can slice the data.

Data Types, Operational Definitions and the Data Collection Plan

Check your work

Two people applying your operational definitions to the same records produce identical values, and your plan captures stratification factors alongside the main measure.

Sampling Techniques and Sample Size Calculation

LAB 10

Measuring the whole population is rarely affordable, and a biased sample invalidates every conclusion that follows. Choose the right sampling technique, then calculate how much data is actually needed using the Green Belt sample size formula rather than guessing.

You'll build

A justified sampling plan and calculated sample sizes for continuous and discrete data.

Tools & techniques: Simple random, stratified, systematic and cluster sampling, sample size formulas

Sampling Techniques and Sample Size Calculation

STEPS 1-4 OF 9

1

Review the four sampling techniques: simple random, stratified (sample within subgroups), systematic (every Nth), and cluster.

2

Choose stratified sampling if your process has distinct subgroups — for Northwind, sample within each shift and each carrier so no group is missed.

3

Note which techniques are non-random (convenience, judgment) and why they must not be used when the data will feed statistical analysis.

4

Open lab10-sampling-frame.xlsx for the population by shift and carrier, then apply $n = (1.96s / d)^2$ using the preliminary standard deviation given in the notes.

Sampling Techniques and Sample Size Calculation

STEPS 5-8 OF 9

5

Worked example: you want to estimate average order cycle time within 5 hours ($d = 5$) and a preliminary estimate of the standard deviation is 10 hours ($s = 10$). Calculate n .

$n = (1.96 \times 10 / 5)^2 = (3.92)^2 = 15.4$, round up to 16 observations

6

For discrete/proportion data, calculate the sample size using the proportion defective and the margin of error given in the deck's formula.

7

Worked example: the proportion defective is 20% and the margin of error is 0.0784. Calculate the required sample size.

8

Compare the calculated n against what is practically collectable. If n is unaffordable, either widen the margin of error d or reduce the confidence level — and record that trade-off.

Sampling Techniques and Sample Size Calculation

STEPS 9-9 OF 9

9

Write the final sampling plan: technique, sample size, sampling interval, period covered and who collects.

Sampling Techniques and Sample Size Calculation

Check your work

Your sample size is calculated from the formula rather than assumed, and your sampling technique is random or stratified — never convenience.

Measurement System Analysis and Gage R&R

LAB 11

This is a defining Green Belt skill. If the measurement system itself varies, you will chase phantom process problems. MSA separates total variation into real process variation and measurement variation, then tests repeatability (same appraiser) and reproducibility (different appraisers).

You'll build

An attribute Gage R&R study with repeatability, reproducibility and accuracy percentages.

Tools & techniques: MSA, components of variation, accuracy, repeatability, reproducibility, resolution, Gage R&R acceptance

Measurement System Analysis and Gage R&R

STEPS 1-4 OF 10

1

Draw the components of variation tree: total observed variation = actual process variation + measurement system variation.

2

Split measurement variation into its parts — repeatability (one appraiser, repeated measures) and reproducibility (between appraisers).

3

Check resolution first using the ten-bucket rule: the measurement device must resolve to about one tenth of the tolerance you need to detect.

4

Set up an attribute Gage R&R: take at least 20 sample records, label them opaquely so appraisers cannot recognise them, and record the known correct attribute for each.

Measurement System Analysis and Gage R&R

STEPS 5-8 OF 10

5

Open lab11-gage-rr-attribute.xlsx — 30 order records already classified On time/Late by three appraisers, twice each, against the known correct attribute.

6

Repeat the exercise with the sample order randomised so appraisers cannot recall their first answer.

7

Calculate repeatability per appraiser from the workbook: the percentage of the 30 samples where that appraiser agreed with THEMSELVES across both trials.

8

Calculate reproducibility: the percentage of samples where all appraisers agreed with each other.

Measurement System Analysis and Gage R&R

STEPS 9-10 OF 10

9

Calculate accuracy: the percentage where each appraiser matched the known correct attribute.

10

Apply the acceptance criteria — a system agreeing only around 50% of the time is not fit for use. Record what must be fixed: the operational definition, training or the device.

Measurement System Analysis and Gage R&R

Check your work

You can state your repeatability, reproducibility and accuracy percentages and give a clear go/no-go verdict on whether the measurement system can be trusted.

Yield, DPU, DPO, DPMO, RTY and the Hidden Factory

LAB 12

Classic yield counts what came out good at the end and hides all the rework that got it there — the hidden factory. Rolled throughput yield multiplies the first pass yield of every step and exposes the true cost of a multi-step process.

You'll build

A metrics worksheet with yield, FPY, RTY, DPU, DPO and DPMO calculated for the process.

Tools & techniques: Classic yield, first pass yield, rolled throughput yield, DPU, DPO, DPMO, DUDO analysis

Yield, DPU, DPO, DPMO, RTY and the Hidden Factory

STEPS 1-4 OF 10

1

Run the DUDO analysis first: define the Defect, the Unit, the Defect Opportunities per unit and the Observed defects — every metric depends on these four definitions.

2

Open lab12-process-step-yields.xlsx. Calculate classic yield: units passing final inspection / units started, as a percentage.

3

Calculate first pass yield for each process step: units passing that step first time without rework / units entering that step.

4

Calculate rolled throughput yield by multiplying the FPY of every step together — $RTY = FPY1 \times FPY2 \times \dots \times FPYn$.

Yield, DPU, DPO, DPMO, RTY and the Hidden Factory

STEPS 5-8 OF 10

5

Compare RTY against classic yield. The gap between them is the hidden factory: the rework you were paying for but not measuring.

6

Calculate $DPU = \text{total defects found} / \text{total units inspected}$.

7

Calculate $DPO = \text{defects} / (\text{units} \times \text{opportunities per unit})$.

8

Calculate $DPMO = DPO \times 1,000,000$, then convert to a sigma level using the conversion table.

Yield, DPU, DPO, DPMO, RTY and the Hidden Factory

STEPS 9-10 OF 10

9

For a five-step process where each step runs at 95% FPY, calculate RTY and note how a 'good' 95% per step collapses to roughly 77% end to end.

$$\text{RTY} = 0.95^5 = 0.7738, \text{ or } 77.4\%$$

10

Record the baseline figures — these are what the Improve phase must beat.

Yield, DPU, DPO, DPMO, RTY and the Hidden Factory

Check your work

Your RTY is lower than your classic yield, you can explain the hidden factory gap between them, and your DPMO converts to a stated sigma level.

Descriptive Statistics, Normality and Baseline Process Capability

LAB 13

Close the Measure phase by describing the data — central tendency, dispersion and shape — testing whether it is normal, and converting it into capability indices. Cp asks whether the process COULD fit inside the specification; Cpk asks whether it actually does, given where it is centred.

You'll build

A descriptive statistics summary, a normality assessment and calculated Cp and Cpk values.

Tools & techniques: Mean, median, range, standard deviation, histogram, normal distribution, Cp, Cpk, sigma level

Descriptive Statistics, Normality and Baseline Process Capability

STEPS 1-4 OF 12

1

Open lab13-order-lead-times.xlsx — all 4,200 baseline orders. Calculate mean, median and mode of the Lead Time column, and note which resists outliers.

2

Calculate the measures of dispersion: range and standard deviation. Variation, not the average, is what the customer feels.

3

Build a histogram of the baseline data and choose the bin count carefully — too few bins show nothing, too many look like a comb.

4

Read the histogram shape: bell-shaped, skewed, or bi-modal. A bi-modal shape usually means you are measuring two processes as if they were one — stratify and re-plot.

Descriptive Statistics, Normality and Baseline Process Capability

STEPS 5-8 OF 12

5

Recall the empirical rule for a normal curve: 68.26% of data within +/-1 standard deviation, 95.46% within +/-2, and 99.73% within +/-3.

6

Assess normality — if the data is clearly non-normal, note that hypothesis tests assuming normality will not be valid in Lab 17.

7

Mark USL = 48 hrs and LSL = 0 from your Lab 4 CTQ onto the histogram, then count observations outside — the Status column should confirm exactly 357.

8

Calculate Cp using the formula from the deck: $C_p = (USL - LSL) / 6s$, that is specification width divided by process spread.

Descriptive Statistics, Normality and Baseline Process Capability

STEPS 9-12 OF 12

9

Worked example: USL = 48 hours, LSL = 0 hours, standard deviation $s = 6$ hours. Calculate C_p .

$$C_p = (48 - 0) / (6 \times 6) = 48 / 36 = 1.33$$

10

Calculate C_{pk} , which also accounts for how far off-centre the process sits, and compare against the C_p value.

11

Interpret the result: $C_p \geq 1$ means potentially capable, $C_{pk} \geq 1.33$ is the usual minimum for customer satisfaction, and many organisations target 2.0.

12

If C_p is acceptable but C_{pk} is poor, record the conclusion: the process spread is fine but the process is off-centre — a very different fix.

Descriptive Statistics, Normality and Baseline Process Capability

Check your work

You can state your baseline C_p and C_{pk} , explain the difference between them, and say whether the problem is spread, centring or both.

Recap — Measure

- Process mapping — Flowcharts and swimlanes expose the handoffs, delays and rework loops where defects are born.
- Value stream mapping — Map material AND information flow, then compare value-added time against total lead time.
- Lean metrics — WIP, lead time, cycle time, throughput and takt time — $\text{takt} = \text{available time} / \text{customer demand}$.
- Types of data — Continuous vs discrete; nominal vs ordinal. The data type dictates which statistical tool is legal.
- Sampling — Simple random, stratified, systematic and cluster — a biased sample invalidates every later conclusion.
- Sample size — $n = (1.96s/d)^2$ for continuous data — how much data is enough, decided before you collect it.

DMAIC · ANALYZE

Analyze — Find and Prove the Root Cause

Variation · Pareto · Run charts · Fishbone · 5 Whys · Multi-voting · Hypothesis testing · p-values · Correlation
· Regression

Key Concepts — Analyze (1)

1

Common vs special cause

Common cause is built into the process; special cause is an external, assignable signal.

2

Pareto analysis

The 80/20 rule separates the vital few causes from the trivial many — attack the vital few.

3

Run charts

Plot the metric over time to reveal trend, shift, cluster, mixture, oscillation and bias.

4

Fishbone (Ishikawa)

Organise candidate causes by 5M+E — Manpower, Method, Machine, Material, Measurement, Environment.

5

5 Whys

Drill from the symptom to an actionable cause — stop when the answer is a process, not a person.

6

Descriptive statistics

Central tendency (mean, median) and dispersion (range, standard deviation) describe the data.

7

Hypothesis testing

State H_0 and H_a , choose the right test, then let the p-value decide — not the loudest voice.

8

p-value & alpha

If $p < \alpha$, reject H_0 . At 95% confidence $\alpha = 0.05$. Type I risks the producer, Type II the consumer.

Key Concepts — Analyze (2)

1

Correlation & regression

Quantify how strongly an X drives Y — and remember correlation never proves causation.

ANALYZE

Analyze finds the proven root cause

Measure told us the process runs at 8.5 percent late. Analyze answers the next question - WHY - and proves the answer with data, not opinion.

What the Analyze phase delivers

1

A short list of validated causes

Not 40 brainstormed guesses - the two or three the data actually supports

2

A stratified problem

Where the defects concentrate: which shift, which carrier, which product family

3

Statistical evidence

Hypothesis tests that separate a real signal from ordinary noise

4

Quantified X to Y relationships

Correlation and regression showing how much of Y each X explains

5

A rejected-causes log

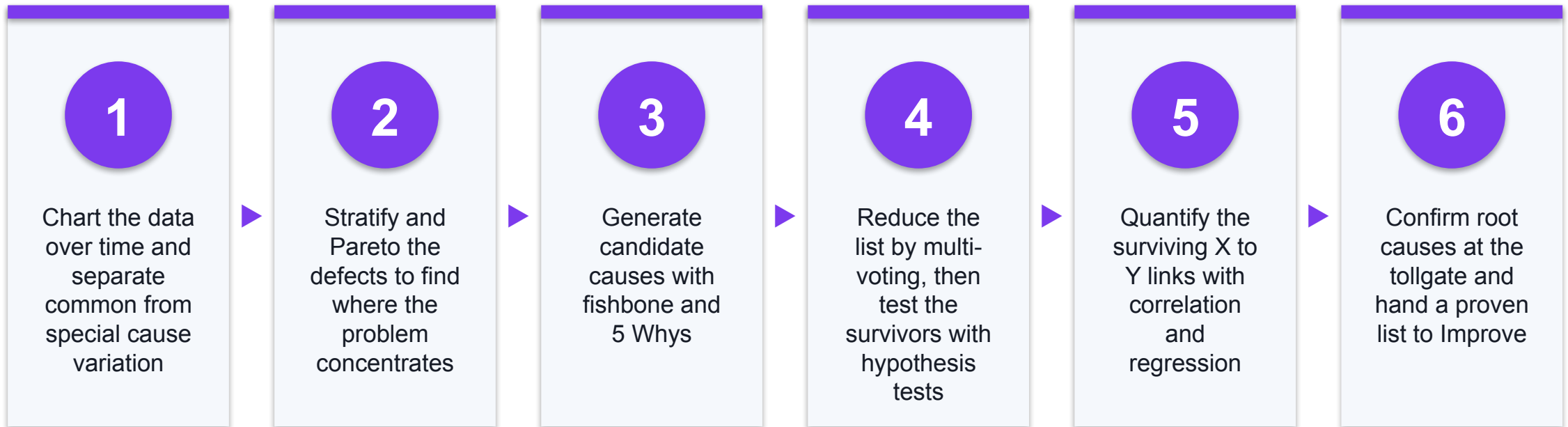
Causes tested and cleared - so nobody re-litigates them in Improve

6

A sponsor-ready story

Statistics translated into money, risk and customer impact

Steps of the Analyze phase



Common cause vs special cause variation

COMMON CAUSE

Inherent, random, always present in the system

- Built into the process design itself
- Many small sources acting together
- Predictable within a range - stable
- Fix requires a PROCESS CHANGE by management
- Northwind: normal 6 to 11 percent weekly late rate

SPECIAL CAUSE

Assignable to a specific identifiable event

- Something changed - not part of the system
- One traceable source you can name
- Unpredictable - shows as a signal on the chart
- Fix is LOCAL: find it, remove it, prevent recurrence
- Northwind: 23 percent late the week the WMS server failed

Why the distinction decides your whole strategy

1

Wrong diagnosis, wrong owner

Special cause is a local fix. Common cause needs management to change the system.

2

Blame follows the error

Treating common cause as special makes you hunt for a guilty operator who does not exist.

3

Roughly 85 percent is systemic

Most defects come from the process, not the person running it.

4

It sets the tool

Special cause: investigate the event. Common cause: redesign with DMAIC.

5

Stability comes first

You cannot judge capability or test causes on an unstable process.

6

It stops tampering

Knowing the variation is common cause is what gives you licence to leave it alone.

Tampering - what it looks like at Northwind

THE TRIGGER

A single ordinary point moves

- Tuesday late rate rises from 8 to 11 percent
- Both values sit inside normal common cause range
- Nothing actually changed in the process
- But the number is visible on a daily report

THE REACTION

The supervisor adjusts

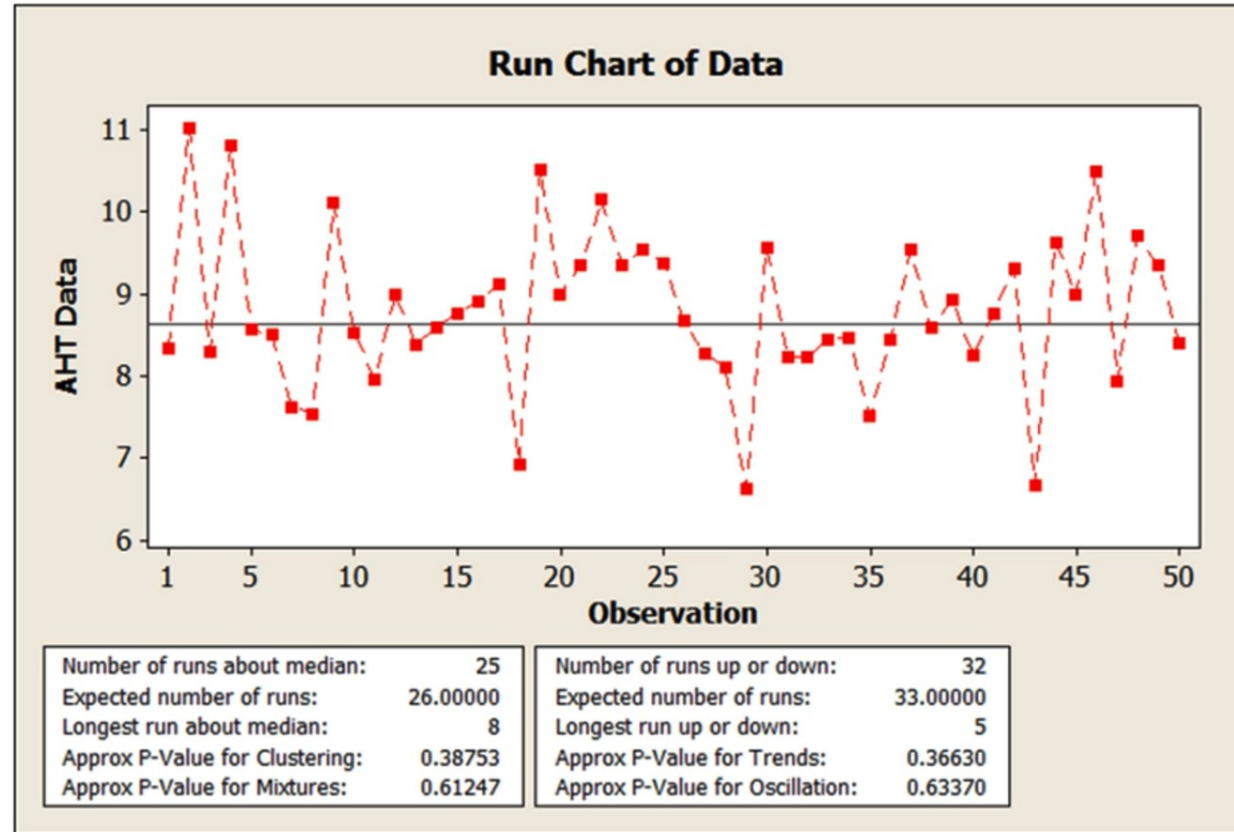
- Two pickers moved from zone C to zone A
- Carrier cut-off pulled forward by 30 minutes
- The adjustment is made with good intentions
- No data supports the change - only the one point

THE RESULT

Variation roughly doubles

- Zone C now short-staffed - it starts running late
- The next swing is bigger, prompting a bigger correction
- The process oscillates around the target
- Rule: never adjust a stable process on one data point

Anatomy of a run chart



Time on the horizontal axis, the measure on the vertical axis, and a MEDIAN centre line - not a mean, because the median resists outliers.

How to build a run chart correctly

1

Plot in TIME ORDER

Sequence is the whole point - sorting the data destroys the information

2

Use the MEDIAN centre line

The median is not distorted by the extreme points you are hunting

3

At least 15 to 20 points

Fewer points cannot reliably show a trend, shift or oscillation

4

Keep the interval constant

Daily or weekly - do not mix, and do not skip periods

5

Annotate known events

Mark the server outage or the new carrier so signals can be explained

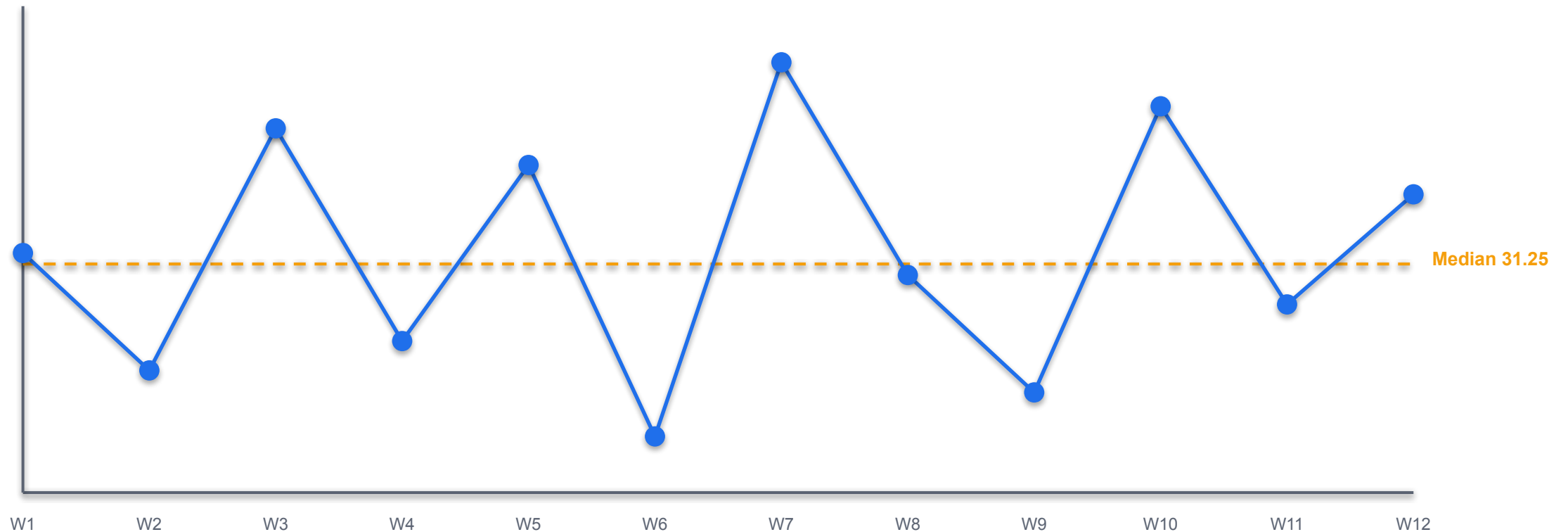
6

Read patterns, not points

One high point is noise. A PATTERN is a signal worth investigating.

ANALYZE

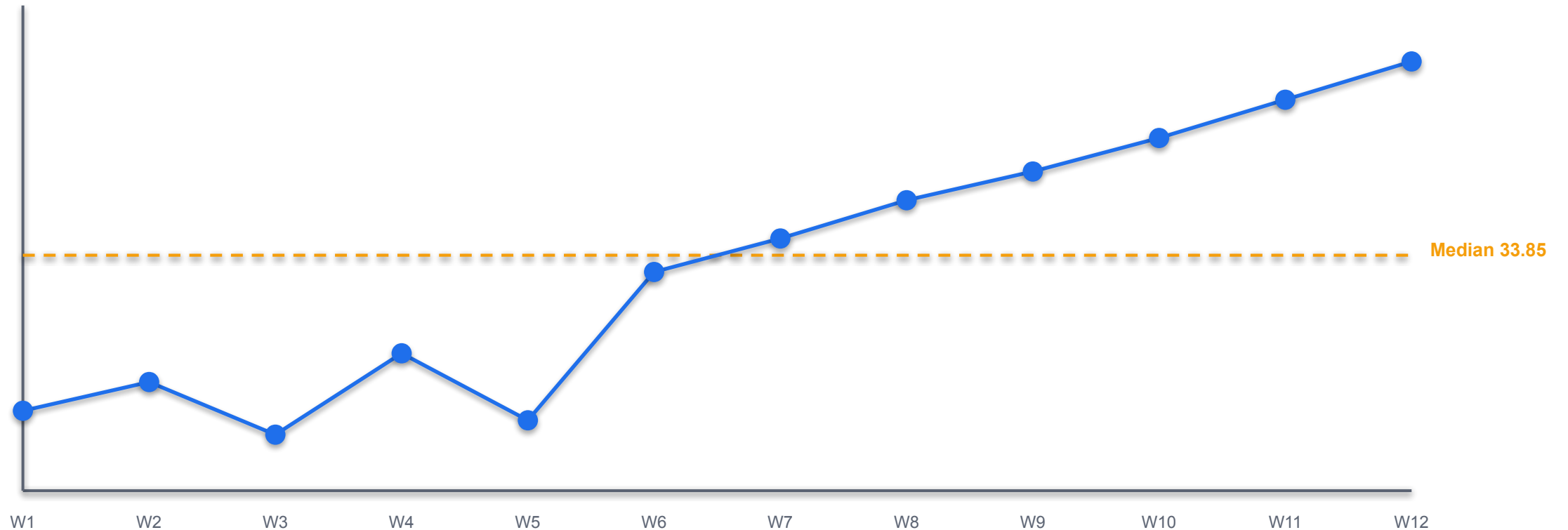
Northwind order cycle time - 12 weeks (hours)



Points scatter randomly either side of the median with no run, trend or oscillation - this is a STABLE process showing only common cause variation.

ANALYZE

Northwind picking cycle time - a signal appears (hours)



Six consecutive rising points from W6 - a TREND. Something changed at W6. Go and find the event; do not average it away.

The six non-random run chart patterns

1

TREND

6 or more consecutive points all increasing, or all decreasing - drift or wear

2

SHIFT

8 or more consecutive points on ONE side of the median - the level has moved

3

CLUSTER

Points bunched together in one area - a batch, a shift or a single operator

4

MIXTURE

Points AVOID the median - usually two processes plotted as one

5

OSCILLATION

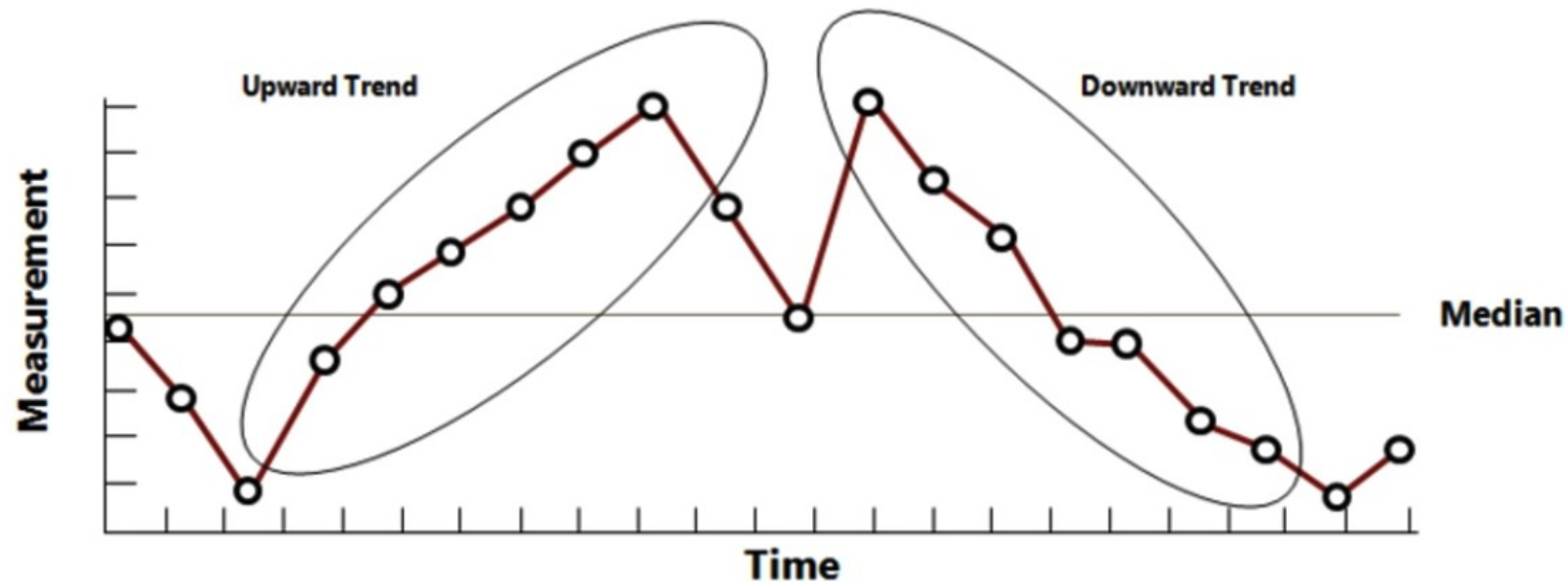
14 or more points alternating up and down - classic over-adjustment

6

BIAS

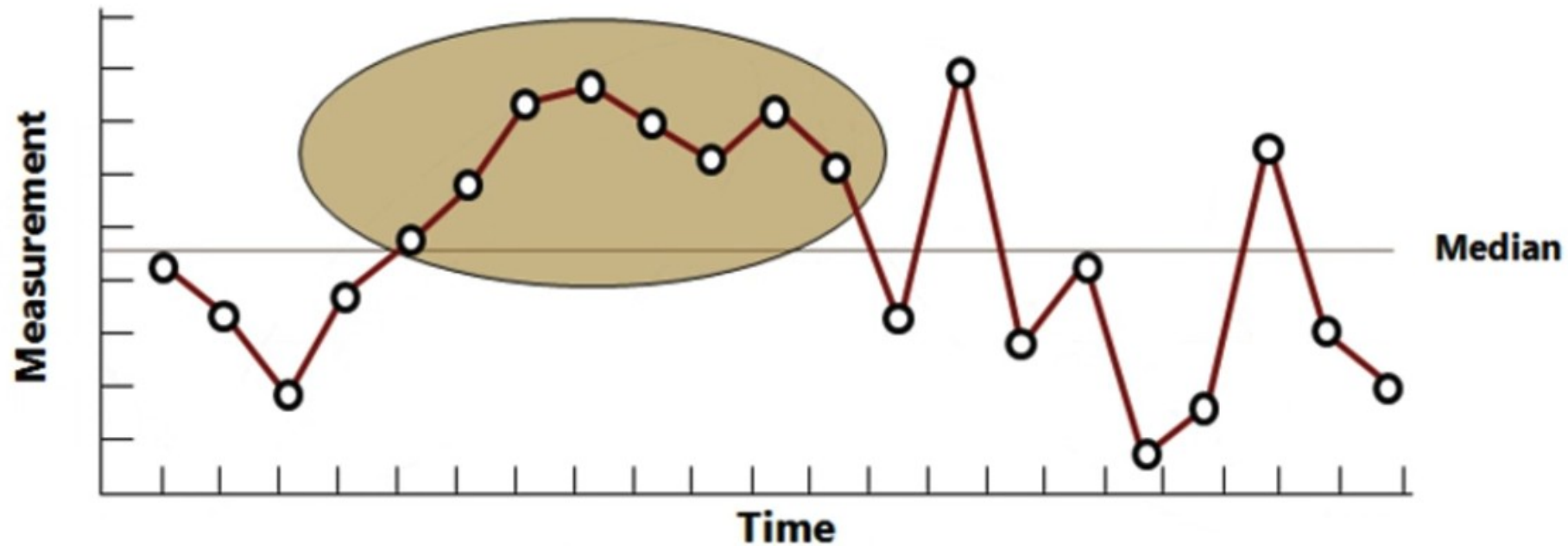
Too few or too many runs across the median versus what chance predicts

Pattern 1 - TREND



Six or more consecutive points moving in the same direction. Tool wear, gradual staffing loss or a slowly filling queue all read like this.

Pattern 2 - SHIFT



Eight or more consecutive points on one side of the median. The process level has moved - a new method, new carrier or new system release.

Using the Pareto principle properly

1

Rank descending

Categories on the horizontal axis, biggest first - always

2

Add the cumulative line

The 80 percent crossing point defines your project scope

3

One category, one bar

Categories must be mutually exclusive or the chart lies

4

Beware the Other bar

If Other is the tallest bar, your categories are wrong

5

Pareto by COST too

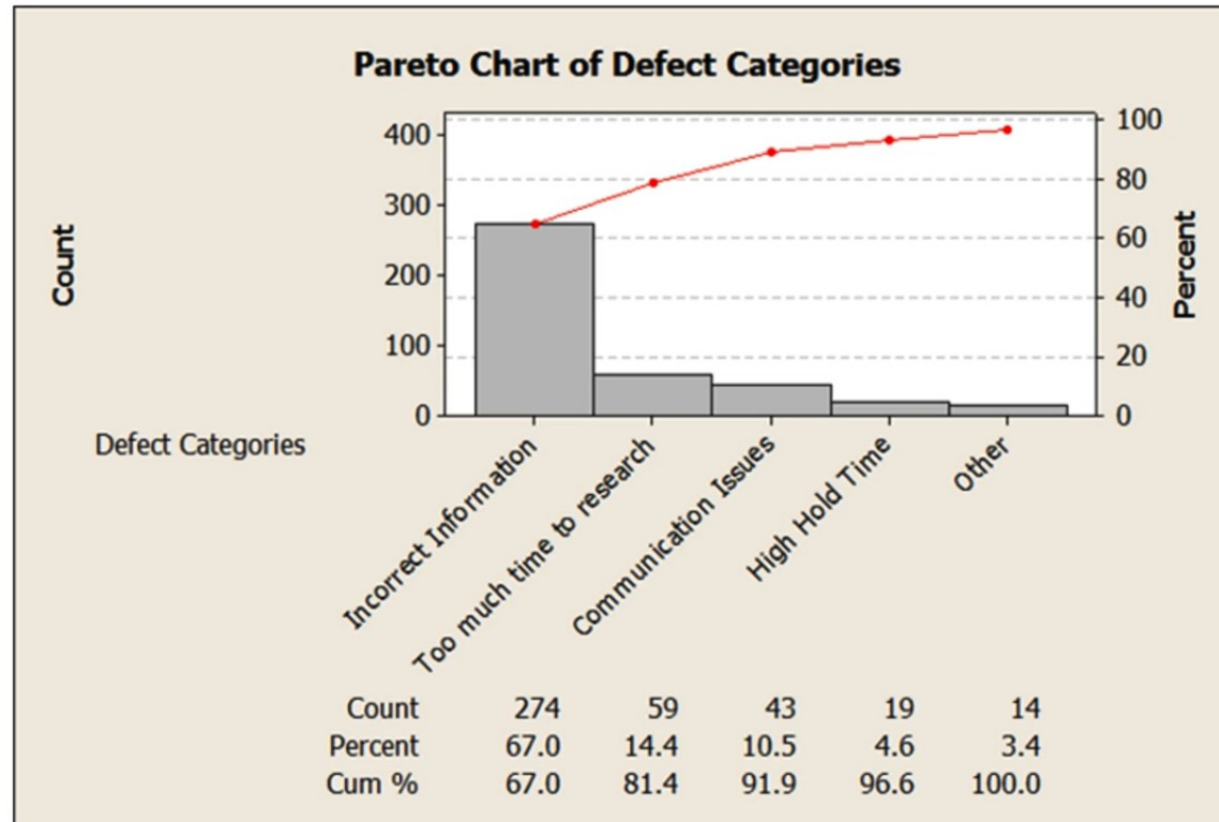
The most frequent defect is not always the most expensive one

6

Re-Pareto after Improve

The vital few change once you have fixed the first one

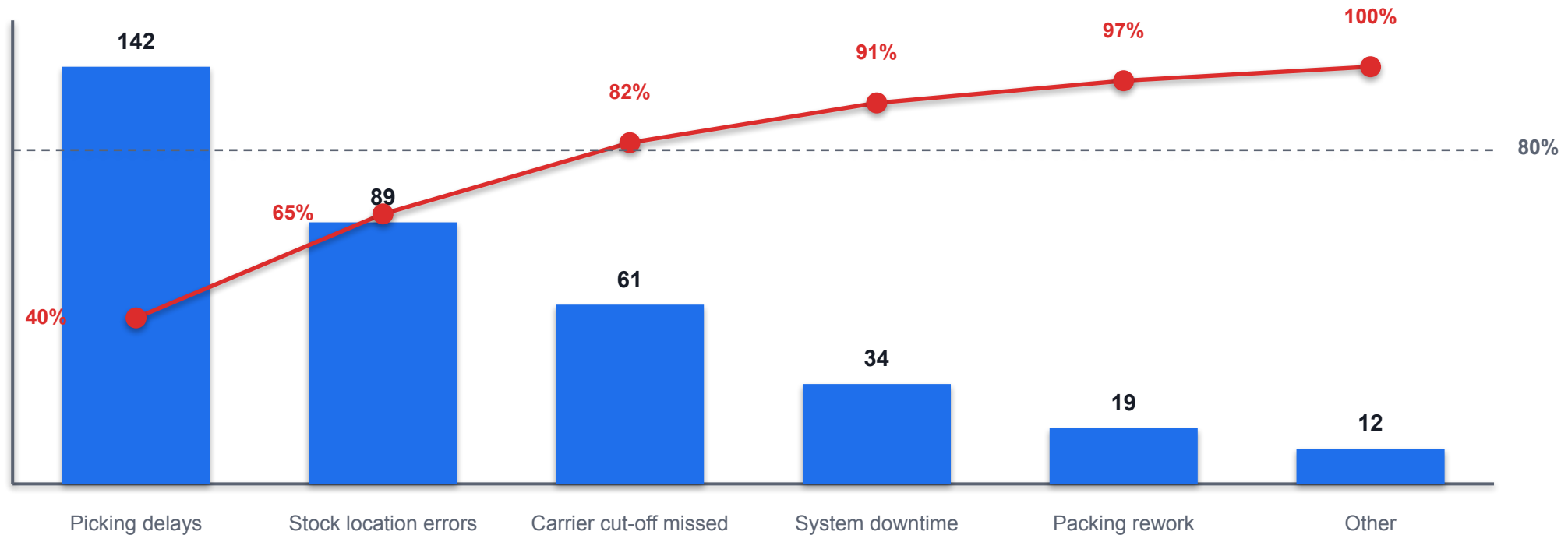
Reading a Pareto chart



Bars descend left to right; the red cumulative line climbs to 100 percent. Where it crosses 80 percent, draw the line around your project.

ANALYZE

Northwind late deliveries by cause - 357 late orders



Picking delays and stock location errors are 231 of 357 late orders - about 65 percent from just two causes. Add carrier cut-off and you pass 80 percent.

Stratifying the Northwind picking delays

BY SHIFT

Night shift is the concentration

- Day shift: 38 of 142 picking delays
- Afternoon shift: 31 of 142
- Night shift: 73 of 142 - over half
- Night runs the same volume with 60 percent of the staff

BY PRODUCT FAMILY

Bulk beverages dominate

- Bulk beverages: 61 of 142 delays
- Chilled goods: 44 of 142
- Ambient dry goods: 37 of 142
- Bulk beverages sit furthest from despatch

BY CARRIER

Essentially flat - not a driver

- Carrier A: 49, Carrier B: 47, Carrier C: 46
- No meaningful concentration by carrier
- So carrier is NOT the picking-delay lever
- A flat stratification is a valid, useful answer

Concentrated versus universal problems

1

Concentrated is good news

Night shift plus bulk beverages is a small, targeted, affordable fix

2

Universal is a system issue

Even spread across every stratum means the process design is the cause

3

Plan strata BEFORE collecting

You cannot stratify by a field you never recorded

4

Common strata

Shift, operator, machine, location, product, day of week, customer

5

Flat is still an answer

Ruling a factor OUT saves the team weeks of wasted effort

6

Stratify then re-Pareto

Every stratum deserves its own Pareto before you test causes

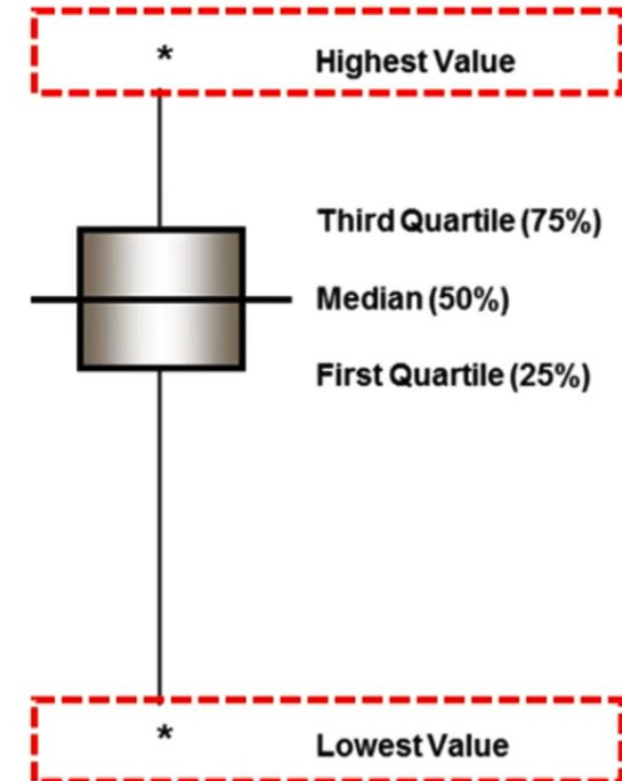
Boxplots - comparing groups visually

Box: The box represents the middle 50% values of the process data

Median: The center line of the box is called the median. It represents the point for which 50% of the data points are above and 50% are below the line.

Q1, Q3: Q1 represents the point for which 75% of the data points are above and 25% are below the line. While Q3 represents the point for which 25% of the data are above and 75% are below the line.

Aestrix: Aestrix represents an outlier and is a point which is more than 1.5 times the inter-quartile range i.e. $Q3 - Q1$ in the data (I hope you remember our discussion on Quartiles in the Introduction Phase)



The box spans the interquartile range with the median inside it; the whiskers reach the last non-outlier points and dots mark outliers.

How to read a boxplot

1

The median line

The 50th percentile - half the data above, half below

2

The box = IQR

Q1 to Q3 - the middle 50 percent of the data sits inside the box

3

The whiskers

Reach to the furthest point within $1.5 \times \text{IQR}$ of the box edge

4

The outlier dots

Points beyond the whiskers - investigate them, never delete them

5

Box position = centring

Boxes at different heights suggest the group means differ

6

Box width = spread

A wide box is an inconsistent group, even if the median looks fine

Boxplot of Northwind picking time by shift

DAY SHIFT

Median 26 min, IQR 22 to 31

- Tight box - consistent performance
- Whiskers 17 to 38 minutes
- No outliers in 34 observations

AFTERNOON SHIFT

Median 28 min, IQR 24 to 34

- Slightly higher and slightly wider
- Whiskers 19 to 44 minutes
- One outlier at 61 minutes

NIGHT SHIFT

Median 41 min, IQR 33 to 52

- Clearly higher AND much wider box
- Whiskers 24 to 74 minutes
- Three outliers above 80 minutes

Brainstorming ground rules

1

NO criticism

Not a word, not a face, not a sigh - judgement kills contribution instantly

2

Quantity over quality

Aim for volume first; the good ideas hide inside the long list

3

Build on ideas

Yes, and - hitch-hiking on someone else's idea is actively encouraged

4

Wild ideas welcome

It is far easier to tame a wild idea than to invigorate a dull one

5

Everyone contributes

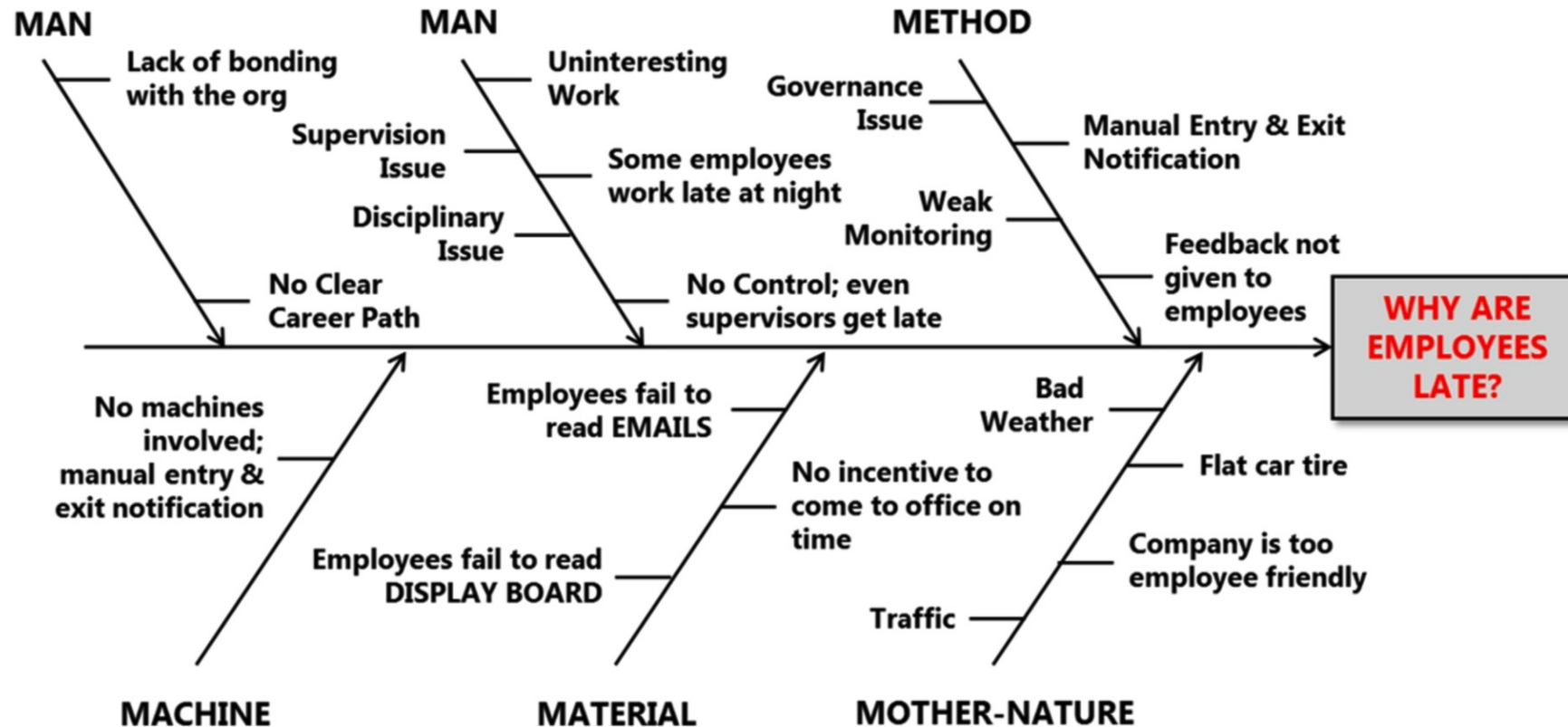
Round-robin or silent written rounds stop one loud voice dominating

6

Include the doers

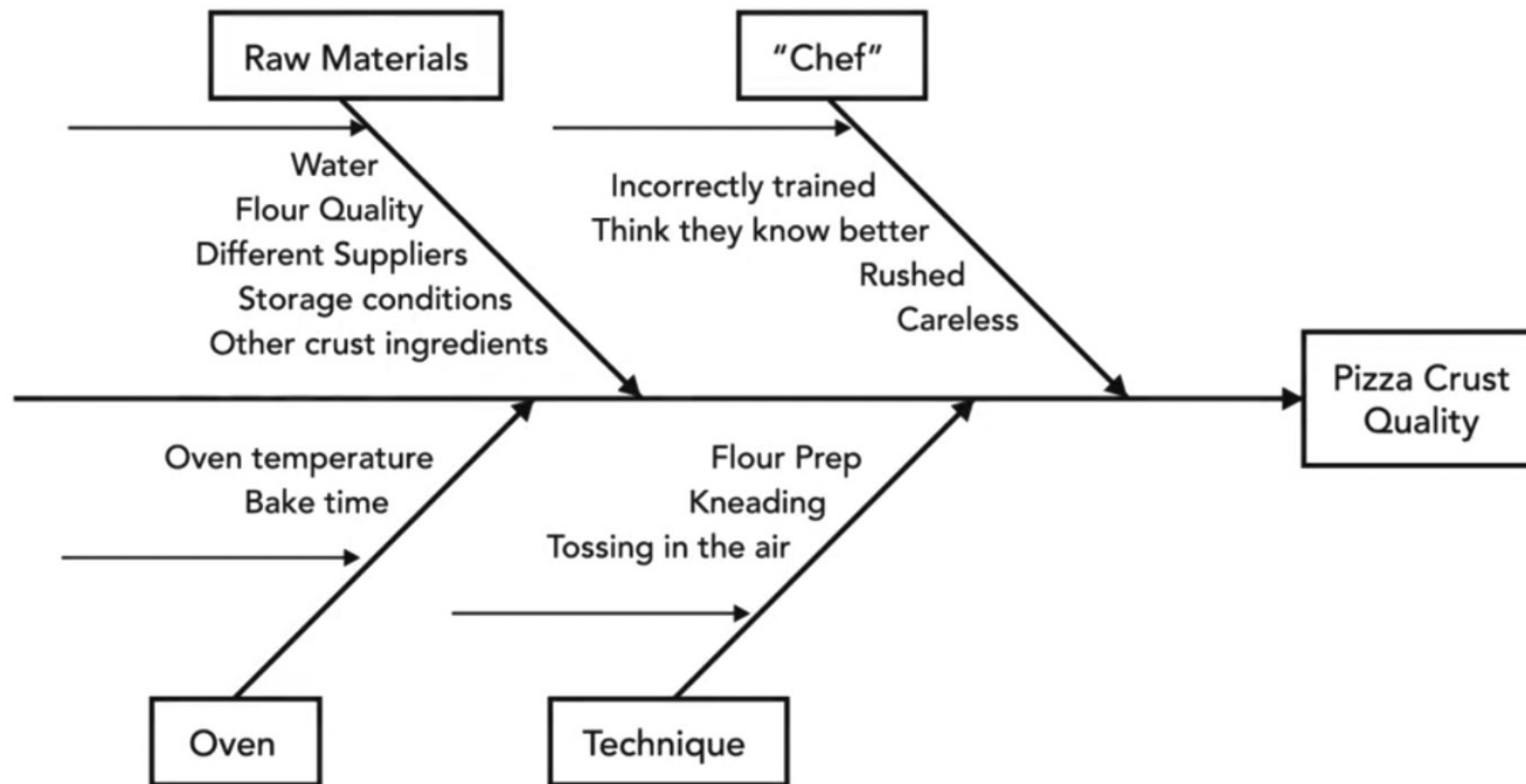
Pickers and drivers know causes no manager will ever think of

The fishbone / Ishikawa skeleton



The effect sits in the head of the fish; each bone is a cause category, and each twig off a bone is a specific potential cause.

Worked fishbone example



A completed fishbone on a familiar everyday effect - note how each bone is drilled two or three levels deep, not just labelled.

The 5M plus E categories

1

MANPOWER

People - skill, training, staffing level, fatigue, motivation, turnover

2

METHOD

The process - SOPs, sequence, batch rules, approvals, handoffs

3

MACHINE

Equipment - scanners, conveyors, the WMS, forklifts, printers

4

MATERIAL

Inputs - packaging, labels, stock condition, supplier quality

5

MEASUREMENT

The data - gages, definitions, reporting timing, system accuracy

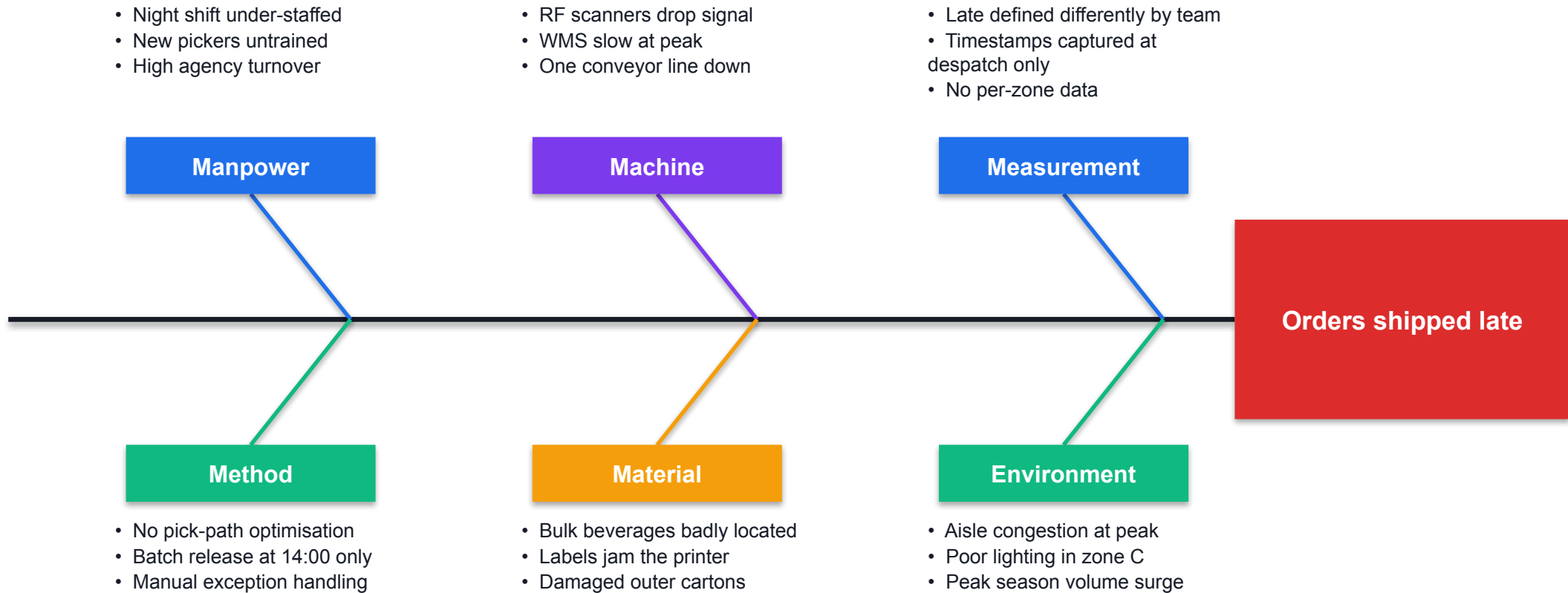
6

ENVIRONMENT

Conditions - layout, temperature, lighting, congestion, season

ANALYZE

Northwind fishbone - orders shipped late



Running a fishbone session that works

1

Write the effect precisely

Orders shipped late beats poor performance - vague effects give vague bones

2

Ask why on every twig

One level of cause is a label. Three levels is an investigation.

3

Do it at the gemba

Run it in the warehouse with the map and the data on the wall

4

Circle the likely few

Dot-vote to mark the causes worth testing with data

5

Causes only, not solutions

Someone will say we need a new WMS - park it, that is Improve

6

It produces HYPOTHESES

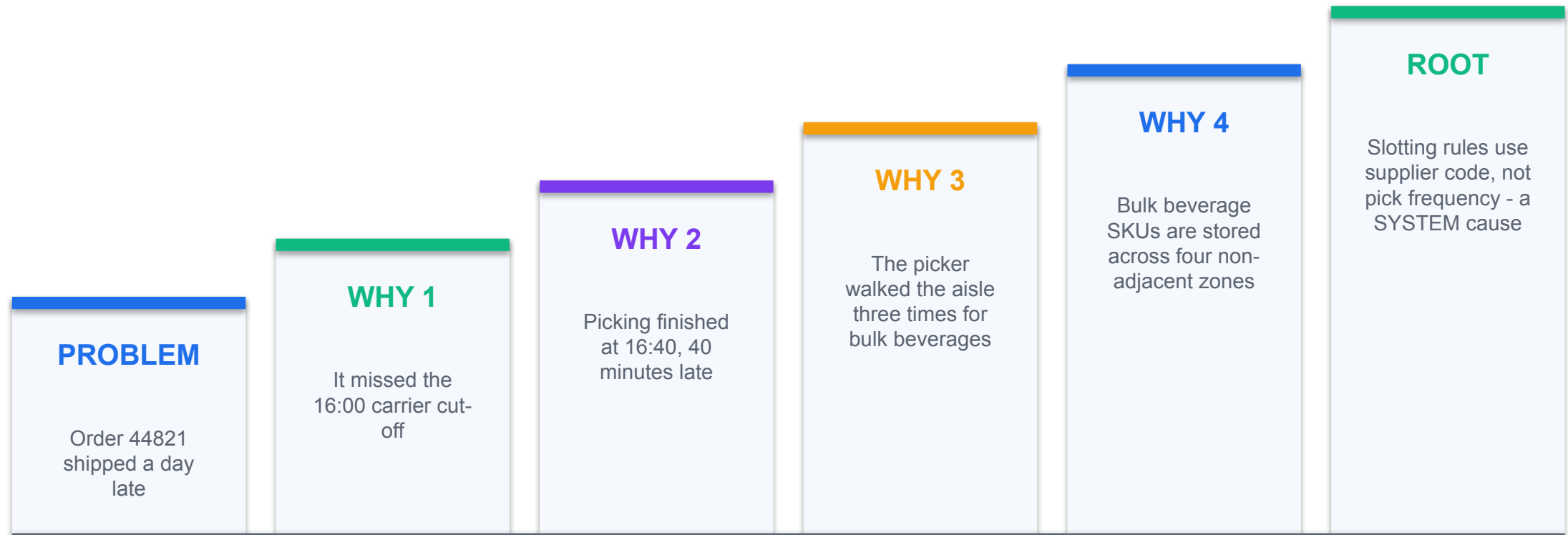
Nothing on the fishbone is a root cause until data confirms it

5 Whys - worked example

1	Why was the request rejected?	The request had missing information
2	Why was information missing?	Relevant information was not provided
3	Why was the information not provided?	There was no clarity on the information required
4	Why was there no clarity?	New requirements were not included in training
5	Why were new requirements not included in Training?	SOP's were not up-to-date

Each answer becomes the next question. The chain stops at the level where a countermeasure can actually be installed.

5 Whys - Northwind late shipment



The root cause is a slotting RULE, not a slow picker. That is fixable, ownable and will not recur once changed.

Rules that make 5 Whys reliable

1

Stop at process or system

If the answer is a person's name, you asked the wrong why

2

Validate backwards

Read the chain up using therefore - if it does not flow, a link is missing

3

Five is a guide

Sometimes three whys is enough; sometimes seven is needed

4

Branch when needed

One why can have two valid answers - follow both chains

5

Each link needs evidence

An unverified link makes every link below it worthless

6

Beware the lazy root

Human error and lack of training are almost never real root causes

Converging - multi-voting and NGT

1

Why converge at all

A fishbone yields 30 to 50 causes; you can only test a handful

2

Each voter gets N/3 votes

30 causes on the wall means 10 votes per participant

3

Vote silently first

Silent voting removes seniority pressure from the room

4

Rank the top vote-getters

Take the top five to eight forward for a second round

5

Nominal group technique

Structured rounds so quiet experts carry equal weight

6

Votes select what to TEST

The vote picks the test order - it never proves a cause

From suspected cause to proven root cause

SUSPECTED

It came out of the fishbone

- Night shift picking is slower
- Based on team experience and observation
- Plausible - and completely untested
- Acting on it now is a gamble

TESTED

A hypothesis test is run

- H_0 : night and day mean picking times are equal
- 34 observations from each shift
- 2-Sample T test, $\alpha = 0.05$
- $p = 0.003$ - reject the null

PROVEN

Evidence plus process logic

- Difference is statistically significant
- Slotting explains WHY it is slower
- Now it earns a place in the Improve plan
- And the rejected causes go into the log

Descriptive vs inferential statistics

DESCRIPTIVE

Summarises the data you actually have

- Mean, median, mode, range, standard deviation
- Histograms, boxplots, Pareto and run charts
- Makes NO claim beyond the data in hand
- Northwind: these 34 orders averaged 31.2 hours

INFERENTIAL

Generalises from the sample to the population

- Hypothesis tests, confidence intervals, regression
- Carries a stated risk of being wrong - alpha
- Requires a properly random, representative sample
- Northwind: ALL orders average 31.2 hours, +/- 1.4

Population and sample - the vocabulary

1

Population

Every item of interest - all 4,200 Northwind orders in the month

2

Sample

The subset you actually measure - 34 randomly selected orders

3

Parameter

A number describing the POPULATION - μ for mean, σ for standard deviation

4

Statistic

A number describing the SAMPLE - $\bar{x}$ for mean, s for standard deviation

5

Sampling distribution

The distribution of the sample mean across many repeated samples

6

Standard error

The spread of that sampling distribution - s divided by the root of n

Sampling - the two numbers that matter

Standard error

$$SE = s / \sqrt{n}$$

s = 6.2 hrs, n = 34: SE = 6.2 / 5.83 = 1.06 hours

95% confidence interval

$$\bar{x} \pm 1.96 \times SE$$

31.2 $\pm$ 1.96 x 1.06 = 29.1 to 33.3 hours

Effect of sample size

n x 4 halves the SE

n = 34 to n = 136 takes SE from 1.06 to 0.53

Precision improves with the SQUARE ROOT of n - four times the data buys only twice the precision.

ANALYZE

Hypothesis testing is THE Green Belt skill

It is the mechanism that converts a difference you can see on a chart into a difference you can defend in a boardroom.

H0 and Ha

H0 - THE NULL

Always an EQUALS statement

- There is NO difference, NO effect, NO relationship
- Example: $\mu(\text{night}) = \mu(\text{day})$
- Example: $p(\text{late}) = 0.05$ target
- It is the default we assume true until data says otherwise

Ha - THE ALTERNATIVE

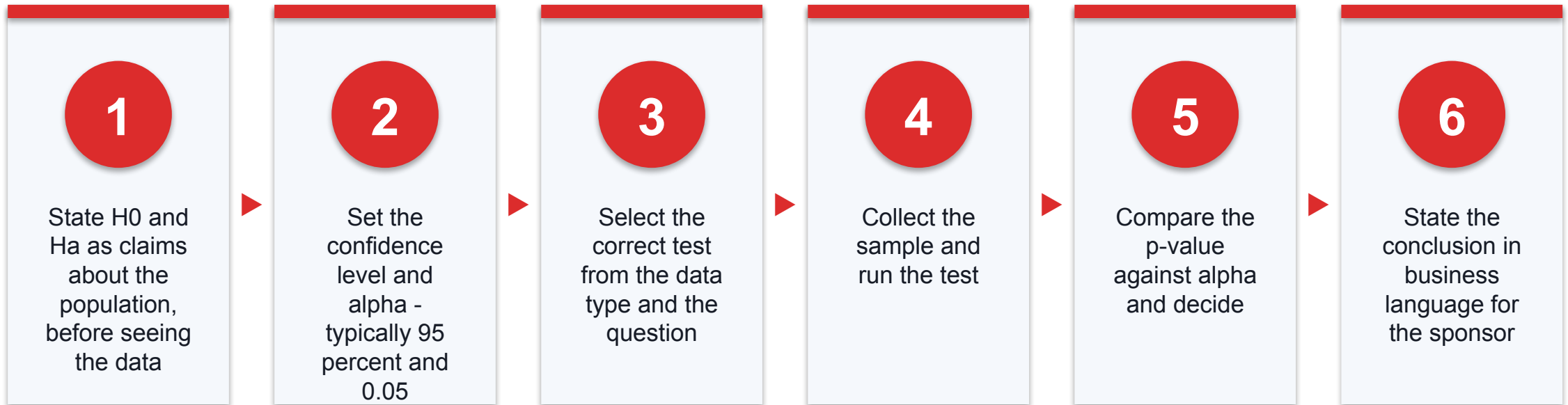
Not equal, greater than, or less than

- What you actually suspect is going on
- Two-tailed: $\mu(\text{night})$ is NOT EQUAL to $\mu(\text{day})$
- One-tailed: $\mu(\text{night})$ is GREATER THAN $\mu(\text{day})$
- Choose the tail BEFORE you look at the data

Writing hypotheses correctly at Northwind

- WRONG: H_0 - the 34 night orders we sampled took the same time as the 34 day orders.
- RIGHT: H_0 - the mean picking time of ALL night orders equals that of ALL day orders.
- The sample comparison needs no test - just subtract the two averages and you are done.
- The population claim is the one that carries risk, and therefore the one that needs a test.
- H_0 always uses equals; the inequality (not equal, greater, less) always lives in H_a .
- State both hypotheses in writing BEFORE collecting or looking at the data.

The hypothesis testing method



Confidence level and alpha

95% confidence

alpha = 0.05

The standard for most business and service processes

99% confidence

alpha = 0.01

Used where a false alarm is costly - safety, regulatory

99.9% confidence

alpha = 0.001

Reserved for critical medical and life-safety decisions

Alpha is the risk you accept of crying wolf - of declaring a difference that is not really there.

Type I and Type II error

The TRUTH about H0	CORRECT decision H0 is TRUE and you fail to reject it. No difference exists and you correctly report none.	TYPE I ERROR - alpha H0 is TRUE but you REJECT it. A false alarm. PRODUCER RISK, measured by alpha. You chase a cause that is not there.
	TYPE II ERROR - beta H0 is FALSE but you ACCEPT it. A missed signal. CONSUMER RISK, measured by beta. A real cause goes undetected.	CORRECT decision H0 is FALSE and you reject it. A real difference exists and you detect it. Probability = POWER = 1 - beta.
The DECISION you make		

Type I, Type II and POWER at Northwind

1

TYPE I - PRODUCER RISK

Reject H_0 when H_0 is TRUE. Alpha = 0.05 is a 5 percent false alarm rate - you re-slot for no gain.

2

TYPE II - CONSUMER RISK

Accept H_0 when H_0 is FALSE. Beta is typically 0.10 to 0.20 - night shift IS slower and you miss it.

3

POWER = 1 - beta

Beta = 0.20 gives power = 0.80, the usual target. Power rises with sample size and effect size.

4

Set n to reach power

Low power is why weak studies find nothing. Size the sample for power, never by habit.

The four test selection questions

1

1. What DATA TYPE is it?

Continuous, or discrete / attribute - this splits the whole map first

2

2. How many LEVELS of X?

One group, two groups, or more than two groups being compared

3

3. Is the data NORMAL?

Normal points to the parametric tests; non-normal to the non-parametric

4

4. What are you COMPARING?

Means, medians, variances or proportions - each has its own test

The hypothesis test selection map

		Output (Dependent Variable) – Y		
		Continuous	Discrete	
Input (Independent Variable) – X's	Continuous	Simple Linear Regression Correlation		Logistic Regression (Out of Scope)
	Discrete	Parametric Tests • 1-Sample t test • 2-Sample t test • Paired t test • One Way ANOVA	Test of Variances • 1-Variance test • 2-Variance test • Test of equal variances	Non-Parametric Tests • 1-Sample Sign test • Mann Whitney test and • Moods Median test
		Chi-Square Test		

Walk the map from the data type downwards. Every Green Belt should be able to reach the right test in under thirty seconds.

Test selection - the discrete and one-sample tests

1

1-PROPORTION

One factor, one level - comparing a RATE against a target. Late rate vs 5 percent.

2

2-PROPORTION

Comparing PROPORTIONS between two samples. Carrier A late rate vs Carrier B.

3

1-SAMPLE T

Comparing a sample MEAN to a TARGET. Mean cycle time vs the 24-hour promise.

4

CHI-SQUARE / 1-VARIANCE

Comparing a VARIANCE to a target or to another sample's variance.

Test selection - comparing groups

1

PAIRED T

SAME subjects measured BEFORE and AFTER a change. Same pickers, old vs new route.

2

2-SAMPLE T

Comparing means from TWO DIFFERENT populations. Night pickers vs day pickers.

3

ANOVA

Comparing means of MORE THAN TWO groups. Day vs afternoon vs night shift.

4

ONE-SAMPLE WILCOXON

Comparing a MEDIAN to a target when the data is NOT normal.

5

MANN-WHITNEY

Comparing MEDIANS between two groups when the data is NOT normal.

6

Choose before you collect

The test dictates the sample structure - decide it in the data plan.

ANALYZE

The classic exam trap: paired or 2-sample?

The SAME children before and after a vaccine is a PAIRED T. Vaccinated versus unvaccinated children is a 2-SAMPLE T. Same subjects means paired.

Paired T vs 2-Sample T - tell them apart

PAIRED T

Same subjects, two measurements each

- Same children before and after the vaccine
- Same pickers on the old route then the new route
- Data comes in natural PAIRS - it tests the differences
- More powerful, because it removes person-to-person variation

2-SAMPLE T

Two different, independent groups

- Vaccinated children vs unvaccinated children
- Night shift pickers vs day shift pickers
- No pairing - the two samples need not even be equal in size
- Tests whether two population MEANS differ

The p-value

CONFIDENCE LEVEL

95%

70%

80%

99%

SIGNIFICANCE LEVEL

5%

30%

20%

1%

The p-value is compared against alpha - and against alpha only. It is never compared against another p-value or against the confidence level.

The p-value decision rule

p LESS THAN alpha

The signal is real

- REJECT the null hypothesis
- ACCEPT the alternative hypothesis
- With $\alpha = 0.05$: $p = 0.02$ - REJECT
- Business reading: the difference is significant

p MORE THAN alpha

Not enough evidence

- FAIL TO REJECT the null hypothesis
- You have NOT proved the null is true
- With $\alpha = 0.05$: $p = 0.13$ - FAIL TO REJECT
- Business reading: no significant difference detected

ANALYZE

Worked example 1 - 1-Proportion test

Setup

$n = 142, x = 38$ failing

Hypothesised proportion 0.20; alternative:
GREATER THAN

Result

$p = 0.031$

0.031 is LESS than $\alpha = 0.05$ - so
REJECT the null

Conclusion

Reject H_0

The failure rate IS significantly higher than 20
percent

One factor, one level, a rate compared against a target - that is the 1-Proportion test.

Worked example 2 - 2-Sample T test

Setup

Two teams, $n = 34$ each

Comparing mean handling time between two independent teams

Result

$p = 0.255$

0.255 is GREATER than $\alpha = 0.05$ - so
FAIL TO REJECT

Conclusion

Fail to reject H_0

13.76 vs 12.82 minutes: NOT significantly
different

The means differ by 0.94 minutes on the sample - but not by enough to claim the populations differ.

Failing to reject does NOT prove H0

1

Check the power

Under 80 percent power and the test was unlikely to find anything

2

Check the sample size

$n = 10$ will miss differences that $n = 100$ would find easily

3

Look at the effect size

A large practical gap with a large p means underpowered, not equal

4

Check the measurement system

Gage variation inflates the noise and buries the signal

5

Report it honestly

Say no significant difference detected, never no difference exists

6

Consider re-testing

Larger n , or a paired design, may reveal what the first test missed

Translating statistics into sponsor language

WHAT YOU RAN

The statistical statement

- 2-Sample T, $\alpha = 0.05$, $n = 34$ per shift
- Night mean 41.3 min vs day mean 26.4 min
- $p = 0.003$, well below α
- Reject H_0 - the population means differ

WHAT YOU SAY

The business statement

- Night shift takes about 15 minutes longer per order
- That gap drives roughly half of all late deliveries
- There is under a 1 percent chance this is a fluke
- Fixing night slotting is worth about S\$180k a year

Read the scatter plot before any number

1

DIRECTION

Positive rises left to right; negative falls. Establish the sign first.

2

SHAPE

Straight or curved - a curve makes the linear R badly misleading

3

STRENGTH

A tight band means a strong relationship; a wide cloud means a weak one

4

OUTLIERS

A single far point can create or destroy an apparent correlation

5

CLUSTERS

Two separate clouds mean two processes - split and re-plot

6

RANGE

A relationship proven over 5 to 25 says nothing at all about 50

Correlation coefficient R

$r = +1$	Perfect +ve Correlation
$r = +0.8$ to $+1$	Strong +ve Correlation
$r = +0.2$ to $+0.8$	Possible +ve Correlation
$r = +0.2$ to -0.2	No Correlation
$r = -0.2$ to -0.8	Possible -ve Correlation
$r = -0.8$ to -1	Strong -ve Correlation
$r = -1$	Perfect -ve Correlation

R runs from -1 (perfect negative) through 0 (no linear relationship) to +1 (perfect positive). The sign is direction; the magnitude is strength.

Interpreting R

1

R = +1

Perfect positive - every point sits exactly on an upward straight line

2

R = -1

Perfect negative - every point sits exactly on a downward straight line

3

R = 0

No LINEAR relationship - though a curved relationship may still exist

4

THRESHOLD: 0.4

Correlation is considered to occur at R of 0.4 or greater, or -0.4 or less

5

Between -0.4 and 0.4

Too weak to treat as a working relationship - do not build a fix on it

6

Sign is not strength

R = -0.85 is far stronger than R = +0.30, negative or not

Coefficient of determination - worked example

Correlation

$$R = 0.860624$$

A strong positive correlation, well above the 0.4 threshold

R-squared

$$r^2 = R \times R = 0.74$$

0.860624 squared is approximately 0.74

Interpretation

74% explained

About 74 percent of the variation in Y is related to X

The remainder

26% unexplained

26 percent comes from other Xs or from measurement noise

A single X explaining 74 percent of Y is a strong lever - but 26 percent still lies elsewhere.

Linear regression - fitting the line

Assume your Regression equation is:

$$Y = 2x + 69000$$

- where Y is the \$ amount generated after selling products
- X is the advertisement cost

X = investment of \$490,000

$$Y = 2*(490,000) + 69000$$

Y = \$1.049 million dollar of sales

R²

> 95% indicates your confidence in this equation is very high

Regression fits the line that minimises the squared vertical distances from the points, giving an equation you can predict with.

Regression - the fitted line and predictions

The model

$$y = mx + c$$

m is the slope - the change in Y per one unit change in X

Fitted line

$$y = 3.3497x + 27.561$$

Slope 3.3497, intercept 27.561, from the regression output

Predict at x = 5

$$y = 3.3497(5) + 27.561$$

$$y = 16.7485 + 27.561 = 44.30$$

Predict at x = 25

$$y = 3.3497(25) + 27.561$$

$$y = 83.7425 + 27.561 = 111.30$$

Every extra unit of X adds 3.3497 to Y; at X = 0 the model predicts Y = 27.561.

Using regression in an improvement project

1

Predict Y from X

Set X and the equation gives the expected Y - a planning tool

2

Solve in reverse

Set the Y you need and solve for X to get the OPERATING WINDOW

3

Northwind example

Need cycle time under 24 hrs, so solve for the maximum pick-path length

4

Never extrapolate

Fitted over $x = 5$ to 25 ? Then $x = 60$ is a guess, not a prediction

5

Check the residuals

Residuals must scatter randomly - a pattern means the model is wrong

6

Confirm with a pilot

The equation sets the target; the pilot proves it in the real process

Does your data qualify for regression?

QUALIFIES

Both variables quantitative

- Pick-path metres vs picking minutes
- Order line count vs cycle time hours
- Staff on shift vs orders despatched
- Both X and Y are continuous or ratio data

DOES NOT QUALIFY

One variable is a category

- Carrier name vs late orders - carrier is a label
- Product family vs defects - family is a label
- Shift name vs picking time - shift is a label
- Use Pareto, or a 2-Sample T / ANOVA instead

ANALYZE

Correlation does not prove causation

But it does not negate it either. Correlation STARTS a causal argument - confirm it with process knowledge, a designed experiment, or a pilot.

Confirming that a correlation is causal

1

Process knowledge

Can an engineer explain the physical mechanism from X to Y?

2

Temporal order

X must change BEFORE Y changes - effects never precede their causes

3

Look for a lurking X

Ice cream and drownings both track summer - the season is the cause

4

Designed experiment

Deliberately set X at levels and observe Y - the strongest evidence

5

Run a pilot

Change X in one zone, hold the others - the Green Belt's practical DOE

6

Dose-response

More X producing proportionately more Y strengthens the causal case

Northwind - the Analyze phase conclusion

- Stratification: 73 of 142 picking delays sit on night shift, concentrated in bulk beverages.
- 5 Whys: bulk beverage SKUs are slotted by supplier code rather than by pick frequency.
- 2-Sample T: night 41.3 min vs day 26.4 min, $p = 0.003$ - a real and significant difference.
- Regression: pick-path length explains about 74 percent of the variation in picking time.
- Carrier was tested and CLEARED - a flat stratification, so it is logged as a rejected cause.
- Proven root cause: the slotting rule. That is what Improve will change and Control will hold.

Common Analyze phase mistakes

1

Skipping straight to solutions

The team names a fix in week one and spends Analyze justifying it

2

Testing without stratifying

Pooled data hides the very concentration you are looking for

3

Wrong test chosen

2-Sample T on paired data - a correct calculation of the wrong thing

4

Treating p as proof of size

A tiny p on a trivial difference is significant but worthless

5

Correlation read as cause

The most expensive single error in the whole Analyze phase

6

Stopping 5 Whys at a person

Human error is a symptom - keep drilling to the system

Analyze phase - checkpoint

- Distinguish common cause from special cause, and give the correct response to each.
- State the numeric rule for a TREND and for a SHIFT on a run chart.
- Same children before and after a vaccine - which test, and why not the other one?
- Alpha is 0.05 and $p = 0.13$. What is your decision, and what have you NOT proved?
- Define Type I and Type II error, name each risk, and state what power equals.
- $R = 0.860624$. What is r-squared, and what exactly does that number mean?

Hands-On Labs — Analyze

1

Lab 14 — Variation, Run Charts and Stability Analysis

A run chart of the baseline data with the six non-random patterns asse

2

Lab 15 — Pareto Analysis, Stratification and Boxplots

A Pareto chart with cumulative line, stratified Pareto charts and comp

3

Lab 16 — Fishbone, 5 Whys, Multi-Voting and Cause Prior

A Fishbone diagram, three 5 Whys chains and a multi-voted shortlist of

4

Lab 17 — Hypothesis Testing — Test Selection, p-values

Stated hypotheses, a justified test selection, and a documented statis

5

Lab 18 — Correlation, Regression and Quantifying the X-

A scatter plot, a correlation coefficient, a regression equation and a

Variation, Run Charts and Stability Analysis

LAB 14

Before hunting root causes, establish whether the process is stable. Common cause variation is built into the process and requires a process change; special cause variation is an assignable external event. Confusing the two leads to tampering — reacting to noise and making the process worse.

You'll build

A run chart of the baseline data with the six non-random patterns assessed.

Tools & techniques: Common vs special cause, run chart, median line, trend, shift, cluster, mixture, oscillation, bias

Variation, Run Charts and Stability Analysis

STEPS 1-4 OF 11

1

Open lab14-daily-mean-lead-time.xlsx and plot the 30 daily means in time order — time order matters, so never sort this data.

2

Draw the median line through the plotted points.

3

Test for a TREND: six or more consecutive points steadily increasing or decreasing.

4

Test for a SHIFT: eight or more consecutive points on the same side of the median, indicating the process level changed.

Variation, Run Charts and Stability Analysis

STEPS 5-8 OF 11

5

Test for CLUSTERING: groups of points bunched together, suggesting a batch or intermittent effect.

6

Test for MIXTURES: points avoiding the median line, which usually means two different processes are being plotted as one.

7

Test for OSCILLATION: fourteen or more points alternating up and down, often over-adjustment by operators.

8

Test for BIAS: too few or too many runs above and below the median compared with what randomness would produce.

Variation, Run Charts and Stability Analysis

STEPS 9-11 OF 11

9

Classify your process: stable with common cause variation only, or unstable with special causes present.

10

If special causes are present, investigate and record what changed at that point in time before proceeding — an unstable process cannot be meaningfully capable.

11

Explain the tampering trap: adjusting a stable process in response to common cause variation always increases variation.

Variation, Run Charts and Stability Analysis

Check your work

You can state whether your process is stable, name every non-random pattern you tested for, and explain what tampering is and why it makes things worse.

Pareto Analysis, Stratification and Boxplots

LAB 15

The 80/20 rule states that roughly 80% of the effect comes from 20% of the causes. Build a Pareto chart to find the vital few, then stratify the data by shift, carrier and product family to test whether the problem is universal or concentrated — a concentrated problem is far easier to fix.

You'll build

A Pareto chart with cumulative line, stratified Pareto charts and comparative boxplots.

Tools & techniques: Pareto principle, Pareto chart, cumulative percentage, stratification, boxplot, quartiles, outliers

Pareto Analysis, Stratification and Boxplots

STEPS 1-4 OF 10

1

Open lab15-late-causes-pareto.xlsx — all 357 late orders already assigned to a root-cause category with their cost impact.

2

Count the frequency of each category and sort the categories in descending order of frequency.

3

Calculate the cumulative percentage across the sorted categories.

4

Draw the Pareto chart: descending bars on the left axis, cumulative percentage line on the right axis.

Pareto Analysis, Stratification and Boxplots

STEPS 5-8 OF 10

5

Read where the cumulative line crosses 80% — the categories to the left of that point are your vital few.

6

Open lab15-late-by-stratum.xlsx and rebuild the Pareto separately for each shift and each carrier.

7

Compare the stratified charts. If one shift or carrier dominates, the problem is concentrated and your project scope should narrow to it.

8

Build a boxplot of cycle time by stratification factor — read the median, the interquartile box, the whiskers and any outliers.

Pareto Analysis, Stratification and Boxplots

STEPS 9-10 OF 10

9

Use the boxplots to compare groups visually: if the boxes barely overlap, the groups are probably genuinely different — a hypothesis you will test formally in Lab 17.

10

Write your prioritisation conclusion: which categories you will pursue and which you are explicitly deferring.

Pareto Analysis, Stratification and Boxplots

Check your work

Your Pareto identifies the vital few crossing 80% cumulative, and your stratified charts show whether the problem is universal or concentrated in a subgroup.

Fishbone, 5 Whys, Multi-Voting and Cause Prioritisation

LAB 16

Structured cause generation prevents the team jumping to a favourite theory. Use a Fishbone to organise causes by category, 5 Whys to drill from symptom to actionable cause, and multi-voting to converge on the few worth testing with data.

You'll build

A Fishbone diagram, three 5 Whys chains and a multi-voted shortlist of causes to test.

Tools & techniques: Ishikawa/Fishbone, 5M+E categories, 5 Whys, brainstorming, multi-voting, nominal group technique

Fishbone, 5 Whys, Multi-Voting and Cause Prioritisation

STEPS 1-4 OF 11

1

Open lab16-cause-investigation-log.xlsx — ten candidate causes raised by the team. Write the effect in the fish head as a measurable problem, not a vague complaint.

2

Draw the main bones using the 5M+E categories: Manpower, Method, Machine, Material, Measurement and Environment.

3

Brainstorm causes onto each bone. Set brainstorming ground rules first: no criticism, quantity over quality, build on others' ideas.

4

For each major bone, ask 'why does this happen?' to add sub-causes — a bone with no sub-causes has not been explored.

Fishbone, 5 Whys, Multi-Voting and Cause Prioritisation

STEPS 5-8 OF 11

5

Select the three most promising causes and run a 5 Whys chain on each, asking why repeatedly until you reach an actionable process cause.

6

Stop each 5 Whys chain when the answer becomes a process or system, not a person — 'the picker was careless' is a symptom, not a root cause.

7

Check each chain for logical validity by reading it backwards with 'therefore' — if it does not read logically, the chain is broken.

8

Run multi-voting in the workbook's Team Votes column: each participant gets N/3 votes to distribute across the candidate causes.

Fishbone, 5 Whys, Multi-Voting and Cause Prioritisation

STEPS 9-11 OF 11

9

Rank the causes by votes and select the top three to five for statistical validation.

10

For each shortlisted cause, write the hypothesis you will test in Lab 17 and name the data you need to test it.

11

Record explicitly that a cause is only a ROOT cause once data supports it — until then it remains a theory.

Fishbone, 5 Whys, Multi-Voting and Cause Prioritisation

Check your work

Every Fishbone bone has sub-causes, each 5 Whys chain ends at a process cause rather than a person, and each shortlisted cause has a testable hypothesis written for it.

Hypothesis Testing — Test Selection, p-values and Conclusions

LAB 17

The defining Green Belt skill. Instead of asserting that the night shift is slower, state it as a hypothesis, choose the correct test for your data type and question, and let the p-value decide. This is how a Green Belt replaces opinion with evidence.

You'll build

Stated hypotheses, a justified test selection, and a documented statistical conclusion.

Tools & techniques: H_0 and H_a , alpha, p-value, Type I and Type II error, t-tests, paired t, chi-square, ANOVA, Mann-Whitney

Hypothesis Testing — Test Selection, p-values and Conclusions

STEPS 1-4 OF 15

1

Write the null hypothesis H_0 as a statement of NO difference or no effect — it always contains an equals relationship.

2

Write the alternative hypothesis H_a as the difference you suspect — not equal, greater than, or less than.

3

Remember hypotheses are statements about the POPULATION, not the sample. You can simply calculate the sample mean; you infer about the population.

4

Set the confidence level and hence alpha: 95% confidence gives $\alpha = 0.05$; 99% gives 0.01.

Hypothesis Testing — Test Selection, p-values and Conclusions

STEPS 5-8 OF 15

5

Understand the two error types: Type I rejects a true H_0 (producer risk, measured by α); Type II accepts a false H_0 (consumer risk, measured by β).

6

Select the correct test using four questions: what data type, how many levels of X, is the data normal, and are you testing means, medians, variances or proportions.

7

Apply the selection rules — 1-Sample t compares a sample mean to a target; 2-Sample t compares means of two different populations; Paired t compares the same subjects before and after.

8

Note the classic trap: the same team measured before and after training needs a PAIRED t-test, while team A versus team B needs a 2-SAMPLE t-test.

Hypothesis Testing — Test Selection, p-values and Conclusions

STEPS 9-12 OF 15

9

For non-normal data comparing medians, use the non-parametric equivalents: One-Sample Wilcoxon or Mann-Whitney.

10

For comparing proportions use 1-Proportion or 2-Proportion; for comparing variances use the chi-square or F-test; for more than two groups use ANOVA.

11

Open lab17-shift-comparison-samples.xlsx — 40 night-shift and 40 day-shift orders. Test whether night-shift mean lead time is significantly greater. State H_0 , H_a and your chosen test.

12

Run the test and read the p-value. Then repeat the exercise on lab17-cause-count-contingency.xlsx, where the discrete data forces a chi-square test instead. If $p < \alpha$, reject H_0 .

Hypothesis Testing — Test Selection, p-values and Conclusions

STEPS 13-15 OF 15

13

Worked interpretation: with alpha set at 0.05, a returned p-value of 0.031 means reject H_0 — the difference is statistically significant.

14

Translate the statistical result into business language for your sponsor — never present a p-value without saying what it means for the process.

15

Record the caution: failing to reject H_0 does not prove H_0 is true; it means you lack sufficient evidence to reject it, which may simply mean too small a sample.

Hypothesis Testing — Test Selection, p-values and Conclusions

Check your work

For each tested cause you can state H_0 , H_a , the test selected with justification, the p-value, the decision against alpha, and the business conclusion in plain language.

Correlation, Regression and Quantifying the X-Y Relationship

LAB 18

Hypothesis testing tells you whether a difference exists; correlation and regression tell you how strongly two variables move together and let you predict Y from X. This is how a Green Belt identifies which Xs are worth controlling — while never forgetting that correlation does not prove causation.

You'll build

A scatter plot, a correlation coefficient, a regression equation and a prediction with its limits.

Tools & techniques: Scatter plot, Pearson correlation coefficient R, coefficient of determination r-squared, linear regression, prediction

Correlation, Regression and Quantifying the X-Y Relationship

STEPS 1-4 OF 13

1

Open lab18-replen-vs-leadtime.xlsx. Build a scatter plot with replenishment backlog (X) on the horizontal axis and mean lead time (Y) on the vertical axis.

2

Read the plot visually first: is the pattern positive, negative, or absent? Is it linear or curved? Are there outliers?

3

Calculate the correlation coefficient R. Recall R ranges from -1 to +1: +1 is perfect positive, -1 perfect negative, 0 no relationship.

4

Apply the decision threshold from the reference: correlation is considered to occur when R is 0.4 or greater, or -0.4 or less.

Correlation, Regression and Quantifying the X-Y Relationship

STEPS 5-8 OF 13

5

Calculate the coefficient of determination r-squared by squaring R.

6

Interpret r-squared as the proportion of variation in Y explained by X. If $R = 0.86$ then $r\text{-squared} = 0.74$, so about 74% of the variation in Y relates to X and 26% is unexplained.

7

Confirm the data type requirement: both correlation and regression need continuous or ratio data. Category names against outputs do not qualify — use Pareto instead.

8

Fit the regression line to the workbook data and record the equation in the form $y = mx + c$ — the slope tells you how many hours each extra backlog task adds.

Correlation, Regression and Quantifying the X-Y Relationship

STEPS 9-12 OF 13

9

Use the equation to predict Y at two specific X values, and check the predictions against actual observed data at those points.

10

Solve the equation in reverse to find the X range that delivers your target Y — this becomes the operating window you will control in the Control phase.

11

Test the model's honesty: for a low r-squared, show a point where the prediction badly misses the actual value and explain why the model must not be used there.

12

Write the causation caution explicitly: strong correlation does not prove X causes Y. Confirm causation with process knowledge, a designed experiment, or a pilot.

Correlation, Regression and Quantifying the X-Y Relationship

STEPS 13-13 OF 13

13

Combine your evidence: list which Xs are now supported by BOTH a significant hypothesis test and a meaningful correlation — these are your validated vital few.

Correlation, Regression and Quantifying the X-Y Relationship

Check your work

You can state R, r-squared and the regression equation, use it to predict Y, and explain in one sentence why the correlation alone does not prove causation.

Recap — Analyze

- Common vs special cause — Common cause is built into the process; special cause is an external, assignable signal.
- Pareto analysis — The 80/20 rule separates the vital few causes from the trivial many — attack the vital few.
- Run charts — Plot the metric over time to reveal trend, shift, cluster, mixture, oscillation and bias.
- Fishbone (Ishikawa) — Organise candidate causes by 5M+E — Manpower, Method, Machine, Material, Measurement, Environment.
- 5 Whys — Drill from the symptom to an actionable cause — stop when the answer is a process, not a person.
- Descriptive statistics — Central tendency (mean, median) and dispersion (range, standard deviation) describe the data.

DMAIC · IMPROVE

Improve — Select, De-Risk and Pilot the Fix

Solution generation · Benchmarking · Solution selection matrix · 5S · Poka-Yoke · Pull/JIT · FMEA · DOE · Piloting

Key Concepts — Improve (1)

1

Generating solutions

Brainstorming, brainwriting, six thinking hats and anti-brainstorming — diverge before converging.

2

Benchmarking

Learn from who already does it best, inside or outside your industry.

3

Solution selection matrix

Score candidate solutions against weighted criteria — feasibility, cost, impact and time.

4

5S & visual workplace

Sort, Set in order, Shine, Standardise, Sustain — for physical and digital work alike.

5

Poka-Yoke

Mistake proofing makes the error difficult, obvious or impossible in the first place.

6

Pull, JIT & Heijunka

Let demand pull the work, level the load, and stop pushing inventory into the process.

7

FMEA & RPN

Severity x Occurrence x Detection = RPN — de-risk the change before it goes live.

8

Design of Experiments

Vary factors together, not one at a time, so interactions between Xs become visible.

Key Concepts — Improve (2)

1

Piloting

Test at small scale, measure against the baseline, then decide to scale, adjust or stop.

What the Improve phase delivers

1

A generated solution set

Many candidate solutions, each traced to a statistically proven root cause

2

A selected solution

Chosen by the team with a weighted selection matrix, not by the loudest voice

3

A risk assessment

FMEA on the new process, with actions taken on the highest RPNs

4

Piloted evidence

The solution tested at small scale against the baseline before full rollout

5

An implementation plan

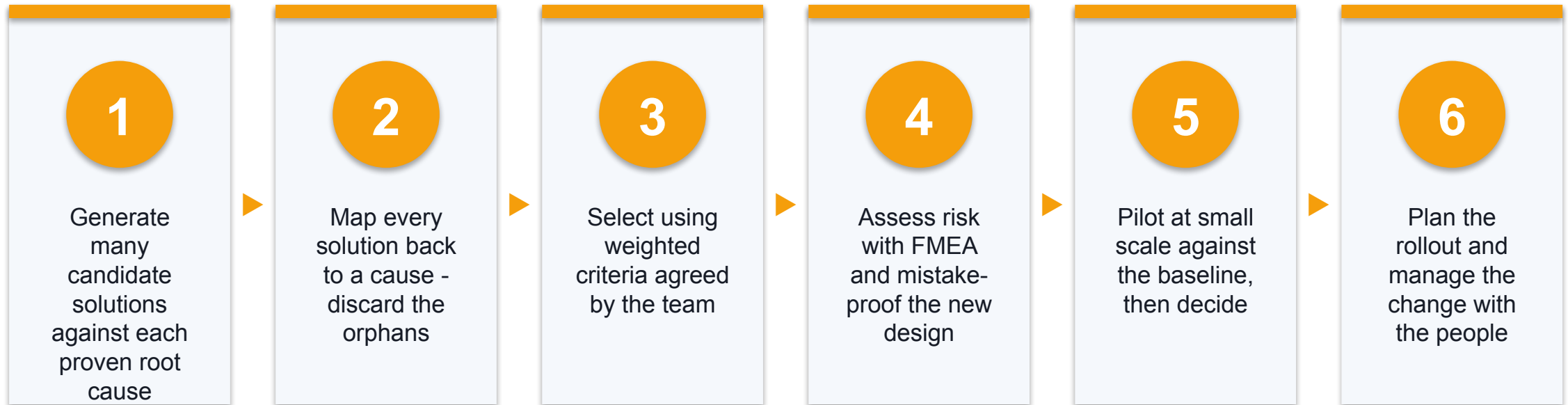
Who does what, by when, with the change management wrapped around it

6

A demonstrated improvement

Northwind late deliveries moving from 8.5% toward the 3.0% target

Steps of the Improve phase



Legitimate solution vs pet project

Legitimate

Traceable to proven data

- Cause proved by a hypothesis test at $p < 0.05$
- Solution attacks that specific cause
- Expected effect on Y can be estimated
- Measurable against the 8.5% baseline

Pet project

Traceable to somebody's opinion

- Cause never tested, only asserted
- Solution was chosen before the analysis
- Effect on Y cannot be described
- Usually expensive and usually IT

Solution generation techniques

1

Classic brainstorming

Group generates aloud; quantity first, no criticism, build on others' ideas

2

Brainwriting

Each person writes 3 ideas silently, then passes the sheet on to be built upon

3

Six Thinking Hats

Deliberately switch modes so every angle gets equal airtime

4

Anti-brainstorming

Ask how to make the problem WORSE, then invert every answer

5

SCAMPER

Substitute, Combine, Adapt, Modify, Put to other use, Eliminate, Reverse

6

Benchmarking

Borrow proven ideas from inside, from rivals and from other industries

Anti-brainstorming at Northwind

Make it worse

- How could we make late deliveries MUCH worse?
 - Give pickers no route sequence at all
 - Release all orders to the floor at 4pm
 - Let anyone override the dispatch priority
 - Never tell drivers about traffic or road closures
 - Train new pickers by watching a colleague once

Invert it

- Now invert each answer into a solution
 - Print an optimised pick route on every pick list
 - Level order release across the whole shift
 - Lock dispatch priority to a documented rule
 - Push live traffic alerts to the driver app
 - Certify pickers against standard work before solo picking

Three types of benchmarking

Internal

Another site or shift in your own firm

- Cheapest and fastest to arrange
- Data is comparable and accessible
- Northwind: the Tuas DC runs 3.1% late
- Risk: you only copy your own habits

Competitive

A direct competitor in your market

- Most relevant performance comparison
- Hardest to get real data from
- Use public reports and industry surveys
- Risk: legal and confidentiality limits

Functional

Best-in-class in ANY industry

- Compare the FUNCTION, not the product
- Fulfilment: study a pizza chain or courier
- Usually the source of breakthroughs
- Risk: needs the most adaptation work

The four benchmarking steps



Plan what to benchmark, collect the comparison data, analyse the gap, then ADAPT - never simply copy.

Northwind: solutions mapped to proven causes

Cause

- Proven root cause (from Analyse)
 - Late order release after the 4pm cut-off
 - Pick errors on the top 12 fast-moving SKUs
 - No documented dispatch priority rule
 - New pickers untrained on the scanner
 - Unmapped: 'the WMS is old'

Solution

- Candidate solution
 - Level order release across the shift (heijunka)
 - Poka-yoke: scan-verify SKU before the tote closes
 - Standard work: one documented dispatch rule
 - Certification against standard work before solo picking
 - DISCARD - no proven cause, this is a pet project

The solution selection matrix

Solution Selection Matrix								
Project Name:		Decision Factor						
Date:								
Relative Importance of each factor	Cost/ Effort/ Risk	Impact on Primary Metric	Impact on CTQ 2	Impact on CTQ 3	Impact on CTQ 4	Overall Score	Take Action Y/N	Why/Why not
	low=10 high=1	low=1 high=10	low=1 high=10	low=1 high=10	low=1 high=10			
	Weight							
Solution A						0	Sum of each rating * its weighting factor	
Solution B						0		
Solution C						0		
Solution D						0		
Solution E						0		
Solution F	Proposed Solution	Rating of solutions on 1 to 10 scale				0		
Solution G						0		
Solution H						0		
Solution I						0		
Solution J						0		
Solution K						0		
Solution L						0		
Solution M						0		

Score each solution against weighted criteria, then total. This is the tool that makes the decision defensible.

The four standard selection criteria

1

Feasibility

Is it practical? Can we do it with the people, skills and systems we have?

2

Cost

Is it cost-effective? Capital plus running cost against the benefit

3

Impact

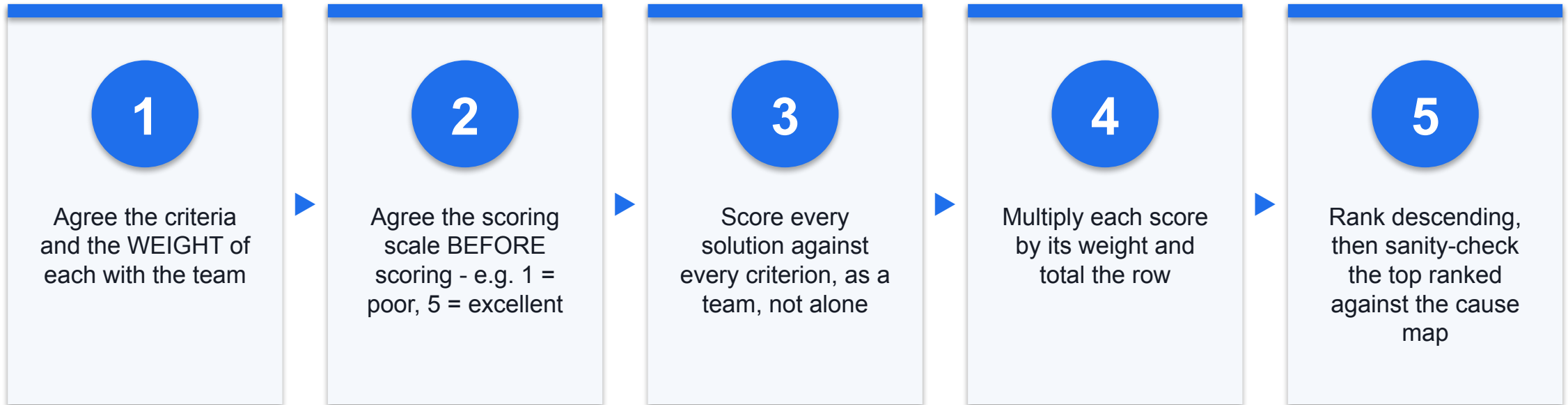
Does it significantly reduce the defect? How much of the 8.5% does it remove?

4

Time to implement

How quickly can it be in place and producing the effect?

How to run the selection matrix



Calculating the weighted score

Weighted score

Sum of (score x weight)

Scan-verify: $(5 \times 3) + (4 \times 2) + (5 \times 4) + (4 \times 1) = 47$

Criterion weight

Importance rating 1 - 5

Impact weighted 4; time to implement weighted 1

Score scale

1 = poor ... 5 = excellent

Agreed and written down BEFORE any solution is scored

Same scale, same weights, every solution - otherwise the ranking is meaningless.

Northwind worked example - weights and scores

Setup

- Weights agreed by the team
 - Impact on late % - weight 4
 - Feasibility - weight 3
 - Cost effectiveness - weight 2
 - Time to implement - weight 1
 - Scale agreed: 1 = poor, 5 = excellent

Result

- Weighted totals
 - Scan-verify poka-yoke = 47 (rank 1)
 - Level the order release = 42 (rank 2)
 - Documented dispatch rule = 39 (rank 3)
 - Picker certification = 35 (rank 4)
 - Replace the WMS = 22 (rank 5, and no proven cause)

Cost-benefit analysis and payback

Net benefit

Annual benefit - annual cost

\$180,000 saved - \$30,000 running =
\$150,000 per year

Benefit-cost ratio

Benefit / Cost

$180,000 / 30,000 = 6.0$ - every \$1 spent
returns \$6

Payback period

Investment / Annual net benefit

$\$75,000 / \$150,000 = 0.5$ years, i.e. about 6
months

Have Finance validate the benefit figure - a saving they will not sign off is not a saving.

The 5S cycle

SORT

- Distinguishes the work that needs to be done, and eliminated the unnecessary steps

STRAIGHTEN

- Follows the maxim, "*A place for everything and everything in its place.*"

SHINE

- Clean and remove the clutter from the workplace

STANDARDIZE

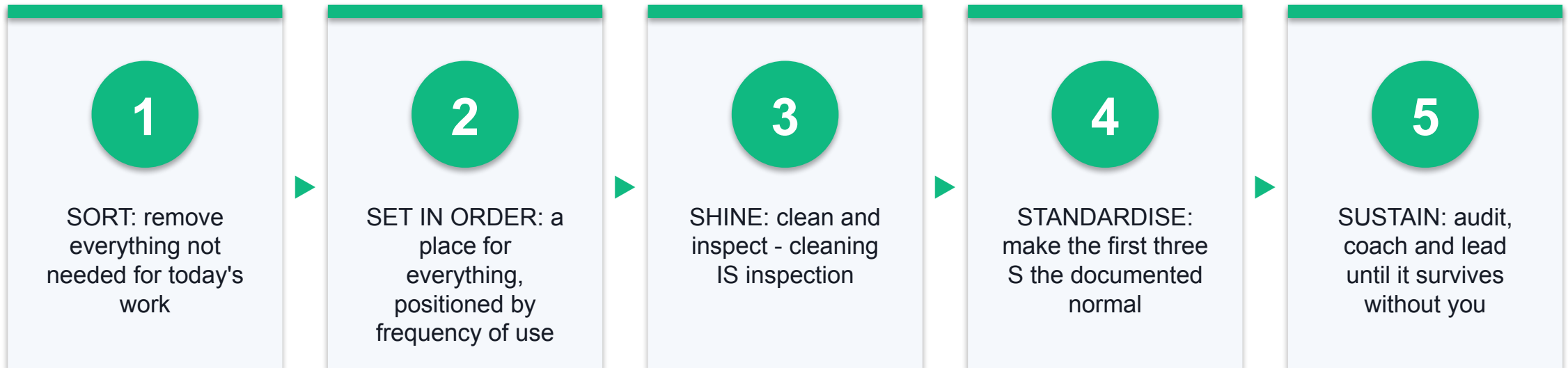
- Refers to efforts to develop and implement standardized procedures
- This will make the work environment more orderly

SUSTAIN

- Reinforces the prior four steps with repeated and continuous performance

Sort, Set in order, Shine, Standardise, Sustain - the fifth S is the one that fails, because it needs audits and leadership.

The five S



5S - physical and digital examples

1

Sort - physical

Clear the obsolete totes and dead stock out of the pick aisle

2

Sort - digital

Archive the 40 dead report files nobody has opened in a year

3

Set in order - physical

Fast-moving SKUs at waist height, slow movers high or low

4

Set in order - digital

One agreed folder tree; the live dispatch file pinned, not searched

5

Shine - physical

Daily scanner and label-printer clean-and-check by the shift team

6

Shine - digital

Weekly data hygiene: purge duplicate customer address records

7

Standardise - both

Shadow boards and aisle labels; a file naming convention that is enforced

8

Sustain - both

A short weekly 5S audit with a score, reviewed by the shift leader

IMPROVE

Poka-yoke: design the error out

Mistake proofing assumes people will err, because they will. Change the process so the error is impossible, or is caught the instant it happens.

IMPROVE

Poka-yoke examples

	A	B	C	D	E	F
	ID #	Type	Equipment Detail	Checked Out	Checked In	Checked Out By
5	1021	Laptop	15"		01-Oct-13	Sofie Ragnar
6	1022	Laptop	15"		16-Aug-13	Hank Sorenson
7	1023	Laptop	15"		15-Aug-13	Jennifer Weiss
8	1025	Laptop	15"		04-Oct-13	Min Seung
9	1031	Laptop	17"			Nick Ortiz
10	1032	Laptop	17"			Stanley Geyer
11	1033	Laptop	17"		26-Sep-13	George D'Agosta
12	1034	Laptop	17"		27-Aug-13	Jay Peralta
26	1011	Tablet	10"			Jay Peralta
27	1012	Tablet	10"			August Zorn

Sort Oldest to Newest

Sort Newest to Oldest

Sort by Color

Clear Filter From "Checked Out"

Filter by Color

Date Filters

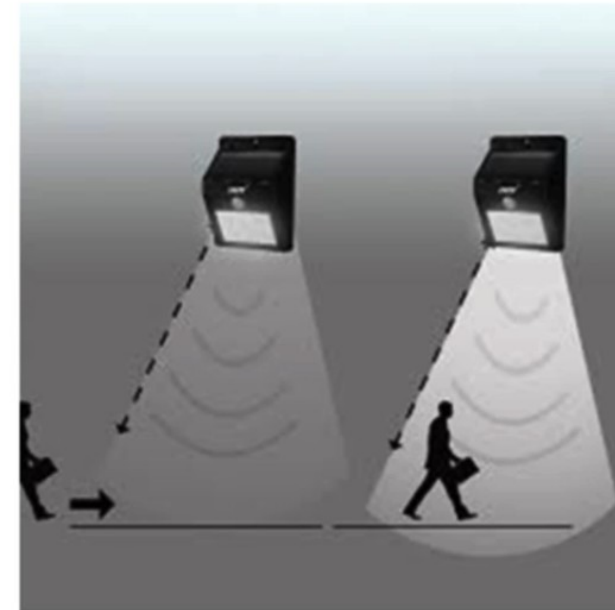
Search (All)

- (Select All)
- 2013
 - ☒ August
 - ☐ September
 - ☐ October

OK

Cancel

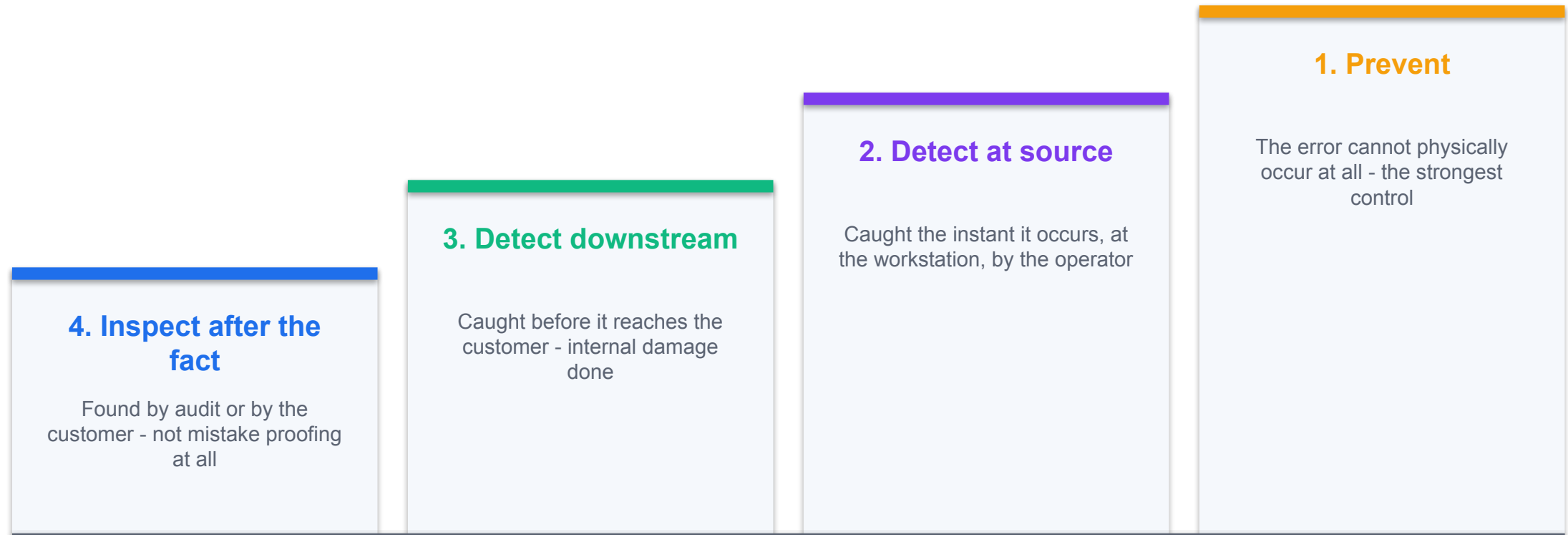
EXCEL FILTER



SENSOR/MOTION LIGHTING

Everyday mistake proofing: the design makes the wrong action physically impossible rather than merely discouraged.

Levels of mistake proofing - climb toward prevention



Always climb toward prevention: a connector that only fits one way beats a checklist.

Prevention beats detection

Prevent

The error is impossible

- USB-C plug fits either way round
- Car will not start unless it is in park
- System blocks dispatch without a scan
- Strongest, and usually cheapest to run

Detect at source

Caught as it happens

- Scanner beeps on the wrong SKU at the tote
- Tote weight check fails at the pack bench
- Form rejects an invalid postcode on entry
- Good - but the error still occurred

Detect downstream

Caught before it escapes

- Final audit at the loading bay
- 100% inspection before the van leaves
- Weakest and the most expensive to sustain
- A checklist lives here, not at level 1

Push versus pull



SIMPLIFY



MINIMIZE SET UP



**REDUCTION OF
HAND-OFFS**



**ELIMINATE
REWORK**



**MANAGING
BOTTLENECKS**



**POSITIONING
INVENTORY**



**REDUCING
REDUNDANCY**

Push is driven by a schedule upstream; pull is triggered by real consumption downstream.

Push vs pull, and the kanban signal

Push

Work released by a schedule

- Built to a forecast, which is always wrong
- Queues and WIP build between steps
- Defects hide inside the queue for days
- Bottleneck is drowned, upstream is idle

Pull

Work released by a signal

- Downstream consumption triggers upstream
- WIP is capped by the number of signals
- Problems surface fast - the queue is short
- Lead time becomes predictable

Kanban

The signal itself

- A card, a bin, a slot or a screen tile
- No card means no production - WIP is capped
- Removing a card deliberately shrinks WIP
- Northwind: capped pick waves per hour

Northwind: order release before and after levelling

Peaks

- Before - peaked release
 - 09:00-15:00: pickers idle, 40% utilisation
 - 16:00-18:00: 65% of the day's orders released
 - Overtime paid almost every evening
 - Error rate rises sharply under the rush
 - Late deliveries cluster on the evening wave

Levelled

- After - levelled release
 - Orders released in even waves across the shift
 - Peak headcount requirement drops
 - Overtime cost falls, idle time falls too
 - Error rate stabilises - no end-of-day rush
 - Late % falls without adding any headcount

The elements of standard work

1

The work sequence

The exact order of steps, the same way every time, by every operator

2

Takt time

The customer demand rate the sequence must keep pace with

3

Standard WIP

The minimum in-process stock needed to run the sequence without stalling

4

Quality checks

Where the check happens, what is checked and what to do on a fail

FMEA worksheet example

Failure modes and effects analysis (FMEA)																
Project:	FMEA - TECHNOLOGY					Date:	Update the date									
FMEA Team:	Names of Team Members					Prepared by:	Names of Individuals who prepare this worksheet									
SEV = How severe is effect on the customer? OCC = How frequent is the cause likely to occur? DET = How probable is detection of cause? RPN = Risk priority number in order to rank concerns; calculated as SEV x OCC x DET																
Business Function	Process step	Potential failure mode	Potential failure effects	SEV	Potential causes	OCC	Current process controls	DET	RPN	Actions recommended	Responsibility (target date)	Actions taken	NS EWV	NO ECWC	ND EWT	NR EPWN
Which Functional Department?	What is the step?	In what ways can the step go wrong?	What is the impact on the customer if the failure mode is not prevented or corrected?	10	What causes the step to go wrong? (i.e., How could the failure mode occur?)	10	What are the existing controls that either prevent the failure mode from occurring or detect it should it occur?	10	1000	What are the actions for reducing the occurrence of the cause or for improving its detection? You should provide actions on all high RPNs and on severity ratings of 9 or 10.	Who is responsible for the recommended action? What date should it be completed by?	What were the actions implemented? Include completion month/year (then recalculate resulting RPN).	10	10	10	1000
	Connectivity	No calls	• Loss of Business • Client Dissatisfaction • Impact on Repeat Business • Impact on Revenue • Communication Breakdown	8	• CMS link failure • Issues with PG (Peripheral Gateway) server for incoming calls • Issues with outbound calls which are routed through PG. • For outbound calls: Issues with the trunks, either from SP end or from local end.	7	• Continuous Monitoring of CMS link • Monitoring of session running on CMS for PG • Any issue with PG, Technology will escalate to GS telephony team for resolution	3	168	• Voice DR Infrastructure • Appropriate Notification procedure (Escalation Matrix) • Awareness training of Escalation Matrix to second rung of frontline supervisors	• DR Plan (Operations): Dec 2009	• Reviewed Escalation Matrix: 12th Feb				0
		Call Drops	• Loss of Business • Client Dissatisfaction • Impact on Repeat Business • Impact on Revenue • Communication Breakdown	8	• Fluctuation in trunks at local end or from Service Provider end	2	• Monitoring the status of the trunks & alarms on PBX	1	16	• Timely review of Escalation Matrix • Increase Awareness to follow Escalation Matrix	Responsibility: Shiv	Target: 12th Feb				0
		Skills mismatch /Incorrect Mapping	• Customer effected as unable to connect to the agent with specific skills • Agents getting incorrect calls • Incorrect Call Routing • SLA Miss	4	• Improper skills assigned to the agent	2	• Process Mission team to take care of the skill mapping	1	8	• Prompt Notification to the Technology Team	• Team leaders & Mission team	• As and when required				0
				• Customer/Agent dissatisfaction • High Resolution Time		• Faulty headsets • Faulty QD chord		• Supervisors to replace faulty headset/QD chord			• Proper Asset Maintenance (Technology Team to follow Checklist) • Escalate Supervisors as per					

One row per failure mode: effect, cause, current controls, then S, O and D rated 1-10 to give the RPN.

What each FMEA row captures

1

Process step

The specific step being assessed - work down the new process map

2

Potential failure mode

What could go wrong at this step - wrong SKU picked into the tote

3

Potential effect

What the CUSTOMER experiences - wrong item delivered, order returned

4

Potential cause

Why the failure mode occurs - two similar SKUs in adjacent bins

5

Current controls

What exists today to prevent or detect it - a visual check only

6

S, O, D and RPN

The three ratings and their product, used to rank the risk

The three FMEA ratings, each 1 to 10

Severity (S)

How serious for the customer

- 1 = customer would not even notice
- 10 = safety, legal or total loss of the order
- Rate the EFFECT, not the failure mode
- Severity is rarely reduced without redesign

Occurrence (O)

How likely is the cause

- 1 = extremely unlikely, once in years
- 10 = happens almost every shift
- Use real data where you have it
- Poka-yoke is how you drive O down

Detection (D)

SCALE IS INVERTED

- 1 = almost certainly caught before escape
- 10 = escapes to the customer undetected
- A HIGH D is BAD - this trips everybody up
- Better controls lower D, they do not raise it

Risk Priority Number

RPN

$$\text{RPN} = \text{S} \times \text{O} \times \text{D}$$

Wrong SKU picked: S=7, O=6, D=5 gives
RPN = 210

Ranking

Sort RPN descending

Address the highest RPN rows first, then work down the list

Projected RPN

Recalculate S x O x D after action

Scan-verify drops D from 5 to 2: $7 \times 6 \times 2 = 84$

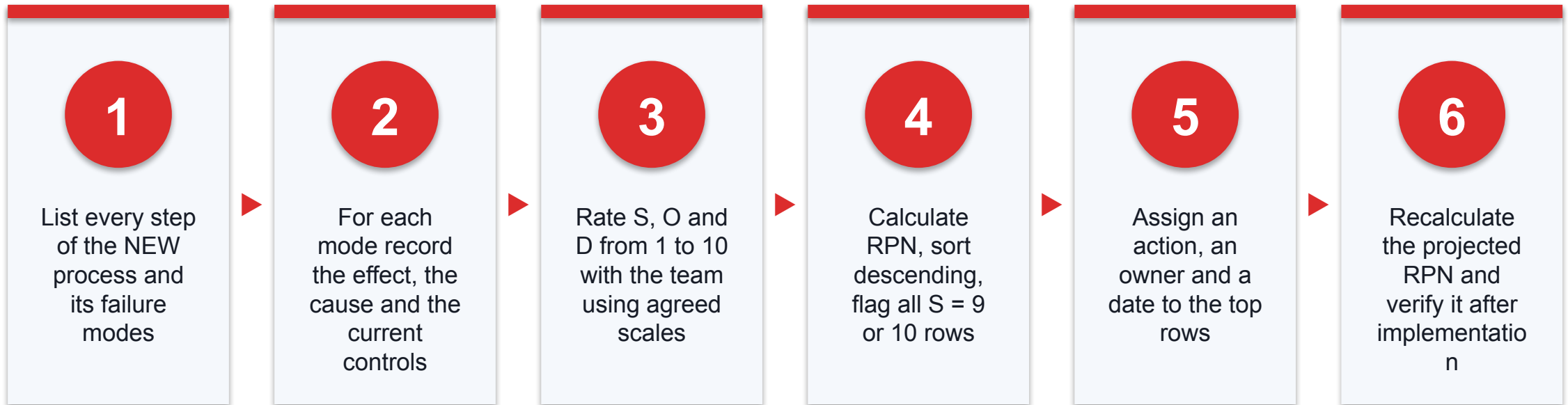
RPN ranges from 1 to 1000. It is a ranking device, not an absolute measure of risk.

IMPROVE

Severity 9 or 10 always requires action

Act on any Severity of 9 or 10 REGARDLESS of its RPN. A rare, well-detected catastrophe still scores a low RPN - and still ends careers.

Running an FMEA



Northwind FMEA - top rows and actions

Before

- Failure mode and initial rating
 - Wrong SKU picked: S=7 O=6 D=5, RPN 210
 - Order missed the 4pm cut-off: S=6 O=7 D=4, RPN 168
 - Van dispatched with wrong priority: S=6 O=5 D=5, RPN 150
 - Chilled item left out of chiller: S=9 O=2 D=3, RPN 54
 - Address record stale: S=5 O=4 D=6, RPN 120

After

- Action and projected RPN
 - Scan-verify at tote close - D 5 to 2, RPN 84
 - Level the release across the shift - O 7 to 3, RPN 72
 - Lock the documented dispatch rule - O 5 to 2, RPN 60
 - ACTED ON DESPITE RPN 54 - Severity is 9
 - Address validation at order entry - D 6 to 2, RPN 40

IMPROVE

DOE: change several factors on purpose

DOE varies factors together so you learn their individual effects AND how they interact - in far fewer runs than trial and error.

Why one-factor-at-a-time is inadequate

OFAT

Change one thing, hold the rest

- Feels rigorous and is easy to explain
- Needs many runs for very little information
- CANNOT reveal INTERACTIONS between factors
- Only ever finds each factor's individual effect

Designed experiment

Vary factors together, by design

- Estimates main effects AND interactions
- More information from fewer total runs
- Effects are estimated with known precision
- Result generalises across the factor space

DOE vocabulary

1

Response (Y)

The output you are trying to improve - delivery time in hours

2

Factor (X)

An input you deliberately set - driver experience, GPS, dispatcher policy

3

Level

The setting a factor is run at - GPS: no or yes; experience: 0 or 1 year

4

Run

One combination of all factor levels, with the response measured

5

Main effect

The average change in Y when one factor moves from low to high

6

Interaction

When one factor's effect on Y **DEPENDS** on the level of another factor

Full factorial run count

2^k design

Runs = 2 raised to power k

k = 4 factors at 2 levels each gives $2^4 = 16$ runs

With replication

Runs = 2^k x replicates

16 runs replicated twice = 32 deliveries measured

Goal for this study

Baseline - target reduction

72 hour baseline, goal to reduce by 24 hours

A full factorial tests every combination, so every main effect and interaction is estimable.

Worked DOE - furniture delivery process

1

The response

Delivery time in hours - baseline 72 hours, goal to reduce by 24 hours

2

Factor 1: driver experience

Two levels: 0 years of experience versus 1 year

3

Factor 2: GPS

Two levels: no GPS fitted versus GPS fitted

4

Factor 3: dispatcher policy

Two levels: no policy versus dispatcher follows the policy

5

Factor 4: pieces of furniture

Two levels: 1 piece versus 5 pieces on the delivery

6

The design

Four factors at two levels = a 16-run full factorial grid

Existing data vs a designed experiment

Existing (historical) data

Already sitting in the system

- Faster and much cheaper to obtain
- No disruption to live operations
- You did NOT control the other variables
- Factors are often confounded - causality is weak

Designed experiment

Run deliberately, to a plan

- You set the levels and randomise the runs
- Other variables are held or balanced out
- Supports a genuine causal conclusion
- Costs money, time and operational disruption

Designing the pilot

1

Scope

One shift, one product family, one bay - small enough to control and reverse

2

Duration

Long enough to cover normal variation, including a peak day and a quiet day

3

Success criteria

Numeric, agreed up front, measured AGAINST THE BASELINE of 8.5%

4

Measurement method

The same operational definition and gage used in Measure - no new metric

5

Rollback plan

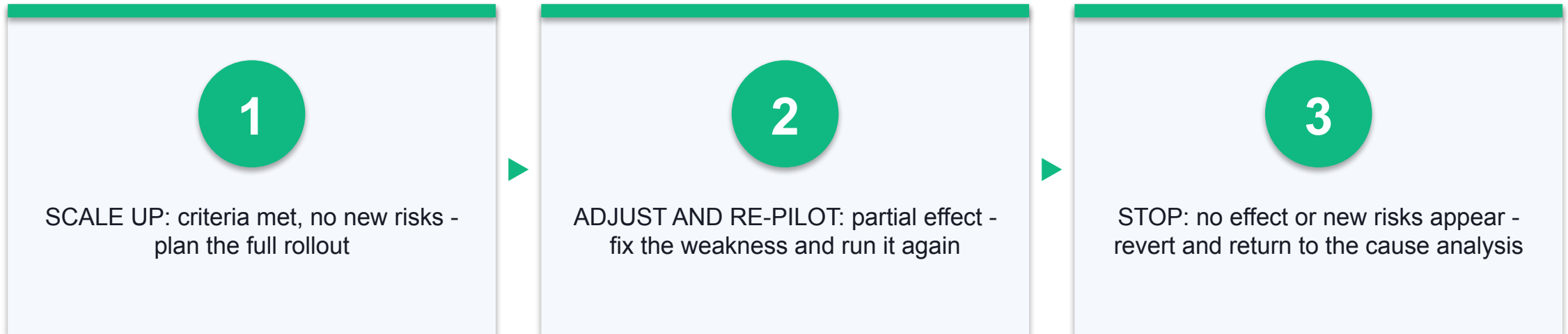
Exactly how to revert, who decides, and how fast - agreed before you start

6

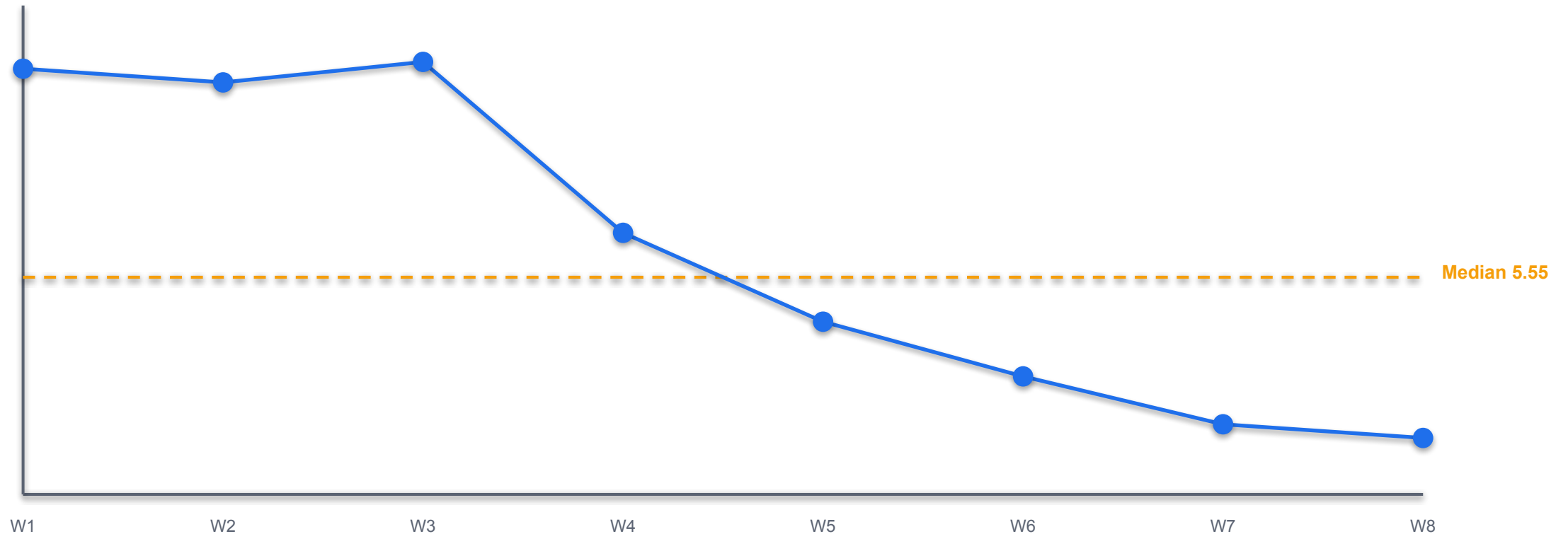
Who is involved

The operators who do the work, briefed and trained before day one

The three pilot outcomes



Northwind pilot: weekly late % on the pilot shift



Scan-verify and levelled release went live at W4. A sustained shift, not a single good week.

The Improve tollgate checklist

1

Solutions map to proven causes

Every selected solution names the cause and the evidence for it

2

Selection was documented

Weighted matrix with agreed criteria, weights and scale, scored as a team

3

Risk assessed

FMEA complete; top RPNs actioned and all S = 9 or 10 rows addressed

4

Piloted against the baseline

Results measured with the Measure-phase definitions, versus 8.5%

5

Benefit validated

Finance has agreed the cost-benefit and the payback period

6

Ready for Control

Implementation plan, standard work and owners in place for full rollout

Checkpoint - Improve

- Why must a solution be traceable to a PROVEN root cause before it is considered?
- In a solution selection matrix, what must be agreed BEFORE any scoring begins?
- Name the three levels of poka-yoke - and say which is strongest, and why.
- In FMEA, why is a Detection score of 10 bad news rather than good news?
- Which FMEA rows must be actioned regardless of their RPN?
- What can a designed experiment reveal that one-factor-at-a-time cannot?

WHAT YOU'LL DO

Hands-On Labs — Improve

1

Lab 19 — Solution Generation, Benchmarking and Brainwri

A solution log of at least fifteen candidate solutions mapped to prove

2

Lab 20 — Lean Countermeasures — 5S, Poka-Yoke, Pull, JI

A countermeasure plan applying 5S, poka-yoke, pull and standard work t

3

Lab 21 — Solution Selection Matrix and Cost-Benefit Ana

A weighted solution selection matrix with ranked results and a cost-be

4

Lab 22 — FMEA, Risk Priority Numbers, DOE and Piloting

A completed FMEA with RPN scores, a DOE plan and a pilot plan with suc

Solution Generation, Benchmarking and Brainwriting

LAB 19

Only now — with root causes proven by data — is it legitimate to talk about solutions. Diverge deliberately before converging: structured techniques produce far better solution sets than an open discussion dominated by the loudest voice.

You'll build

A solution log of at least fifteen candidate solutions mapped to proven root causes.

Tools & techniques: Brainstorming, brainwriting, six thinking hats, anti-brainstorming, benchmarking, SCAMPER

Solution Generation, Benchmarking and Brainwriting

STEPS 1-4 OF 9

1

List your validated root causes from Labs 17 and 18, then open lab19-improvement-ideas.xlsx — the team's ideas and benchmark findings, each tagged to the cause it targets. Solutions without a cause are pet projects.

2

Run a structured brainstorm with the ground rules displayed: no criticism during generation, quantity over quality, build on others' ideas, wild ideas welcome.

3

Run brainwriting for the quieter participants: each person writes three ideas silently, then passes the sheet on for others to build upon.

4

Apply anti-brainstorming: ask how you could make the problem WORSE, then invert each answer into an improvement.

Solution Generation, Benchmarking and Brainwriting

STEPS 5-8 OF 9

5

Use six thinking hats to examine the leading ideas from different perspectives — facts, feelings, risks, benefits, creativity and process.

6

Benchmark: identify who performs this process best, internally or in another industry, and record what they do differently.

7

Follow the four benchmarking steps — plan what to benchmark, collect the comparison data, analyse the gap, and adapt rather than copy.

8

Map every generated solution against the root cause it addresses in a two-column table, and discard any that address no proven cause.

Solution Generation, Benchmarking and Brainwriting

STEPS 9-9 OF 9

9

Group similar solutions and remove exact duplicates, keeping at least fifteen distinct candidates.

Solution Generation, Benchmarking and Brainwriting

Check your work

You have at least fifteen distinct solutions, every one traces to a root cause proven with data, and at least three came from benchmarking.

Lean Countermeasures — 5S, Poka-Yoke, Pull, JIT and Standard Work

LAB 20

Lean supplies a catalogue of countermeasures with a strong track record. Rather than inventing a fix from scratch, match the waste type you found in Lab 8 to the countermeasure that reliably addresses it — and prefer poka-yoke, which prevents the error, over inspection, which merely detects it.

You'll build

A countermeasure plan applying 5S, poka-yoke, pull and standard work to your process.

Tools & techniques: 5S, poka-yoke levels, pull system, kanban, JIT, heijunka, jidoka, standard work

Lean Countermeasures — 5S, Poka-Yoke, Pull, JIT and Standard Work

STEPS 1-4 OF 10

1

Open lab20-5s-waste-audit.xlsx — 5S scores for each warehouse area plus the waste walk log. Score each area out of 25; any area under 15 needs a 5S event first.

2

Design a poka-yoke for your highest-severity defect. Work through the three levels: prevent the error occurring, detect it as it occurs, or detect it before it passes downstream.

3

Prefer prevention over detection — a connector that only fits one way beats a checklist asking the operator to check the orientation.

4

Evaluate whether a pull system would help: does work get pushed into the process faster than it can be consumed, creating queues?

Lean Countermeasures — 5S, Poka-Yoke, Pull, JIT and Standard Work

STEPS 5-8 OF 10

5

Design a kanban signal that lets a downstream step pull work only when it has capacity.

6

Consider heijunka (load levelling) if demand arrives in peaks — levelling the load reduces both overtime and idle time.

7

Apply jidoka: build in the ability to stop the process automatically when a defect is detected, rather than continuing to produce defects.

8

Write standard work for the improved method: the sequence, the takt time, the standard WIP and the quality checks.

Lean Countermeasures — 5S, Poka-Yoke, Pull, JIT and Standard Work

STEPS 9-10 OF 10

9

Make the standard work visual — a photo or one-page diagram at the workstation beats a document in a shared drive.

10

Map each countermeasure back to the specific waste from your DOWNTIME analysis in Lab 8.

Lean Countermeasures — 5S, Poka-Yoke, Pull, JIT and Standard Work

Check your work

Every countermeasure traces to a specific waste, and your poka-yoke prevents or detects the error rather than relying on someone remembering to check.

Solution Selection Matrix and Cost-Benefit Analysis

LAB 21

Score every candidate solution against weighted criteria so the decision is transparent and defensible — feasibility, cost, impact and time to implement — then confirm the selection with a cost-benefit analysis.

You'll build

A weighted solution selection matrix with ranked results and a cost-benefit analysis.

Tools & techniques: Solution selection matrix, weighted criteria, feasibility, cost, impact, effort-impact grid, cost-benefit analysis

Solution Selection Matrix and Cost-Benefit Analysis

STEPS 1-4 OF 11

1

Open lab21-solution-selection-scoring.xlsx — the eight shortlisted solutions with their cost and annual benefit, and blank columns to score.

2

Define the selection criteria across the columns: Feasibility (is it practical to implement?), Cost (is it cost-effective?), Impact (does it significantly reduce the defect?), and Time to implement (how quickly?).

3

Assign a weight to each criterion reflecting what matters to the sponsor — impact and cost usually carry the highest weights.

4

Agree the scoring scale before scoring anything, and write down what a 1 and a 5 mean for each criterion so scores are comparable.

Solution Selection Matrix and Cost-Benefit Analysis

STEPS 5-8 OF 11

5

Score every solution against every criterion with the team, not alone — divergent scores usually reveal a hidden assumption worth discussing.

6

Calculate the weighted score for each solution: multiply each score by its criterion weight and sum across the row.

7

Rank the solutions by total weighted score and identify the top three.

8

Cross-check the ranking on an effort-impact grid — look for the high-impact, low-effort quick wins in the top-left quadrant.

Solution Selection Matrix and Cost-Benefit Analysis

STEPS 9-11 OF 11

9

Run the cost-benefit analysis from the workbook: $\text{payback period} = \text{Est. Cost} / (\text{Annual Benefit} / 12)$ in months. Check whether the payback ranking agrees with your weighted-score ranking.

10

Sanity-check the selected solution against the root cause it addresses and confirm it does not simply move the problem downstream.

11

Document the rationale for the selection — a sponsor will ask why the obvious expensive option was not chosen.

Solution Selection Matrix and Cost-Benefit Analysis

Check your work

Your matrix has weighted criteria with a defined scoring scale, every solution is scored, and your top-ranked solution has a payback period calculated.

FMEA, Risk Priority Numbers, DOE and Piloting

LAB 22

A solution that fails in production costs more than the problem it fixed. FMEA systematically asks how the change could fail, how bad that would be and how likely it is to be caught. Then pilot at small scale and measure against the baseline before committing.

You'll build

A completed FMEA with RPN scores, a DOE plan and a pilot plan with success criteria.

Tools & techniques: FMEA, severity, occurrence, detection, RPN, DOE factors and levels, pilot design, rollback plan

FMEA, Risk Priority Numbers, DOE and Piloting

STEPS 1-4 OF 16

1

List every process step of the NEW improved process in the FMEA worksheet.

2

For each step, identify the potential failure modes — the ways this step could go wrong.

3

For each failure mode, record the potential effect on the customer and the potential cause.

4

Open lab22-fmea-worksheet.xlsx — six failure modes for the selected solution with blank Sev/Occ/Det columns. Score Severity 1-10: how serious is the effect on the customer?

FMEA, Risk Priority Numbers, DOE and Piloting

STEPS 5-8 OF 16

5

Score Occurrence 1-10: how likely is this cause to happen?

6

Score Detection 1-10, remembering the scale is inverted — 1 means it is almost certainly caught, 10 means it escapes undetected.

7

Calculate the Risk Priority Number in the workbook's RPN column: $RPN = \text{Severity} \times \text{Occurrence} \times \text{Detection}$.

8

Sort by RPN descending and address the highest scores first. Treat any Severity of 9 or 10 as requiring action regardless of its RPN.

FMEA, Risk Priority Numbers, DOE and Piloting

STEPS 9-12 OF 16

9

Write the recommended action for each high-RPN row, assign an owner and a date, then recalculate the projected RPN after the action.

10

Plan a designed experiment for the settings you must optimise: list the factors, choose two levels (high and low) for each, and note that a 2^k design tests all combinations.

11

Record why DOE beats one-factor-at-a-time testing: OFAT cannot reveal the INTERACTION between two factors, only their individual effects.

12

Design the pilot: define the scope (one shift, one product family), the duration, the success criteria and the measurement method.

FMEA, Risk Priority Numbers, DOE and Piloting

STEPS 13-16 OF 16

13

Set the pilot success criteria against your measured baseline from Lab 12 and 13 — the pilot must beat the baseline by a stated margin.

14

Write the rollback plan: what triggers stopping the pilot and how the process reverts safely.

15

Run the pilot, collect data using the same operational definitions as your baseline, and compare like with like.

16

Decide: scale up, adjust and re-pilot, or stop. Record the decision and the evidence behind it.

FMEA, Risk Priority Numbers, DOE and Piloting

Check your work

Every FMEA row has an RPN, the highest RPNs have owned actions with dates, and your pilot has quantified success criteria measured against the Lab 13 baseline.

Recap — Improve

- Generating solutions — Brainstorming, brainwriting, six thinking hats and anti-brainstorming — diverge before converging.
- Benchmarking — Learn from who already does it best, inside or outside your industry.
- Solution selection matrix — Score candidate solutions against weighted criteria — feasibility, cost, impact and time.
- 5S & visual workplace — Sort, Set in order, Shine, Standardise, Sustain — for physical and digital work alike.
- Poka-Yoke — Mistake proofing makes the error difficult, obvious or impossible in the first place.
- Pull, JIT & Heijunka — Let demand pull the work, level the load, and stop pushing inventory into the process.

DMAIC · CONTROL

Control — Hold the Gain

SPC · Control chart selection · Control limits · Out-of-control rules · Cp/Cpk · Control plan · SOP · Visual management · Handover

Key Concepts — Control (1)

1

Statistical Process Control

Monitor the process with data over time so drift is caught before it becomes a defect.

2

Control chart anatomy

Centre line plus upper and lower control limits at ± 3 standard deviations.

3

Control chart selection

Continuous or attribute? Subgroup size? That decides Xbar-R, Xbar-S, I-MR, p, np, c or u.

4

Out-of-control rules

A point beyond 3 sigma, runs, trends and hugging patterns each signal a special cause.

5

Control limits vs spec limits

Control limits come from the process voice; specification limits come from the customer.

6

Cp and Cpk

Cp = spec width / process spread. Cpk also accounts for centring — aim for Cpk ≥ 1.33 .

7

Control plan

Metric, target, measurement method, frequency, owner and the reaction plan when it drifts.

8

Standard work & SOP

Lock the improved method into written, trainable, auditable standard work.

Key Concepts — Control (2)

1

Handover & closure

Transfer ownership to the process owner, verify the benefit, then formally close the project.

What the Control phase delivers

1

A live control plan

Every CTQ metric with its owner, sample plan, limits and reaction plan

2

Control charts in use

The process monitored on the floor, not in a monthly report

3

Updated SOPs and training

The new way of working written down and taught to every shift

4

Verified capability

Re-measured Cp and Cpk proving the gain is real and statistically significant

5

Finance-validated benefit

Savings signed off so the business recognises the money

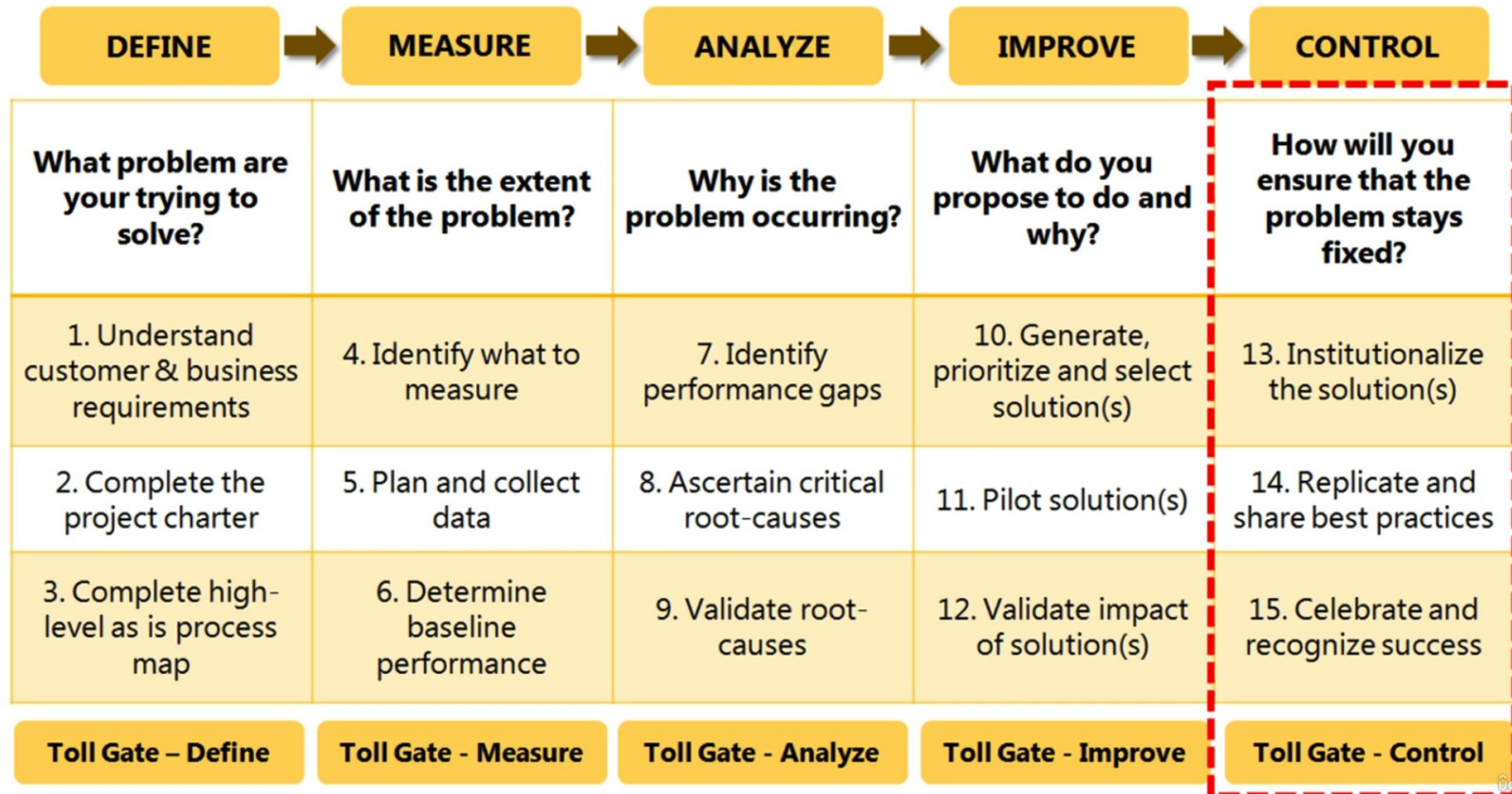
6

Formal handover and closure

Process owner accepts, lessons captured, replication identified

CONTROL

Steps of the Control phase



Control moves from monitoring the improved process, to documenting it, to handing it over with a named owner.

Why gains evaporate - and the antidote

The drift

What happens with no Control phase

- Nobody owns the metric after the team disbands
- Nobody knows what to do when it goes out of limits
- The SOP still describes the old process
- New starters are trained by the person next to them
- Six months later the baseline is back

The antidote

What Control installs

- One named owner per CTQ metric
- A written reaction plan triggered by the chart
- SOPs rewritten and version controlled
- Training plan plus a scheduled audit
- A daily huddle at a visible board

Process CONTROL vs process CAPABILITY

In control

Stable - the voice of the process

- Variation is predictable over time
- Only common cause variation is present
- No signals on the control chart
- Says nothing about the customer spec
- You can forecast it - you may still fail it

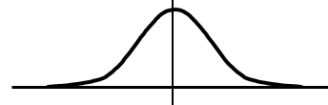
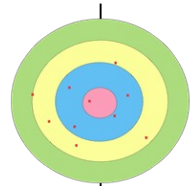
Capable

Stable AND on target for the customer

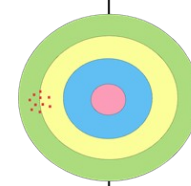
- Low variation AND centred on the requirement
- Output fits comfortably inside the spec limits
- Measured by Cp and Cpk
- Requires control first - it is a precondition
- This is what the customer actually feels

Accuracy vs precision - the target analogy

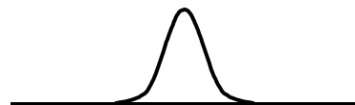
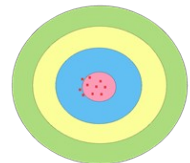
Accurate but is not precise



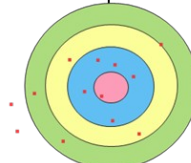
Inaccurate but precise



Accurate and precise



Inaccurate and not precise



Think of shots at a target: where the group sits is accuracy, how tight the group is is precision.

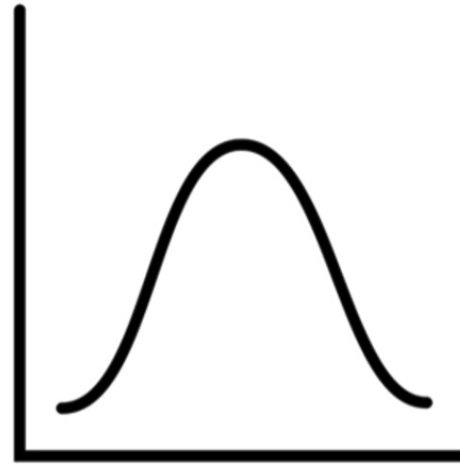
Capability is the curve inside the spec limits

Mean = Median = Mode

Curve is symmetrical on both sides

Area under the curve = 1

Tails are asymptotic



Capability asks a simple question: how much of this distribution falls between LSL and USL?

The Cp formula

$$C_p = \frac{USL - LSL}{6s} = \frac{\text{Spec. Width}}{\text{Process Spread}}$$

$C_p \geq 1$ -----► Potentially capable

$C_p = (USL - LSL) / 6s$ - specification width divided by process spread.

Cp - specification width over process spread

Cp

$$Cp = (USL - LSL) / 6s$$

Specification width divided by process spread

Northwind

$$Cp = 48 / (6 \times 6)$$

USL = 48 hrs, LSL = 0 hrs, s = 6 hrs -> Cp = 48 / 36 = 1.33

Reading it

Cp >= 1 -> potentially capable

Below 1 the spread alone guarantees defects

Cp >= 1 means POTENTIALLY capable - the spread would fit, IF the process were centred.

The Cpk formula

$$C_{pk} = \min \left\{ \frac{USL - \mu}{3\sigma}, \frac{\mu - LSL}{3\sigma} \right\}$$

$$C_{pk} \geq 1 \text{ -----} \blacktriangleright \text{Capable}$$

Cpk uses the distance from the mean to the NEAREST spec limit - so it exposes an off-centre process that Cp would pass.

Diagnosing with Cp and Cpk together

Cp good, Cpk good

Narrow and centred

- Spread fits the spec and sits on target
- The process is genuinely capable
- Action: hold the gain with SPC

Cp good, Cpk poor

Narrow but off-centre

- The spread is fine - the process is OFF-CENTRE
- Do NOT launch a variation reduction project
- Action: shift the mean back onto target

Cp poor

Too much spread

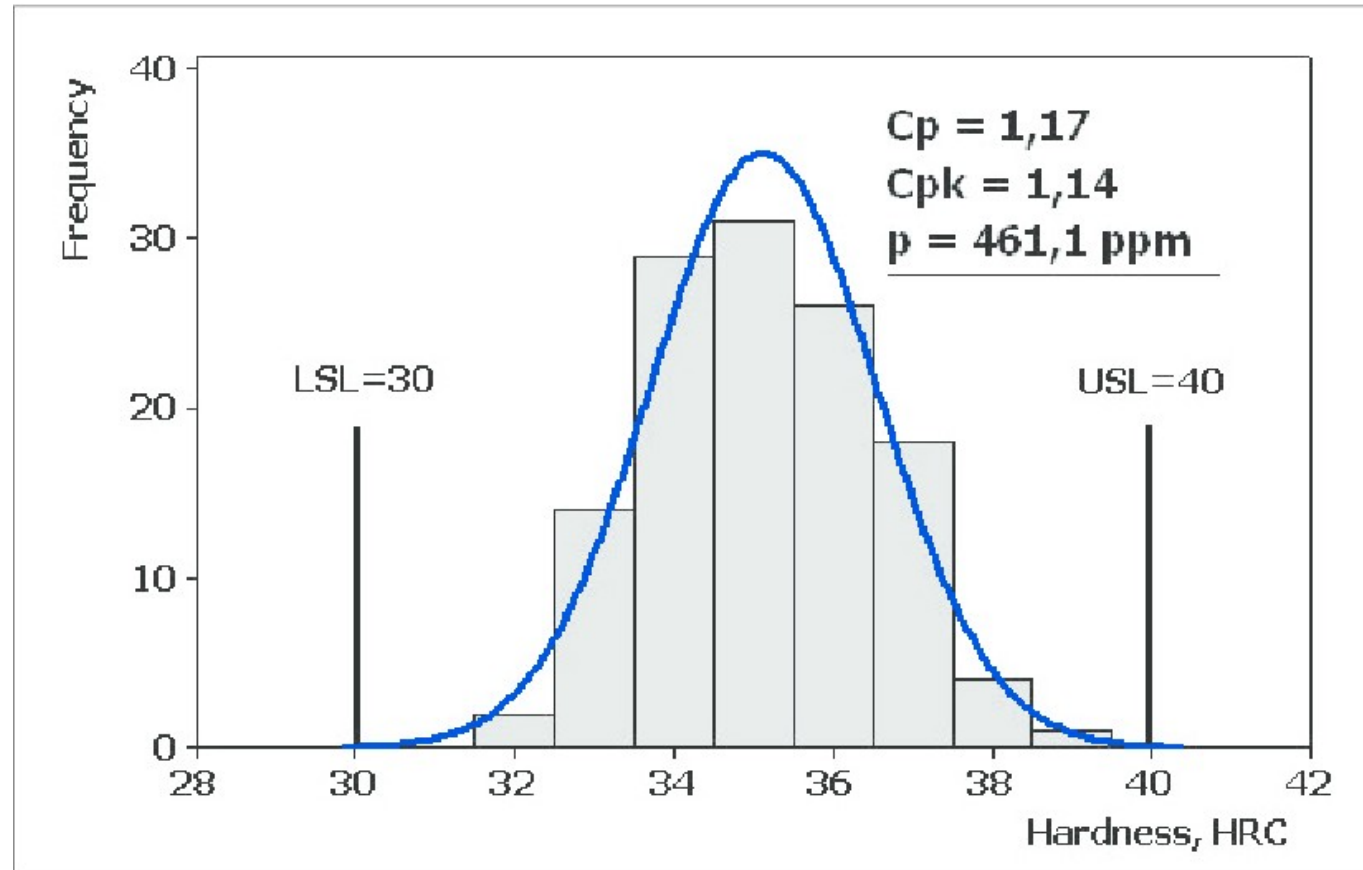
- The spread alone will not fit inside the spec
- Centring cannot rescue it
- Action: reduce variation before centring

CONTROL

Cpk of 1.33 is the usual minimum

Cpk 1.33 is about a 4 sigma process and the usual minimum for customer satisfaction. Many organisations target 2.0, with 1.5 acceptable.

Capability ratio thresholds



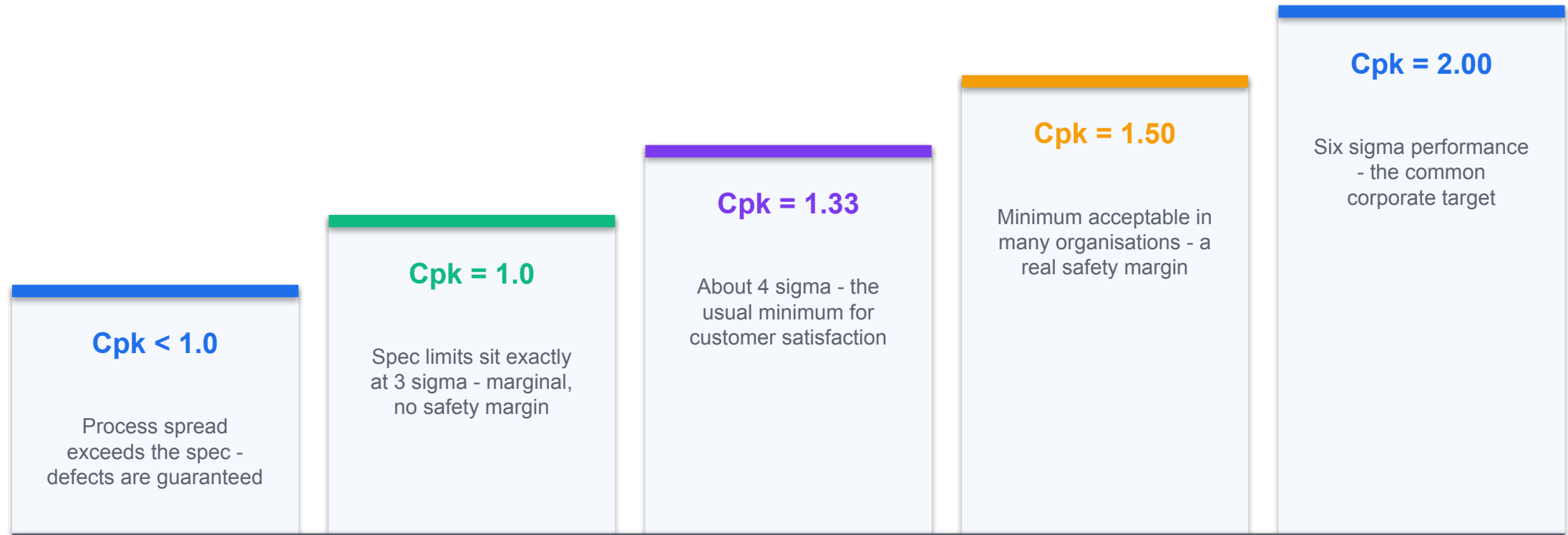
Read the ratio against the threshold your customer or industry demands - automotive and medical typically demand more.

Mapping capability ratios to sigma level

Sigma Conversion Table		
Yield %	Sigma	Defects per million opportunities
90.32	2.8	96800
88.5	2.7	115000
86.5	2.6	135000
84.2	2.5	158000
81.6	2.4	184000
78.8	2.3	212000
75.8	2.2	242000
72.6	2.1	274000

Capability ratio and sigma level are two views of the same thing: how many standard deviations fit between the mean and the spec.

Capability thresholds a Green Belt should know



Northwind's Cp of 1.33 is only POTENTIAL capability - check Cpk before claiming success.

CONTROL

Never plot spec limits on a control chart

Control limits come from the PROCESS - the voice of the process. Spec limits come from the CUSTOMER. Mixing them makes both meaningless.

Control limits vs specification limits

Control limits

The voice of the PROCESS

- Calculated from the process data itself
- Sit at +/- 3 standard deviations from the centre line
- Tell you whether the process is stable
- Move when the process genuinely changes
- Belong on the control chart

Specification limits

The voice of the CUSTOMER

- Set by the customer, contract or regulation
- Have nothing to do with process variation
- Tell you whether output is acceptable
- Change only when the requirement changes
- Belong on a capability histogram, NEVER on the chart

Why mixing them is dangerous

1

You stop reacting to signals

If spec limits are wider, real out-of-control points sit inside them and get ignored

2

You over-react to noise

If spec limits are tighter, you chase common cause variation and make things worse

3

You cannot judge stability

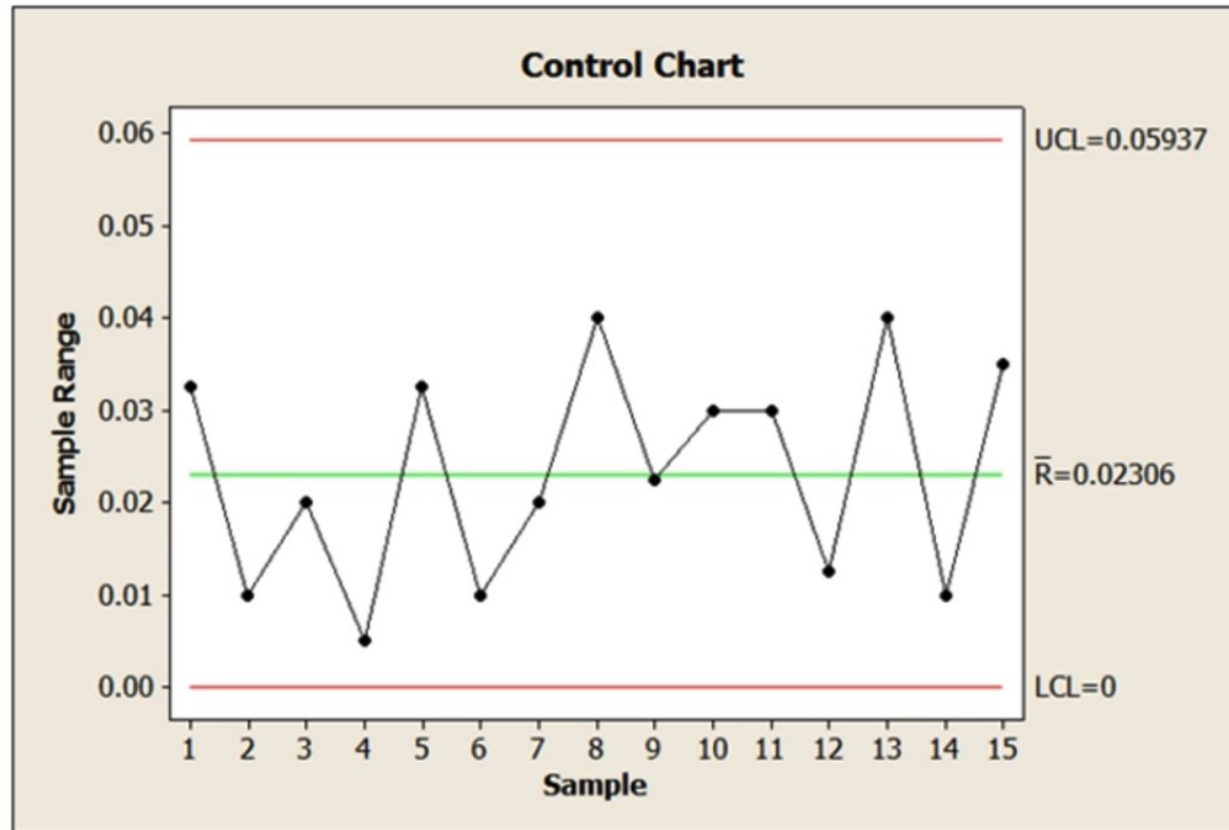
Stability is a property of the process only - the customer's opinion is irrelevant to it

4

You cannot judge capability

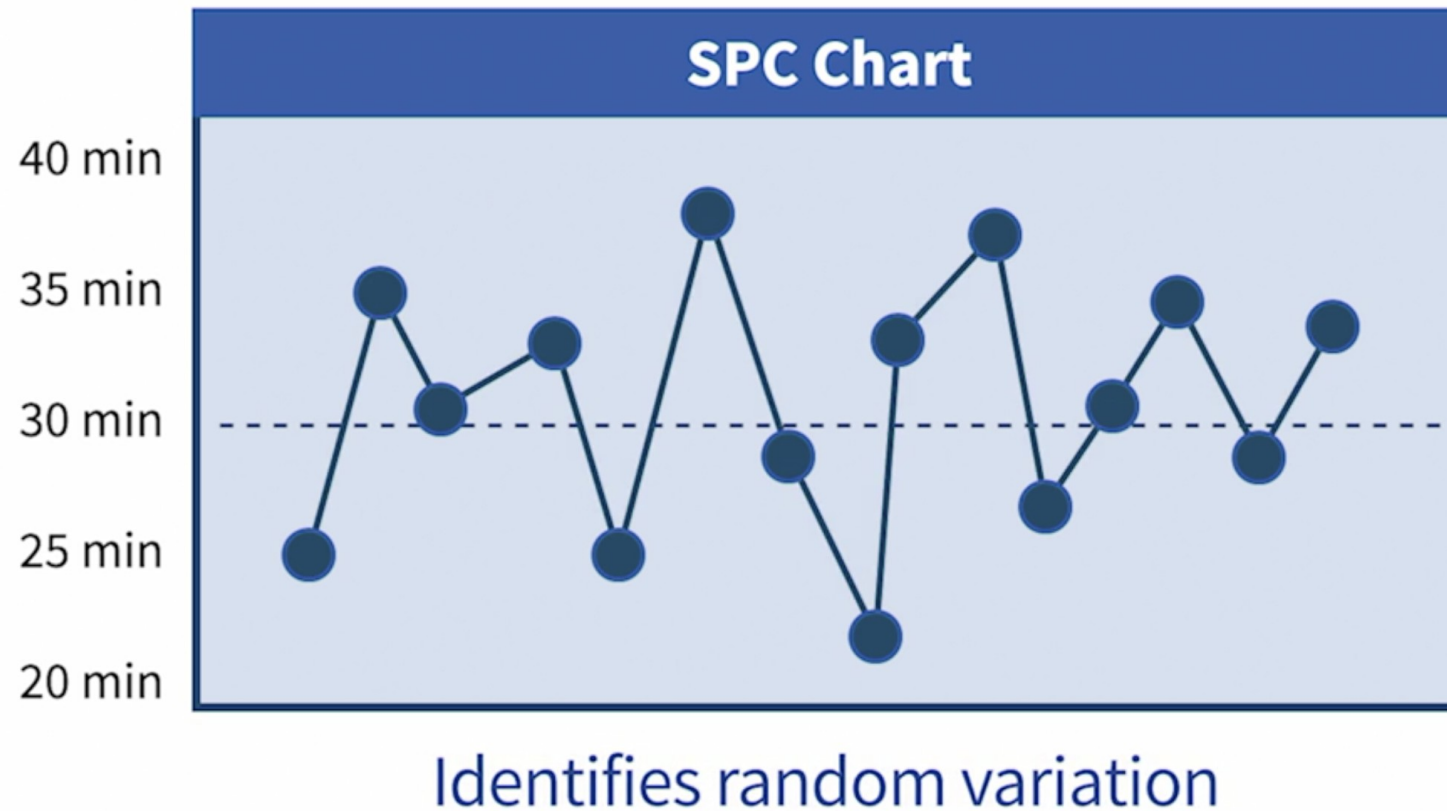
Capability compares the process spread to the spec - two separate charts, two questions

What a control chart is



A control chart is a run chart with statistically calculated limits, so a signal can be distinguished from noise.

Anatomy of a control chart in control



Centre line at the process average, UCL and LCL at ± 3 standard deviations, with the space between divided into zones.

The parts of a control chart

1

Centre line (CL)

The process average - the level the process is currently running at

2

UCL and LCL

Upper and lower control limits at ± 3 standard deviations from the centre line

3

Zone C

Within 1 sigma of the centre line - where most points should fall

4

Zone B

Between 1 and 2 sigma from the centre line - either side

5

Zone A

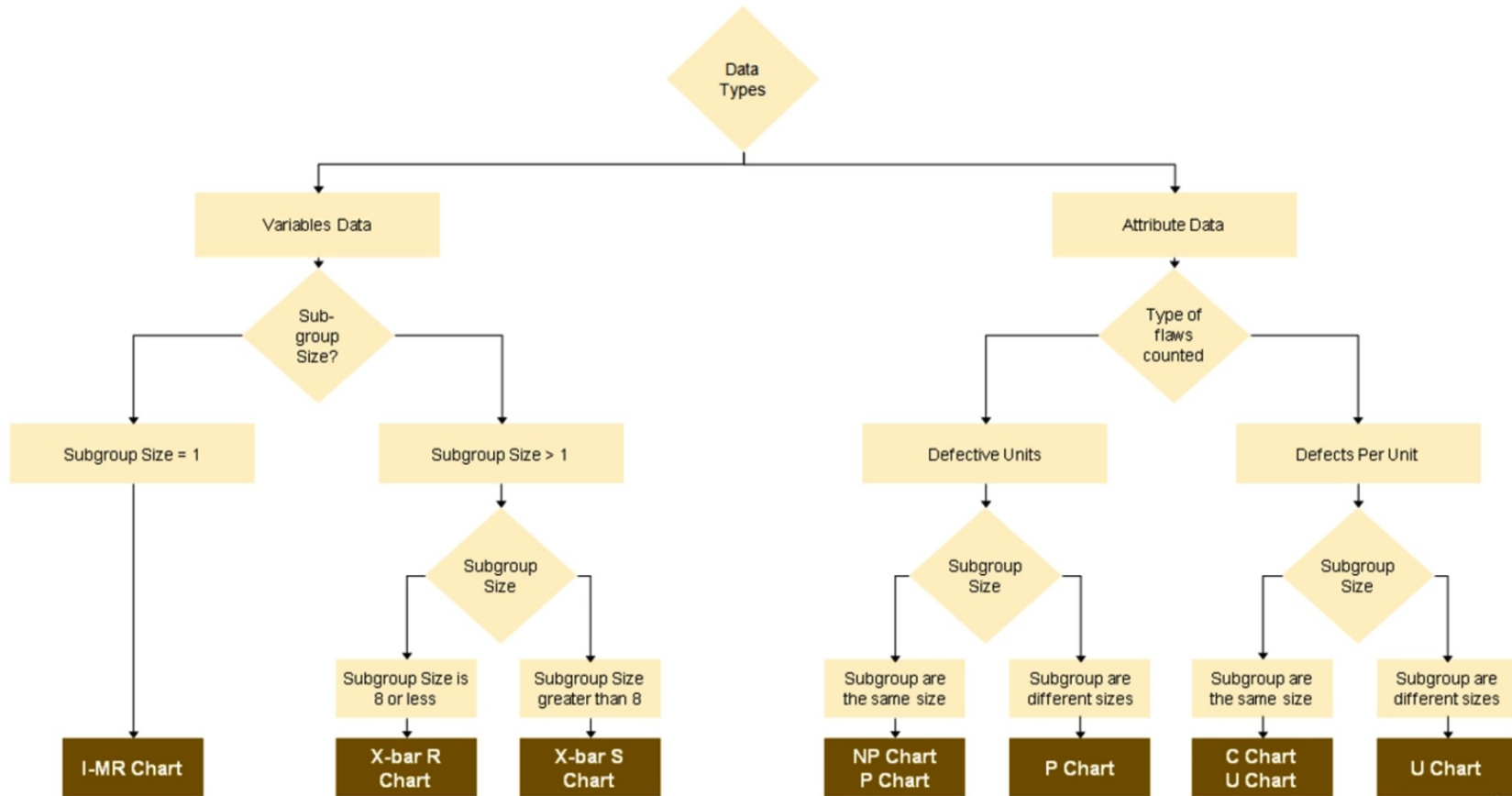
Between 2 and 3 sigma from the centre line - the outer band before the limits

6

The time axis

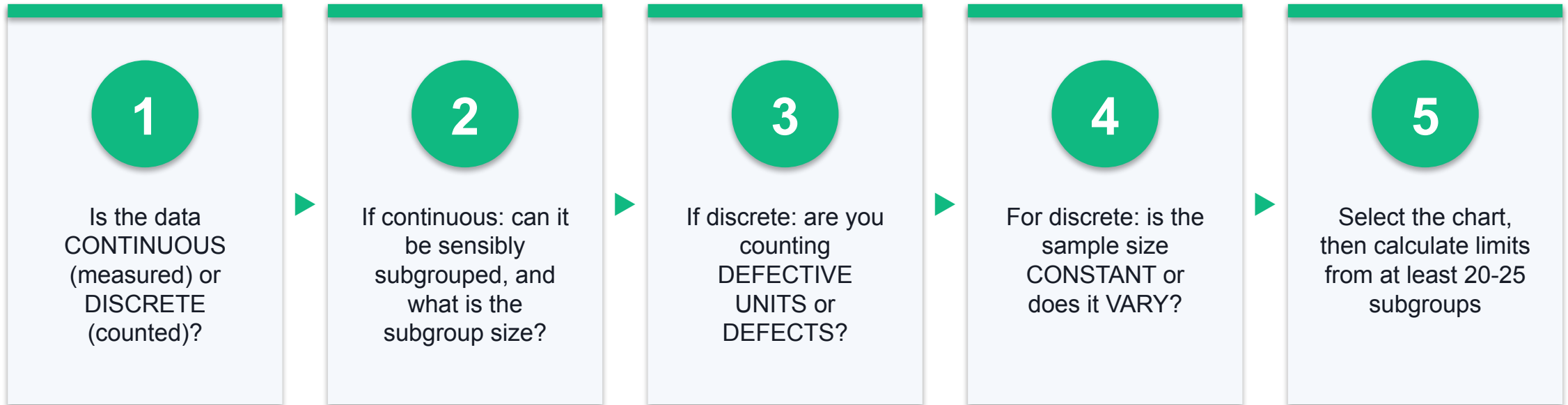
Points plotted in the order produced - sequence is the whole point

The control chart selection decision tree



Start with the data type, then the subgroup size or what you are counting.

How to select a control chart



CONTINUOUS (variable) data - three charts

1

I-MR chart

Individuals and Moving Range - use when the data cannot be sensibly subgrouped

2

Xbar-R chart

Subgroups with size UNDER 8 - plots the subgroup mean and the subgroup range

3

Xbar-S chart

Subgroups with size 8 OR MORE - plots the subgroup mean and standard deviation

4

Why the split at 8

With larger subgroups the standard deviation estimates spread better than the range

DISCRETE data - what are you counting?

1

Defective UNITS, size VARIES

p-chart - the proportion of units that are defective in each sample

2

Defective UNITS, size CONSTANT

np-chart - the raw number of defective units per constant-size sample

3

Counts of DEFECTS, size VARIES

u-chart - defects per unit, where sample size changes between samples

4

Counts of DEFECTS, size CONSTANT

c-chart - the raw count of defects per constant-size sample

Defective units vs defects - what decides the chart

Defective UNITS

Pass or fail per unit

- One unit is either defective or it is not
- Northwind: an order is late or on time
- Sample size varies -> p-chart
- Sample size constant -> np-chart

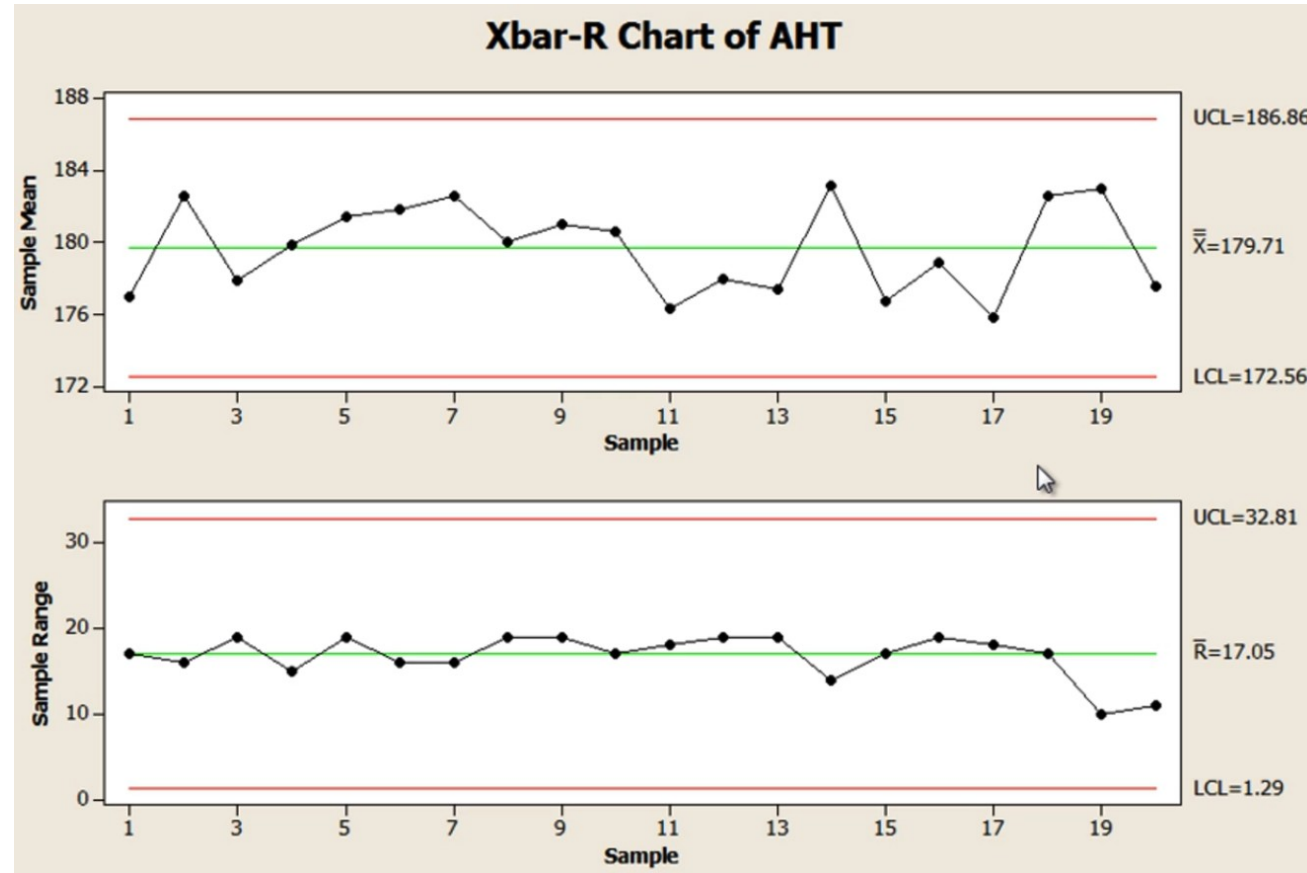
Counts of DEFECTS

A unit can carry several

- One unit can have many defects at once
- Northwind: errors found on one shipment
- Sample size varies -> u-chart
- Sample size constant -> c-chart

CONTROL

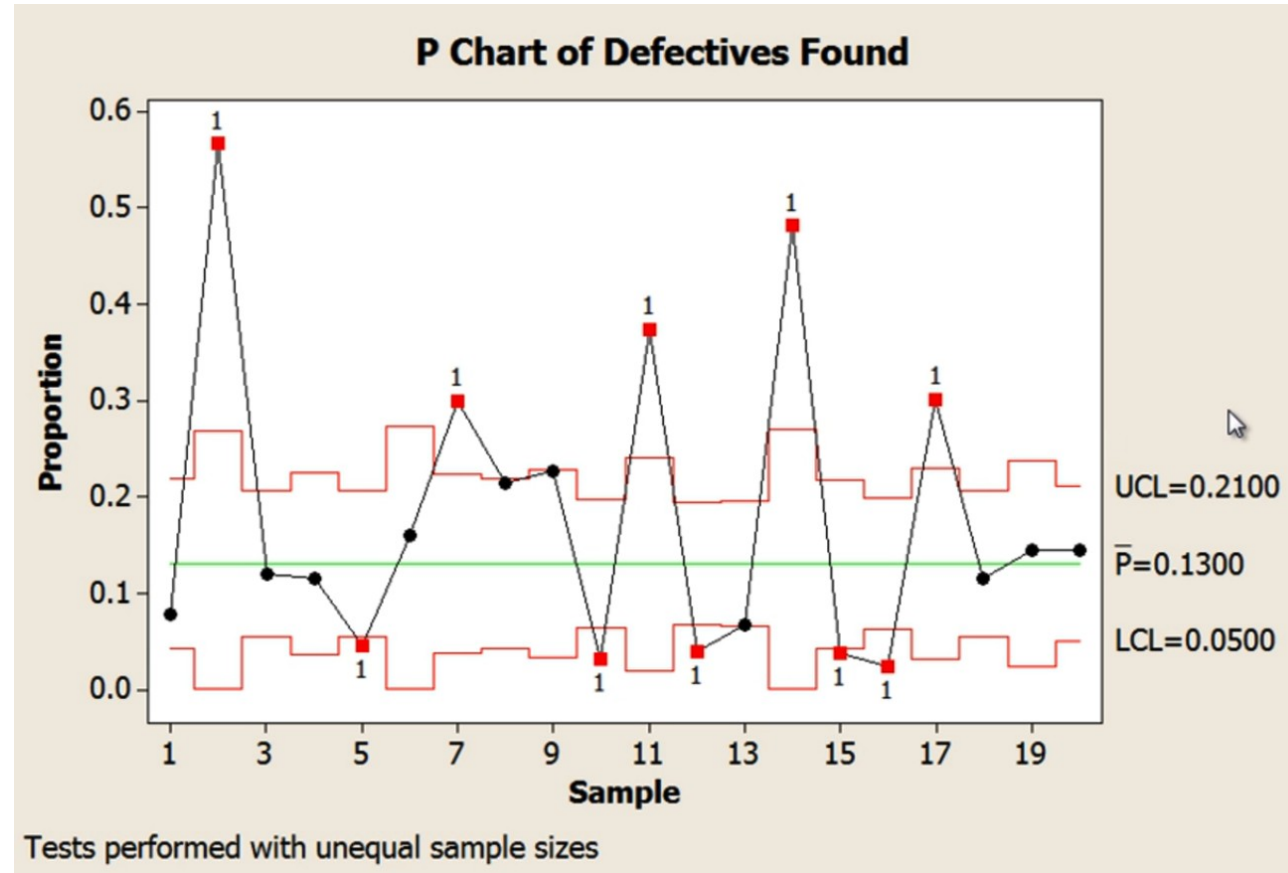
Xbar-R chart - continuous data in small subgroups



Two charts read together: the Xbar chart tracks the subgroup average, the R chart tracks the within-subgroup spread.

CONTROL

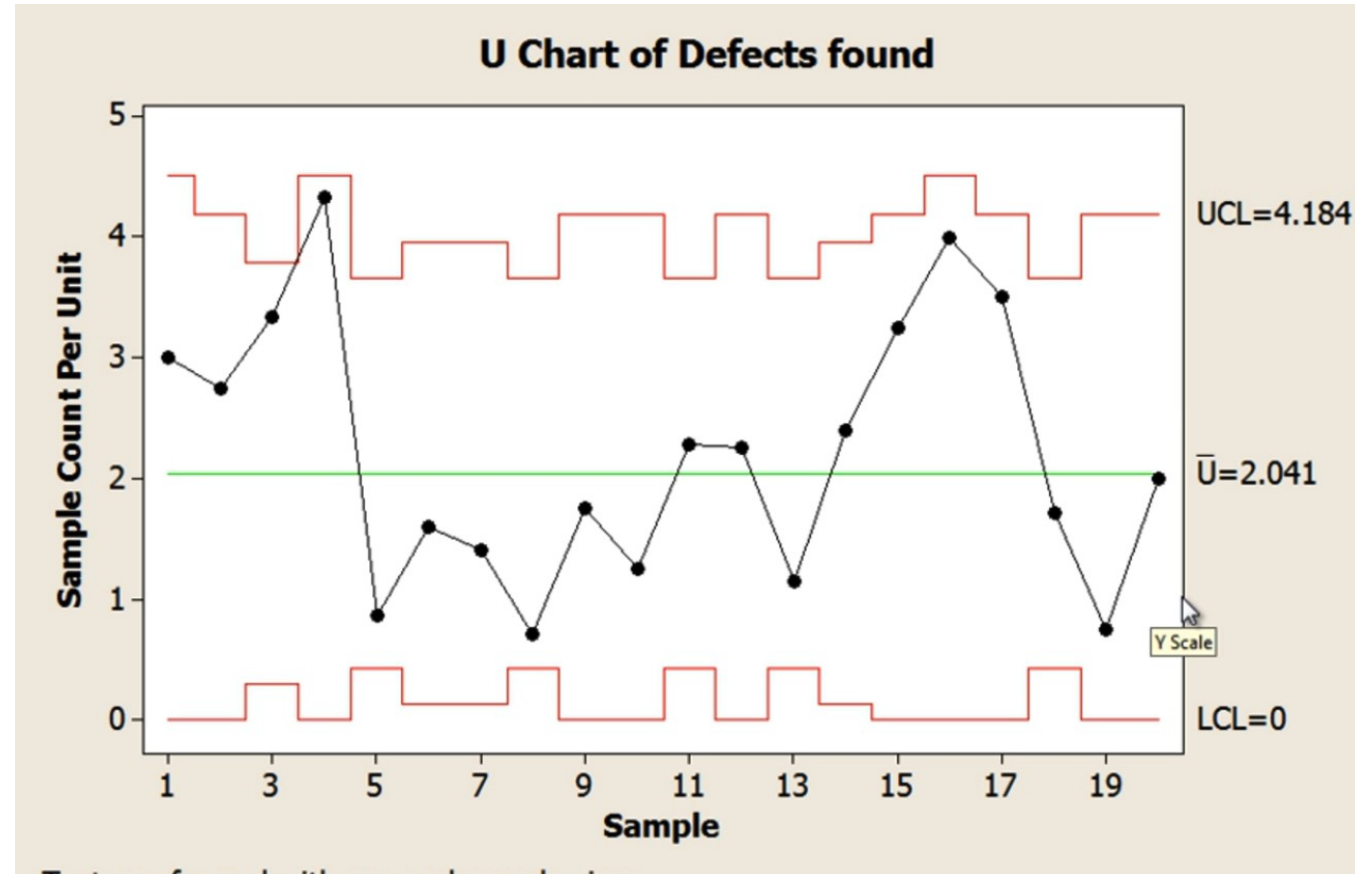
p-chart - proportion defective, varying sample size



Northwind's natural chart for late orders: the proportion late each day, with limits that widen when the daily volume is low.

CONTROL

u-chart - defects per unit with varying sample size



Use when one unit can carry several defects - for example picking errors per shipment across days of differing volume.

Calculating the control limits

- Upper Control Limit (UCL):

$$UCL = \bar{c} + 3\sqrt{\bar{c}}$$

- Lower Control Limit (LCL):

$$LCL = \bar{c} - 3\sqrt{\bar{c}}$$

- If LCL is negative, set $LCL = 0$.

- Example:

For $\bar{c} = 3.4$:

$$UCL = 3.4 + 3\sqrt{3.4} = 3.4 + 3(1.8439) = 3.4 + 5.5317 = 8.93$$

$$LCL = 3.4 - 3\sqrt{3.4} = 3.4 - 5.5317 = -2.13 \approx 0$$

- **UCL = 8.93, LCL = 0**

Centre line = the process average. UCL and LCL sit 3 standard deviations either side of it.

Control limits at +/- 3 standard deviations

Centre line

$$\text{CL} = \text{process average}$$

The mean of all the plotted subgroup values

Upper limit

$$\text{UCL} = \text{CL} + 3 \times \text{sigma}$$

Three standard deviations above the centre line

Lower limit

$$\text{LCL} = \text{CL} - 3 \times \text{sigma}$$

Three standard deviations below the centre line

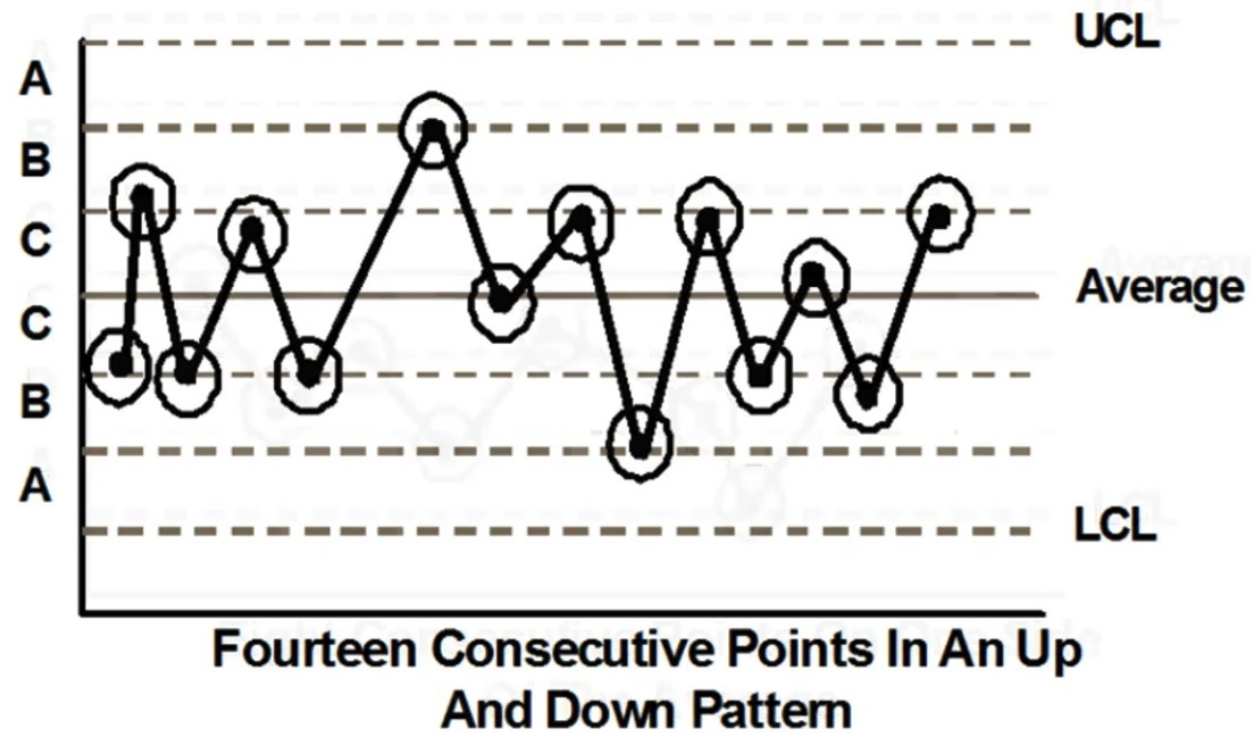
Use at least 20-25 subgroups so the limits are stable, and recalculate only when the process genuinely changes.

CONTROL

Eight rules turn a chart into a signal

A point outside the limits is only rule 1. Seven more patterns show the process changed even when every point sits inside the limits.

The out-of-control patterns



Each pattern has a statistical meaning - learn to read the shape, not just the last point.

The eight out-of-control rules (1 to 4)

1

Rule 1 - one point beyond a limit

A single point outside UCL or LCL. Act immediately - the chance of this at random is about 3 in 1,000.

2

Rule 2 - nine points one side

Nine points in a row on one side of the centre line - the process level has shifted.

3

Rule 3 - six points trending

Six points in a row steadily increasing or decreasing - a trend, such as tool wear or fatigue.

4

Rule 4 - fourteen alternating

Fourteen points alternating up and down - often over-correction, or variation between machines or shifts.

The eight out-of-control rules (5 to 8)

1

Rule 5 - two of three in zone A

Two out of three points in a row in zone A or beyond - a special cause creating sudden high variation.

2

Rule 6 - four of five in zone B

Four out of five points in a row in zone B or beyond - a shift or a major causation problem.

3

Rule 7 - fifteen in zone C

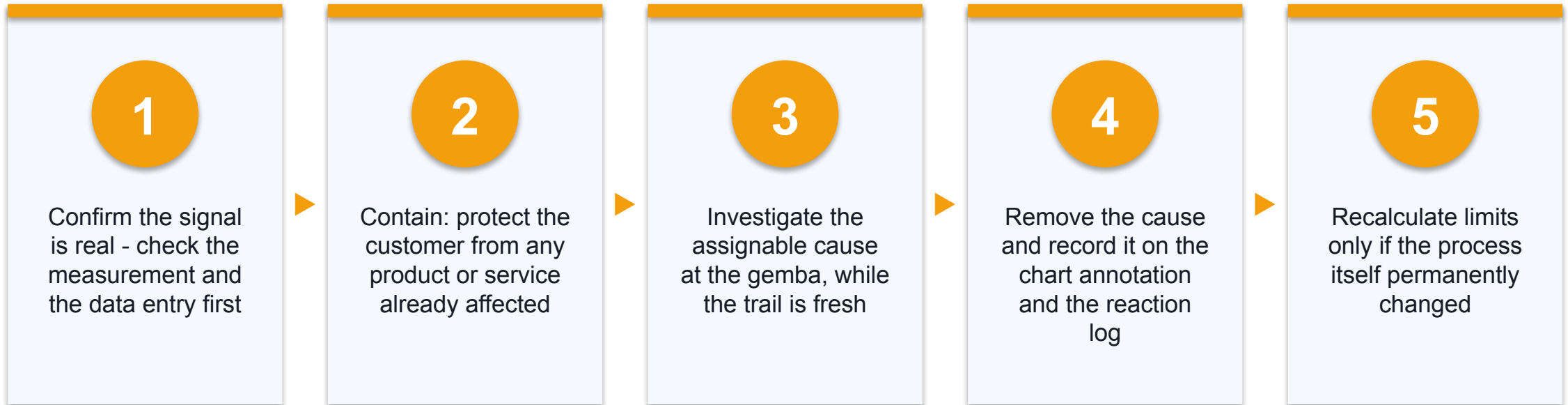
Fifteen points in a row within zone C - limits may be stale; recalculate if variation genuinely improved.

4

Rule 8 - eight with none in zone C

Eight points in a row either side, NONE in zone C - you may be measuring two processes as one.

Responding to an out-of-control signal



Common vs special cause - two responses

Common cause

The process behaving normally

- Random variation inherent in the design
- Every point inside the limits, no pattern
- Wrong response: adjust after each point
- Right response: change the process design
- Requires management action, not operator action

Special cause

Something new has entered

- A signal under one of the eight rules
- Assignable to a specific event or condition
- Wrong response: ignore it as noise
- Right response: investigate and remove it
- Usually actionable locally, at the process

The control plan format

[illegible]

Every row is one CTQ metric, carried across to a named owner and a written reaction plan.

The eight columns of a control plan

1

Process step

Which step in the improved process this control sits on

2

CTQ metric

The critical-to-quality characteristic being monitored

3

Specification

The customer requirement or target the metric must meet

4

Measurement method

How the value is obtained, with the operational definition

5

Sample size

How many units or transactions are measured each time

6

Frequency

How often the measurement is taken - hourly, per shift, daily

7

OWNER

The named role accountable - never a department, always a person

8

REACTION PLAN

The specific action taken when the metric breaches its limit

Northwind control plan - the four tracked metrics

1

Order processing time

Time from order received to release for picking - measured per order, sampled daily

2

Number of late orders

Orders delivered after the promised date - p-chart on the daily proportion late

3

Picking time

Time to pick a complete order in the warehouse - sampled per shift by zone

4

Shipping errors

Wrong item, wrong quantity or wrong address - counted per shipment, u-chart

Standard Operating Procedures that people actually use

1

Purpose

Why this procedure exists and which CTQ it protects

2

Scope

Which process, which shifts, which sites it applies to

3

Step-by-step instructions

The new standard way of working, in the order it is done

4

Quality checks

What to verify at each step and what good looks like

5

Escalation

Who to call, when, and what to do while waiting

6

Short and visual

One page with photos beats ten pages of prose - nobody reads the ten

Huddles and Gemba - two habits that hold the gain

The team huddle

Short daily stand-up at the board

- Ten to fifteen minutes, standing, same time daily
- Read yesterday's chart, name anything red
- Assign one owner and one action per issue
- Surfaces problems while they are still small

Go to Gemba

Leadership goes to where the work happens

- See the actual process, not the report about it
- Ask questions, do not issue instructions
- Verify the SOP matches what is really done
- Signals that the metric matters to management

Training and the audit schedule

1

Who needs training

Every shift and every relief worker - not just the day team who piloted it

2

What they are trained on

The new SOP, the chart, the reaction plan and their role in it

3

Competency check

Observed once at the process, not a signature on an attendance sheet

4

Audit frequency

Weekly for the first month, then monthly once the control is stable

5

What the audit checks

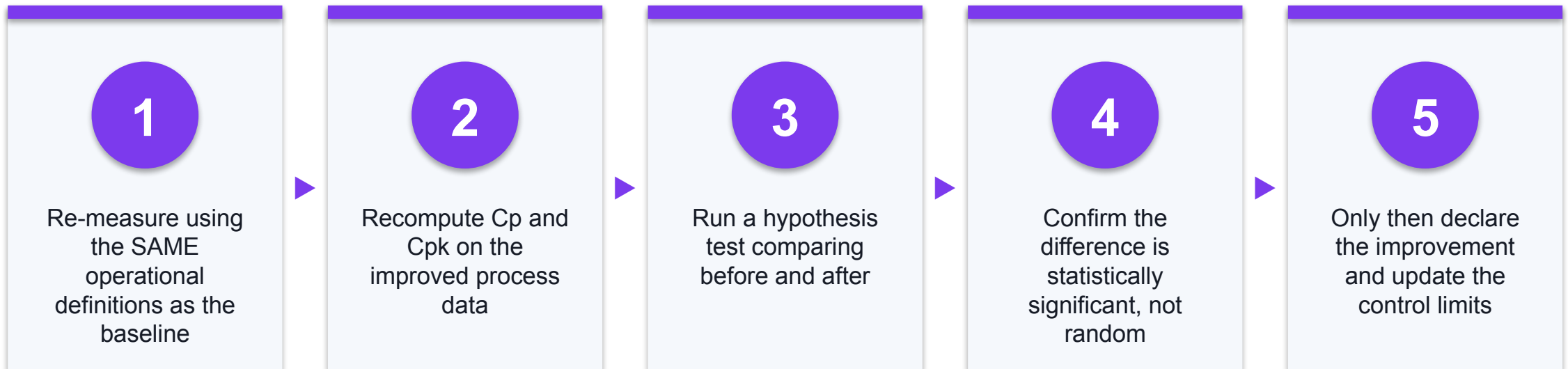
Is the chart current, are limits respected, was the reaction plan followed

6

Who audits

A named role outside the immediate team, reporting to the process owner

Verifying the gain



Verification and benefit validation

1

Same definitions

Change the operational definition and you are comparing two different things

2

Same measurement system

The MSA from Measure must still hold, or the comparison is meaningless

3

Enough data

A handful of good days is not evidence - collect a defensible sample

4

Hypothesis test

Proves the shift is bigger than the noise the process produces anyway

5

Finance sign-off

A benefit Finance has not signed off will not be recognised by the business

6

Hard vs soft benefit

Separate cash savings from capacity released - Finance treats them differently

Presenting results to stakeholders

1

Title as a question

Did late deliveries actually fall? beats Control chart of late orders

2

One graphic per slide

One clear chart that answers the question - not three competing ones

3

Conclusion in plain words

State what it means for the business, not what the statistic is called

4

Label every axis

The graphic must survive being forwarded on without you in the room

5

Leadership audience

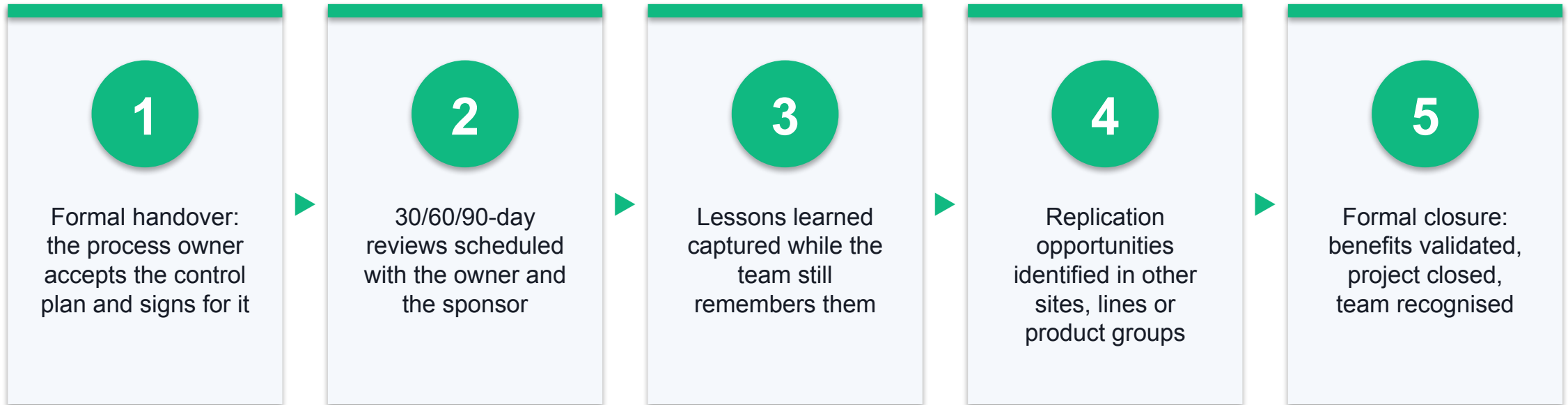
They want the benefit and the evidence it is under control

6

Process team audience

They want the workflow detail and what changes at their step

Handover and project closure



The Control tollgate checklist

1

Improvement verified

Re-measured capability plus a hypothesis test proving significance

2

Control plan complete

Every CTQ metric has limits, frequency, a named owner and a reaction plan

3

Charts live on the floor

Correct chart type selected, limits calculated, being updated daily

4

SOPs and training done

Documents updated, every shift trained, competency observed

5

Benefits validated

Finance has signed off the hard and soft benefits

6

Handover accepted

Process owner has accepted, reviews scheduled, project formally closed

Checkpoint - Control

- A process is in control but producing defects. Is it capable? What does that tell you?
- C_p is 1.5 but C_{pk} is 0.8. What is wrong with the process, and what is the fix?
- Why must specification limits never be plotted on a control chart?
- Northwind counts late orders daily with varying volume - which control chart, and why?
- Which out-of-control rule fires on nine points in a row on one side of the centre line?
- Name the two control plan columns without which the plan will not sustain the gain.

WHAT YOU'LL DO

Hands-On Labs — Control

1

Lab 23 — Statistical Process Control — Chart Selection

A correctly selected control chart with calculated limits and the eigh

2

Lab 24 — Control Plan, SOP, Visual Management and Respo

A complete control plan, an SOP for the improved method and a visual m

3

Lab 25 — Verify the Gain, A3 Storyboard, Handover and P

A before/after capability comparison, a validated benefit, an A3 story

Statistical Process Control — Chart Selection and Control Limits

LAB 23

A control chart is how a process tells you it has drifted before it produces a defect. Choosing the **WRONG** chart for the data type invalidates every signal it gives, so selection comes first — then the limits, then the rules for reading it.

You'll build

A correctly selected control chart with calculated limits and the eight rules applied.

Tools & techniques: SPC, control chart selection tree, Xbar-R, Xbar-S, I-MR, p, np, c, u charts, control limits, zones

Statistical Process Control — Chart Selection and Control Limits

STEPS 1-4 OF 17

1

Establish the difference between control limits and specification limits: control limits come from the process itself (the voice of the process), specification limits come from the customer (the voice of the customer). Never plot spec limits on a control chart.

2

Walk the control chart selection tree. First question: is your data continuous (variable) or discrete (attribute)?

3

For CONTINUOUS data, ask whether it can be sensibly subgrouped. If not, use an Individuals and Moving Range (I-MR) chart.

4

For continuous data in subgroups, use Xbar-R when the subgroup size is under 8, and Xbar-S when the subgroup size is 8 or more.

Statistical Process Control — Chart Selection and Control Limits

STEPS 5-8 OF 17

5

For DISCRETE data, ask whether you are counting DEFECTIVE UNITS or DEFECTS. This distinction decides the next branch.

6

For defective units (pass/fail): use a p-chart when the sample size varies, and an np-chart when the sample size is constant.

7

For counts of defects: use a u-chart when the sample size varies, and a c-chart when the sample size is constant.

8

Select the correct chart for your Northwind data and write down the justification against the tree.

Statistical Process Control — Chart Selection and Control Limits

STEPS 9-12 OF 17

9

Open lab23-spc-subgroups.xlsx — 30 days of post-improvement data, 5 orders per day, with Xbar and Range already computed. Plot in time order and draw the centre line and control limits using A2/D3/D4 for $n=5$.

10

Divide the chart into zones: zone C within 1 sigma of the centre line, zone B between 1 and 2 sigma, zone A between 2 and 3 sigma.

11

Apply out-of-control rule 1: any single point beyond the UCL or LCL. Investigate immediately — the probability of this happening by chance is roughly 3 in 1,000.

12

Apply rule 2: nine consecutive points on the same side of the centre line, indicating the process level has shifted.

Statistical Process Control — Chart Selection and Control Limits

STEPS 13-16 OF 17

13

Apply rule 3: six consecutive points steadily increasing or decreasing, indicating a trend.

14

Apply rule 4: fourteen consecutive points alternating up and down, often caused by over-adjustment.

15

Apply rules 5 and 6: two of three consecutive points in zone A, or four of five in zone B or beyond, indicating a sudden shift.

16

Apply rules 7 and 8: fifteen consecutive points inside zone C (limits may need recalculating), or eight consecutive points with none in zone C (you may be charting two different processes).

Statistical Process Control — Chart Selection and Control Limits

STEPS 17-17 OF 17

17

Record every signal your chart shows and the assignable cause you found for each.

Statistical Process Control — Chart Selection and Control Limits

Check your work

You can justify your chart choice against the selection tree, your limits are calculated at ± 3 sigma, and you have applied all eight out-of-control rules.

Control Plan, SOP, Visual Management and Response Plan

LAB 24

The control plan is the document that keeps the gain after the project team disbands. Without a named owner and a defined reaction plan, processes drift back to their old performance within months.

You'll build

A complete control plan, an SOP for the improved method and a visual management board design.

Tools & techniques: Control plan, control limits, monitoring frequency, reaction plan, SOP, visual management, team huddles, gemba

Control Plan, SOP, Visual Management and Response Plan

STEPS 1-4 OF 13

1

Open lab24-control-plan-template.xlsx — six control points with metrics, targets, owners and a blank Reaction Plan column to complete.

2

Select the key process metrics to track. For Northwind: order processing time, number of late orders, picking time and shipping errors.

3

Define the data collection method for each metric: who collects it, how often, and where it is recorded.

4

Set the control limits — the acceptable performance thresholds that trigger a response.

Control Plan, SOP, Visual Management and Response Plan

STEPS 5-8 OF 13

5

Write the corrective action for each metric: exactly what happens when performance breaches the control limit, and who decides.

6

Assign roles and responsibilities: name the individual accountable for each control point, not a department.

7

Write the SOP for the improved method: purpose, scope, step-by-step instructions, quality checks and escalation path.

8

Make the SOP visual and short. A one-page illustrated work instruction at the point of use beats a twenty-page document nobody opens.

Control Plan, SOP, Visual Management and Response Plan

STEPS 9-12 OF 13

9

Design the visual management board: which metrics are displayed, updated how often, by whom, and visible to whom.

10

Set up the daily team huddle: a short stand-up at the board reviewing yesterday's performance and today's risks.

11

Plan the gemba walk cadence — leadership going to where the work happens to see the process, not the report.

12

Build the training plan so every person performing the process is trained on the new standard, with a record of who was trained and when.

Control Plan, SOP, Visual Management and Response Plan

STEPS 13-13 OF 13

13

Add an audit schedule to verify the control plan is actually being followed — controls that are not audited quietly stop happening.

Control Plan, SOP, Visual Management and Response Plan

Check your work

Every control point has a named owner, a monitoring frequency and a specific reaction plan stating what to do when the metric breaches its limit.

Verify the Gain, A3 Storyboard, Handover and Project Closure

LAB 25

Close the project properly: re-measure capability, prove statistically that the improvement is real rather than random, quantify the financial benefit with Finance, tell the story on a single A3 page, and formally transfer ownership to the process owner.

You'll build

A before/after capability comparison, a validated benefit, an A3 storyboard and a signed handover.

Tools & techniques: Re-measured Cp/Cpk, hypothesis test of improvement, benefit validation, A3 report, handover, lessons learned

Verify the Gain, A3 Storyboard, Handover and Project Closure

STEPS 1-4 OF 15

1

Collect post-improvement data using exactly the same operational definitions and sampling method as your baseline — otherwise the comparison is meaningless.

2

Recalculate the process metrics: yield, RTY, DPMO and sigma level, using the same DUDO definitions from Lab 12.

3

Open lab25-before-after-verification.xlsx — the baseline month against the post-improvement month. Recalculate Cp and Cpk and compare against your Lab 13 baseline.

4

Run a hypothesis test comparing before and after performance — state H0 as 'no difference' and prove the improvement is statistically significant, not random variation.

Verify the Gain, A3 Storyboard, Handover and Project Closure

STEPS 5-8 OF 15

5

Report the p-value and confirm you can reject H_0 at your chosen alpha. An improvement you cannot prove statistically is not yet an improvement.

6

Quantify the financial benefit and have Finance validate it — a benefit the finance team has not signed off will not be recognised by the business.

7

Build the A3 storyboard on one page: background, current state, goal, root cause analysis, countermeasures, results, and follow-up actions.

8

Include the before and after charts on the A3 — the visual comparison communicates faster than any table of numbers.

Verify the Gain, A3 Storyboard, Handover and Project Closure

STEPS 9-12 OF 15

9

Present the results tailored to the audience: leadership wants the benefit and how it is controlled; the process team wants the workflow detail.

10

Follow the presentation rules: ask a question as the slide title, show one clear graphic that answers it, and state the conclusion in plain business language.

11

Hand over formally: walk the process owner through the control plan, the SOP, the charts and the reaction plan, then obtain their signature.

12

Agree the post-project review date — typically 30, 60 and 90 days — to confirm the gain has held.

Verify the Gain, A3 Storyboard, Handover and Project Closure

STEPS 13-15 OF 15

13

Capture lessons learned: what worked, what did not, and what you would do differently on the next project.

14

Identify replication opportunities: where else in the business does this same problem exist, and could this solution be copied across?

15

Close the project formally with the sponsor and release the team.

Verify the Gain, A3 Storyboard, Handover and Project Closure

Check your work

Your after-capability beats the baseline, the improvement is proven with a hypothesis test and a stated p-value, and the process owner has signed the handover.

Recap — Control

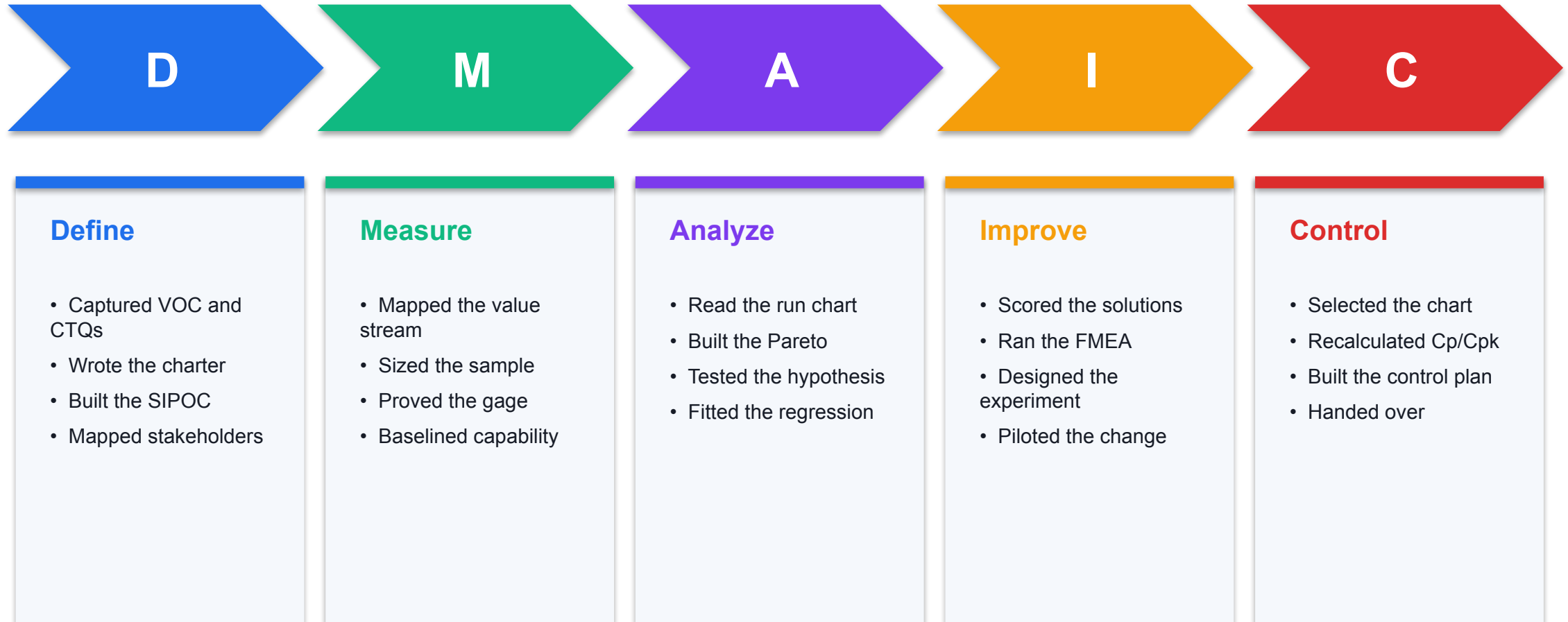
- Statistical Process Control — Monitor the process with data over time so drift is caught before it becomes a defect.
- Control chart anatomy — Centre line plus upper and lower control limits at ± 3 standard deviations.
- Control chart selection — Continuous or attribute? Subgroup size? That decides $\bar{X}$ -R, $\bar{X}$ -S, I-MR, p, np, c or u.
- Out-of-control rules — A point beyond 3 sigma, runs, trends and hugging patterns each signal a special cause.
- Control limits vs spec limits — Control limits come from the process voice; specification limits come from the customer.
- Cp and Cpk — $C_p = \text{spec width} / \text{process spread}$. Cpk also accounts for centring — aim for $C_{pk} \geq 1.33$.



WRAP-UP

Course Summary & Next Steps

What You Achieved — The Full DMAIC Journey



Your Integrated Improvement Package

1

Project selection & charter

Business case, problem statement, goal, scope and RACI.

2

Customer requirements

VOC log, affinity diagram, Kano classification and CTQ tree.

3

Process baseline

SIPOC, swimlane map, VSM, takt time and waste walk.

4

Valid measurement

Sampling plan, calculated sample size and Gage R&R verdict.

5

Performance baseline

Yield, RTY, DPMO, sigma level and baseline Cp/Cpk.

6

Proven root causes

Pareto, Fishbone, 5 Whys, hypothesis tests and regression.

7

Selected improvements

Weighted selection matrix, FMEA with RPNs and a pilot plan.

8

Sustained control

Control chart, control plan, SOP, A3 and signed handover.

CONSOLIDATE YOUR LEARNING

Green Belt Readiness Checklist

1

Can you explain each DMAIC phase and the deliverable that closes it?

2

Can you write a problem statement with no cause and no solution in it?

3

Can you calculate DPMO and convert it to a sigma level?

4

Can you decide whether a measurement system can be trusted?

5

Can you choose the right hypothesis test for a question and data type?

6

Can you apply the p-value rule and state the business conclusion?

Green Belt Readiness Checklist (2)

1

Can you select the correct control chart for your data?

2

Can you calculate Cp and Cpk and say whether the issue is spread or centring?

3

Can you apply the eight out-of-control rules to a control chart?

4

Can you build a control plan with named owners and reaction plans?

5

Can you prove your improvement held using a hypothesis test?

NEXT STEPS

Continuing Your Lean Six Sigma Journey

1

Run a real project

Apply DMAIC to one scoped process in your own area.

2

Black Belt

The next step — leads complex projects and mentors Green Belts.

3

Keep the templates

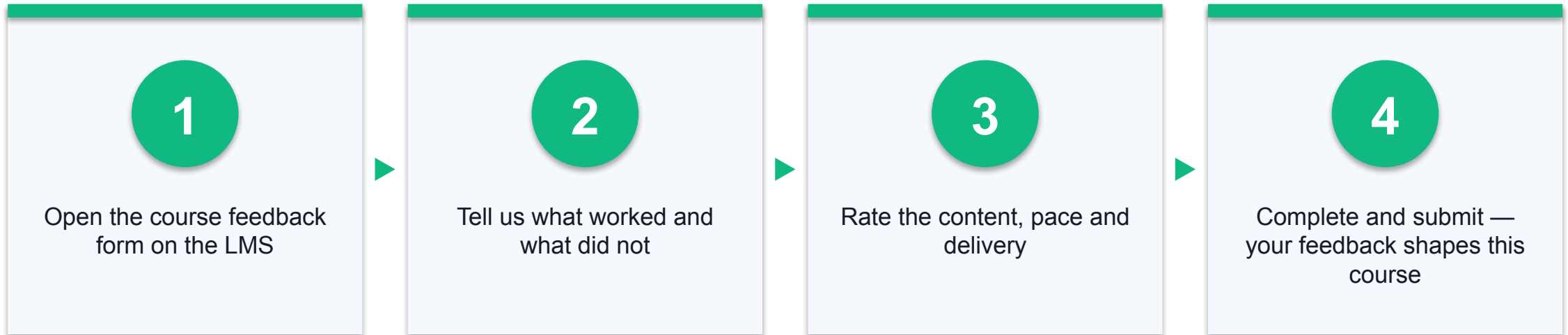
Your lab outputs are reusable templates for real projects.

4

Coach others

Teaching a Yellow Belt is the fastest way to deepen your practice.

Course Feedback



Certificate & Support

- A certificate of completion is awarded on full attendance.
- Email: enquiry@tertiaryinfotech.com
- Tel / WhatsApp: +65 6100 0613

END OF COURSE

Thank You!

Now go and lead one DMAIC project end to end — that is what a Green Belt is for.